2020-21

Computer Networks in 24 Hours

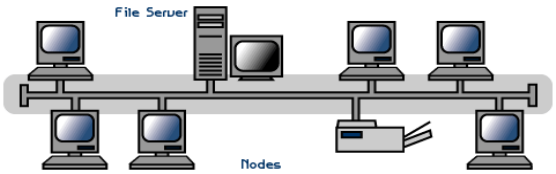
Prepared By:

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Prof. Parinda Prajapati, Asst. Prof. CE Dept, AIT

1. Explain network topologies with advantages and disadvantages.

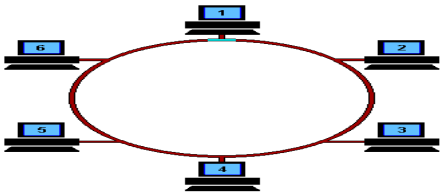
Network topology defines the structure of the network.

- ***Bus Topology:-***



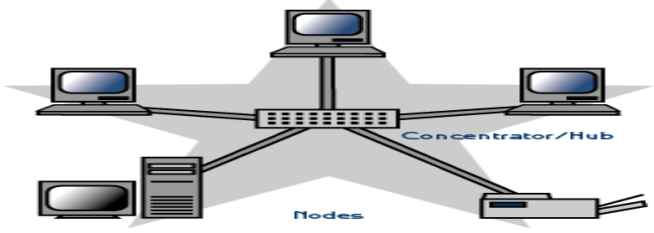
- Uses a single backbone cable.
- All the Nodes connect directly to this backbone.
- Advantage:-
  - Easy to Construct
  - Cost is Low
- Disadvantage:-
  - Traffic is High
  - If Link is Break Then no one can communicate with others.
  - Difficult to add new add Devices.

- ***Ring topology:-***



- Connects one Node to the next and the last Node to the first.
- Use the token.
- Token is passing from 1 Node to another Node.
- Only the node with the token is allowed to send data.
- Advantage:-If Link is broken then that node is reachable from other side of ring.
- Disadvantage:- Un directional Traffic

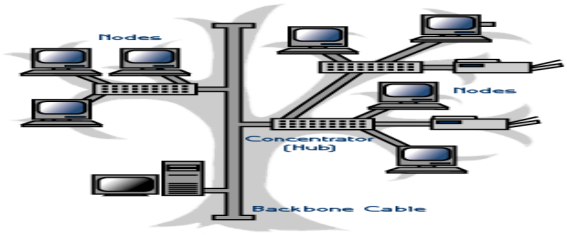
- ***Star topology:-***



- Each Node is directly connected to the central Devices.
- We can use Hub, switch, router as a Central Device.
- Advantage:-
  - If one link is broken then only that particular node is Unreachable.
  - Less Collision.
  - Most Efficient.
  - Simple and easy to identify Fault.
- Disadvantage:-
  - Traffic is more at Central Point.
  - All Nodes are depended on Central devices.

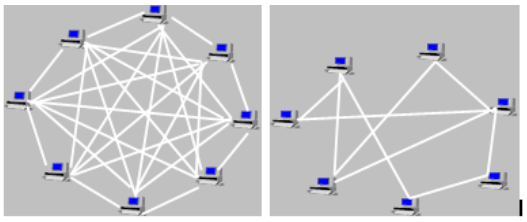
➢ ***Tree Topology:-***

- It's also known as hierarchical topology.
- All the nodes are connected in a tree structure.
- Advantage:-
  - Simple and easy to identify Fault.
- Disadvantage:-
  - If one link is broken then sub branch node can not send data to other side.



➢ ***Mesh topology:-***

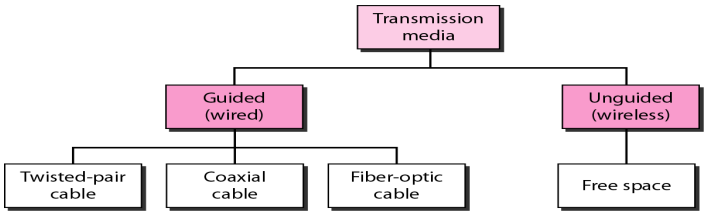
- Two type of Mesh Topology:-
  - Fully Connected
  - Partially Connected
- As seen in the graphic, each host has its own connections to all other hosts.



Fully Connected Partially Connected

- All the node has multiple paths to reach at Particular any one location.
- Advantage:-
    - Reachability is high.
    - Provide much Protection.
    - Used at a nuclear power plant
- Disadvantage:-
    - Cost is high.
    - Very Complex to Build.
    - Maintenance is also high.

2. Write a note on guided and Un-guided transmission medium.



3.3 Guided transmission media

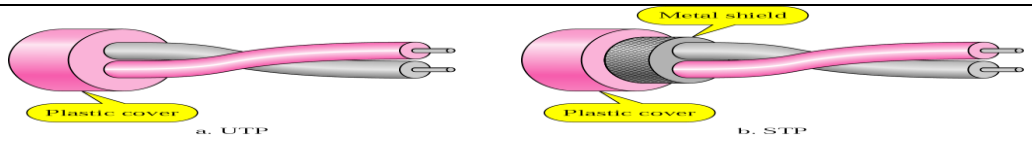
- Guided transmission is where the signal (information or data) is sent through some sort of ***cable, usually copper or optical fiber.***
- There are many different types of cabling:
    - **Twisted Pair**
    - **Coaxial Cable (coax)**
    - **Fiber Optic Cable**

3.3.1 Twisted Pair:

- This consists of two or more insulated wires twisted together in a shape similar to a helix.
- Use metallic conductor
- The cables are twisted around each other to reduce the amount of external interference
- It consist of two conductor (copper), each with it's colored plastic insulation.
- This cable can be used at speeds of several Mb/s for a few kilometers.
- Used for telephone line and lab network.

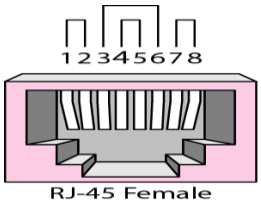


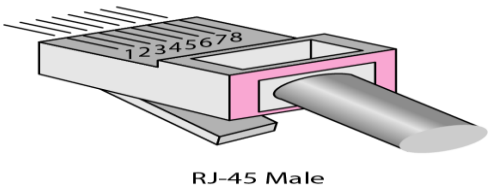
Twisted pair Cable



STP & UTP

- ***UTP connector***



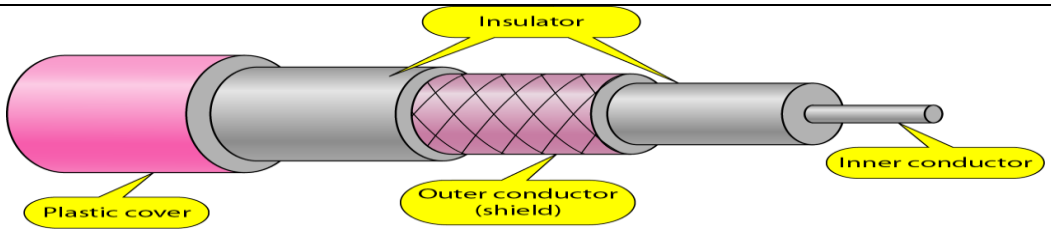


- ***Categories of unshielded twisted-pair(UTP) cables***
  CAT 1, CAT 2, CAT 3, CAT 4, CAT 5, CAT 5E, CAT 6, CAT 7

- **Advantages of UTP**
  - Cost is less
  - Easy to use
  - Easy to install
  - Flexible
- UTP Used in Ethernet and Token ring
- STP has a metal foil or cover.
- Crosstalk (effect of one channel on the other channel) is less in STP.
- STP has the same consideration as UTP
- Shield must connect with ground.
- Disadvantage of STP : cost is high

3.3.2 Coaxial Cable (coax)

- This consists of a copper cable inside a layer of insulating material.
- The insulating material is then inside a braided outer conductor.
- A layer of plastic is on the outermost layer.
- This type of cable was commonly used in the telephone system but has since been replaced by fiber optics on longer routes
- This cable has also been used for Cable TV.



- ***Categories of coaxial cables***

Category	*Impedance*	*Use*
RG-59	75 Ω	Cable TV
RG-58	50 Ω	Thin Ethernet
RG-11	50 Ω	Thick Ethernet

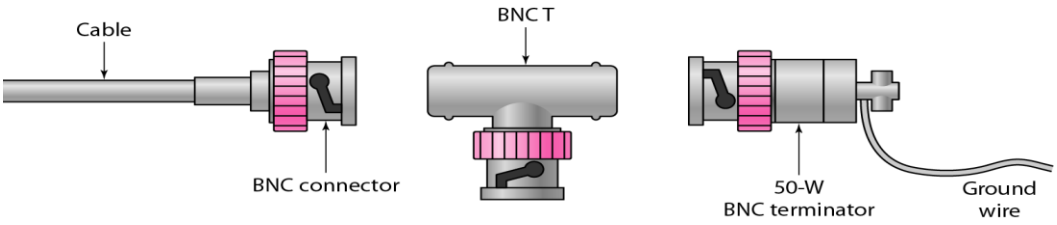
RG-Radio Government., Each cable define by RG ratting.

RG-8 &RG-9 also used in Thin Ethernet.

RG number denotes a unique set of physical Specification, including the wire gauge of the inner conductor.

Impedance

- ***Connector of coaxial cables***



The most common connector is barrel connector.

The most popular type is BNC(Bayonet Network connector)

Two other types are T-connectors and terminators.

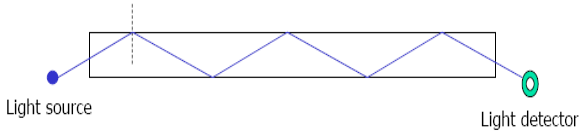
T-connectors and terminators are used in bus topology.

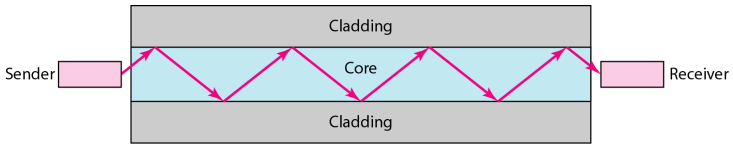
- **Advantages:-**
  - Easy to Install.
  - Inexpensive installation.
  - It is better for Higher Distance at Higher speed than twisted pair.
  - Excellent noise immunity.
- **Disadvantage:-**

- High Cost
- Harder to work

3.3.3 Fiber Optic Cable:

- **Components of Fiber Optics:-**
  - Light Source
  - Transmission Medium
  - Light Detector





- This consists of a central glass core, surrounded by a glass cladding of lower refractive index, so that the light stays in the core (using Total Internal Reflection)
- outside is covered with plastic jacket
- Many fibers may be bundled together surrounded by another plastic cover

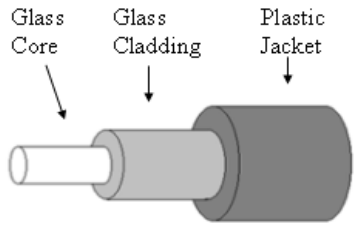


Fig. Single Optical Fiber

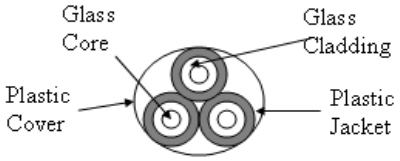


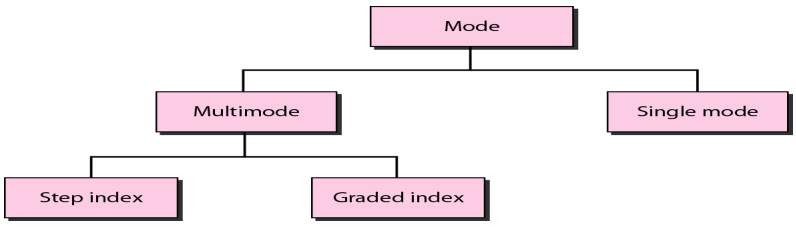
Fig. Bundled Optical Fibers

- **Light traveling of fiber optic cable**
  - The source of light is usually a **Light Emitting Diode** (LED) or a LASER. The light source is placed at one end of the optical fiber.
  - The detector, which is placed at the other end of the fiber, is usually a **Photo Diode** and it generates an electrical pulse when light falls on it.
  - Hence by attaching a light source on one end of an optical fiber and a detector at the other end, we have a unidirectional data transmission system (Simplex)
  - The light source would accepts an electrical signal, converts and transmits it as light

pulses

- The detector at the far end reconverts the light pulses into an electrical signal to be then interpreted as 1 or a 0.
- The limits the data rate is 1Gb/sec (1x109 bits / sec)

- **Propagations Mode:-**

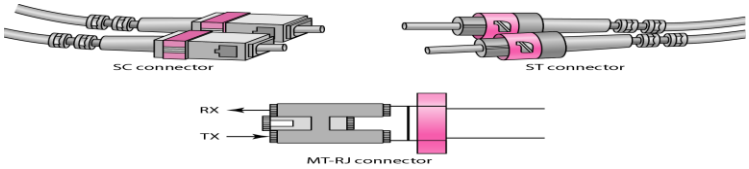


- **Two different light sources – both emit light when voltage applied**
  - LED – Light Emitting Diode – less costly, longer life
  - ILD - Injection Laser Diode – greater data rate
- **Advantages of Fiber Optic over Copper Cable**
  - Fiber can handle much higher data rates than copper(More information can be sent in one second using fiber)
  - Fiber has low loss of signal power (**attenuation**), so repeaters are needed every 100km rather than every 5km for copper
  - Fiber is not affected by power surges, electromagnetic interference or power failure, or corrosive chemicals in the air
  - Fibers are difficult to tap and therefore excellent for security
  - Fibers are thin and lightweight, allowing more cables to fit into a given area.
  - Noise Resistance.
  - 1000 twisted pair cables 1 km long = 800kg
    2 optical fiber cables 1km approx = 100kg allows transfer of more data

- **Disadvantages of Fiber Optic over Copper Cable**
  - Fiber technology is relatively new and certain new skills are required in handling it
  - Optical transmission in a fiber is one way only (Simplex) – if you want two way communication, then you must use two fibers or else use two frequency bands on the one fiber
  - Fiber optic cables and network interface cards to connect a computer to the fiber are an order of magnitude more expensive than their corresponding copper cable equivalents

- Cost is high
- Installation and maintenance.
- Higher Bandwidth.

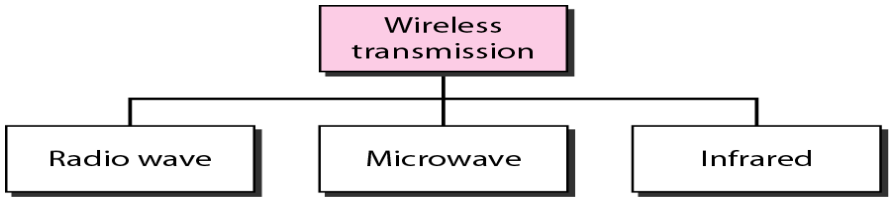
- **Fiber cable Connector**



3.4 Unguided transmission media

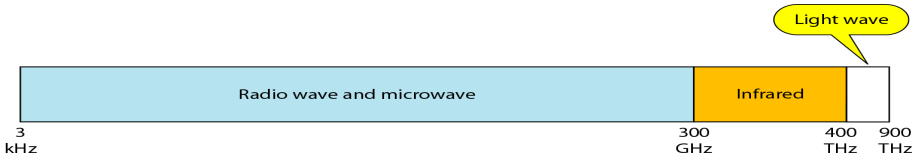
Introduction

Information is usually transmitted by either radio or microwave transmission. Unguided media transport electromagnetic waves without using a physical conductor. Signals are broadcast through air (or in a few cases, water).



3.4.1 Radio Transmission

- Radio waves are easy to generate and can travel long distances and penetrate buildings.
- Radio waves are omni-directional which basically means that they can transmit both ways.
- The transmitter and receiver do not have to be in direct line of sight



- ***Bands***

Band	*Range*	*Propagation*	*Application*
VLF (very low frequency)	3–30 kHz	Ground	Long-range radio navigation
LF (low frequency)	30–300 kHz	Ground	Radio beacons and navigational locators
MF (middle frequency)	300 kHz–3 MHz	Sky	AM radio
HF (high frequency)	3–30 MHz	Sky	Citizens band (CB), ship/aircraft communication
VHF (very high frequency)	30–300 MHz	Sky and line-of-sight	VHF TV, FM radio
UHF (ultrahigh frequency)	300 MHz–3 GHz	Line-of-sight	UHF TV, cellular phones, paging, satellite
SHF (superhigh frequency)	3–30 GHz	Line-of-sight	Satellite communication
EHF (extremely high frequency)	30–300 GHz	Line-of-sight	Radar, satellite

- **Radio Transmission Properties**
  - At low frequencies (<100MHz) radio waves pass through obstacles well but the signal power attenuates (falls off) sharply in air
  - At higher frequencies (>100MHz) radio waves tend to travel in straight lines and bounce of obstacles and can be absorbed by rain (e.g in the 8GHz range)
  - At all frequencies, radio waves are subject to interference from motors and other electrical equipment
    - In very low frequencies (VLF), low frequencies (LF) and medium frequency bands (MF) (<1 MHz) radio waves follow the ground.(The maximum possible distance that these waves can travel is approximately 1000km)

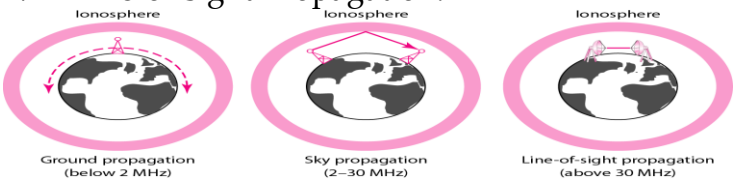
3.4.2 Microwave Transmission

- ***Different types of Propagation***

⇒ Surface Propagation (Ground Propagation)

⇒ Sky Propagation.(Tropospheric Propagation, Ionospheric Propagation, Space Propagation)

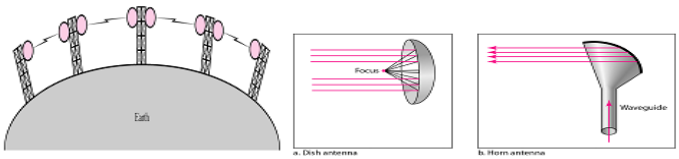
⇒ Line-of-sight Propagation.



- Unlike radio waves, microwaves typically do not pass through solid objects
- Some Waves can be refracted due to atmospheric conditions and may take longer to arrive than direct waves. These delayed waves can arrive out of phase with the direct wave, causing **destructive interference** and corrupting the received signal This effect is called **multipath fading**
- Because of increased demand for more spectrum (range of frequencies used to transmit), transmitters are using higher and higher frequencies

- Microwave communication is widely used for long distance telephone communication and cell phones.
- Microwave signals propagates in one direction at a time which means two different frequencies are necessary for two way communication.
- Transmitter is used to transmit the signal
- Receiver is used to receive the signal
- Transceiver is a equipment, which work as a both Transmitter and Receiver.

Terrestrial Microwave



Terrestrial Microwave **Different Types of Antenna**

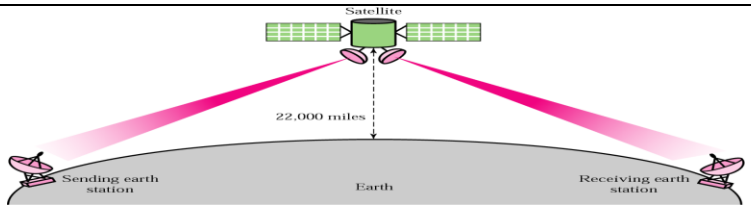
- Two types of antenna are used for terrestrial microwave

Communication: Parabolic dish and Horn.

- Narrow beam – line of sight on towers to avoid obstacles
- Series of towers for long distance
- Repeaters: to increase the distance served by terrestrial microwave a system of repeater install with each antenna.
- Applications:
  - Long telephone line
  - Voice and TV
  - Short point to point between buildings
- Main Source of loss
  - Attenuation – especially with rainfall
  - Repeaters or amplifiers 10 to 100km
  - Interference with overlapping bands

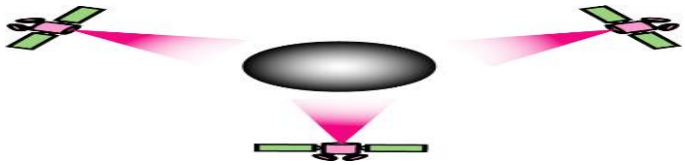
- **Satellite Communication:**

It's Line-of-site Microwave Transmission.



- Operates on a number of frequency bands known as transponders
  - Point to Point (Ground station to satellite to ground station)
  - Multipoint(Ground station to satellite to multiple receiving stations)
- Satellite orbit
  - 35,784 Km, to match earth rotation
  - Stays fixed above the transmitter/receiver station as earth rotates
- Satellites need to be separated by distance
  - Avoid interference
- Applications
  - TV, long distance telephone(satellite phone), private business networks
- Optimum frequency range
  - 1 – 10 GHz
  - Below 1GHz results in noise, above 10GHz results in severe attenuation

- **Satellites in geosynchronous orbit**



- **Frequency bands for satellite communication:**

Each satellite sends and receives over two different bands. Transmission from the earth to the satellite is called **uplink**. Transmission from the satellite to the earth is called **downlink**.

Band	Downlink	Uplink
C	3.7 to 4.2 GHz	5.925 to 6.425 GHz
Ku	11.7 to 12.2 GHz	14 to 14.5 GHz
Ka	17.7 to 21 GHz	27.5 to 31 GHz

- **Advantages of Microwave over Fiber Optics**
  - No need to lay cables
  - This causes less disruption to the areas where the microwave transmitters and receivers are placed
  - This also means that microwave communication is less expensive than fiber optic cable

3.4.3 Infrared:

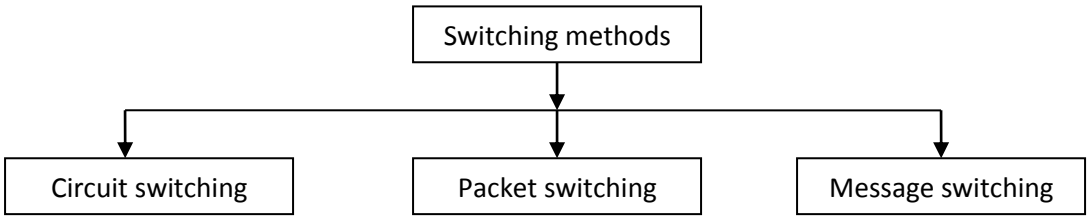
- Infrared signals can be used for short-range communication in a closed area using line-of-sight propagation.
- Transceivers must be within line of sight of each other or via reflection
- Does not penetrate walls like microwave
- No frequency allocation or licensing

3.4.4 Bluetooth

- **Bluetooth** is a wireless technology standard for exchanging data over short distances (2.4 to 2.485 GHz) from fixed and mobile devices and building personal area networks (PANs).
- Invented by telecom vendor Ericsson in 1994, it was originally conceived as a wireless alternative to RS-232 data cables.
- It can connect several devices, overcoming problems of synchronization.
- Penetrate Wall or other objects.
- No Line-of-sight required.

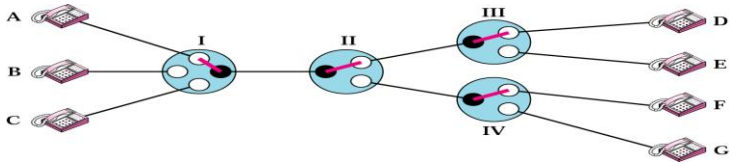
3. Explain Different types of Switching methods with examples.

There are three methods of switching.



A) Circuit Switching

- A circuit switch is a device with *n* inputs and *m* outputs that creates a temporary connection between an input link and an output link. The number of inputs does not have to match the number of outputs.



- It can use either of two technologies:
  (i) Space-division switching,
  (ii) Time-division switches.

- Used in the telephone system: network resources in the telephone system are reserved from your phone to the phone you call when you place the call; they're released when you hang up.

In short, circuit switching..

- *Dedicated communication path* between two stations
- Three phases
  - Establish
  - Transfer
  - Disconnect
- Must have switching capacity and channel capacity to establish connection
- Must have intelligence to work out routing
- Inefficient
  - Channel capacity dedicated for duration of connection
  - If no data, capacity wasted because of dedicated link.
- Set up (connection) takes *time*
- Once connected, transfer is transparent
- Developed for voice communication (phone)
- Resources dedicated to a particular call
- Much of the time a data connection (line) is idle and facilities are wasted.
- Circuit switching is inflexible. Once a circuit has been established, that circuit is the path taken by all parts of the transmission whether or not it remains the most efficient or available.
- Circuit switching sees all transmissions as equal.
- Data rate is fixed-Both ends must operate at the same rate

B) <u>Packet Switching</u>

- Circuit switching designed for voice

So a better solution, For data transmission is **packet switching.**

<u>Basic Operation:</u>

- Data transmitted in small packets
  - Typically 1000 octets
  - Longer messages split into series of packets
  - Each packet contains a portion of user data plus some control info
- Control info

- Routing (addressing) info
- Packets are received, stored briefly (buffered) and past on to the next node
  - Store and forward

Advantages

- Line efficiency
  - Single node to node link can be shared by many packets over time
  - Packets queued and transmitted as fast as possible
- Data rate conversion
  - Each station connects to the local node at its own speed
  - Nodes buffer data if required to equalize rates
- Packets are accepted even when network is busy
  - Delivery may slow down
- Priorities can be used

There are two popular approaches to packet switching:

(i) Datagram approach (Connectionless service)

(ii) Virtual circuit approach. (Connection oriented)

(i) Datagram

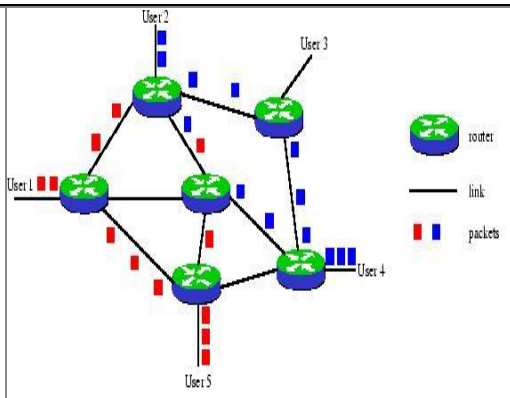
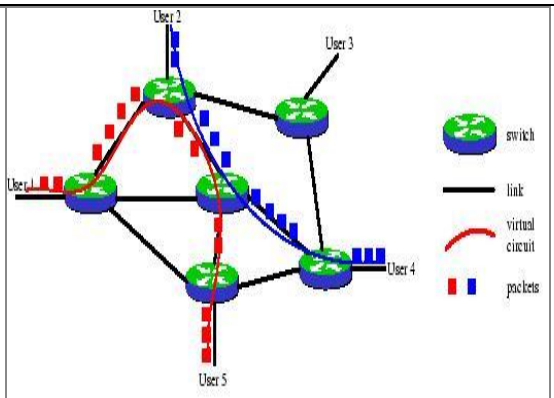
- Every packet contains a complete destination address.
- Switch contains a routing table
- Two successive packets may follow different paths.
- Each switch processes packets independently.

(ii) Virtual Circuit Switching

- Based on the connection oriented model
- Virtual connection has to be established first
- Two stage process:- connection setup, data transfer
- Two type of virtual circuits
  - Permanent virtual circuits (PVC)
  - Switched virtual circuit (SVC)

Virtual circuit and Datagram network Comparison

	Datagram	Virtual Circuit

	Figure		
	Connection Setup	None (Connection setup is initially required prior to sending data)	Required
	Addressing	Packet contains full source and destination address	Packet contains short virtual circuit number identifier.
	State Information	Not Hold	Hold information for Routing
	Routing	Packets routed independently	Route established at setup, all packets follow same route.
	Effect of Router Failure	Only on packets lost during crash	All virtual circuits passing through failed router terminated.
	Congestion Control	**Difficult** since all packets routed independently router resource requirements can vary.	**Simple** by pre-allocating enough buffers to each virtual circuit at setup, since maximum number of circuits fixed.

C) <u>Message Switching</u>

- A store-and-forward network where the block of transfer is a complete *message*.
- Since messages can be quite large, this can cause:
  - buffering problems
  - high mean delay times

4. Explain following terms:

1) **Processing Delay**
2) **Propagation Delay**
3) **Queuing Delay**
4) **Transmission Delay**
5) **Latency**

	• **Processing Delay (Nodel Delay) [D_{proc}]:** Time required examining the packet header and determining where to direct the packet. • **Propagation[D_{prop}]:** This is simply the time it takes for a packet to travel between one place to another at the speed of **light**. IT is a simple measurement of how long it takes for a signal to travel along the cable being tested Propagation time= Distance/ Propagation speed • **Queuing Delay [D_{queue}]** It is a waiting time experienced by packet for transmission across the link. • **Transmission time [D_{trans}](Transmission time Delay):** It is the time between the first bit leaving the sender and the last bit arriving at the receiver. Transmission time = Message size/Bandwidth Transmission time = Packet Length/Transmission Rate=L/R • **Latency (Total Nodel Delay)** Time it takes for a packet of data to get from one designated point to another **Latency** = Propagation time + Transmission time + Queuing time + Processing delay. **Latency =[D_{prop}] + [D_{trans}] + [D_{queue}] +) [D_{proc}]**
5.	**1) Consider a packet of length L which begins at end system A and travels over three links to a destination end system. These three links are connected by two packet switches. Let di, si, and Ri denote the length, propagation speed, and the transmission rate of link i, for i = 1, 2, 3. The packet switch delays each packet by dproc. Assuming no queuing delays, in terms of di, si, Ri, (i = 1, 2, 3), and L, what is the total end-to-end delay for the packet? Suppose now the packet is 1,500 bytes, the propagation speed on all three links is 2.5 · 108 m/s, the transmission rates of all three links are 2 Mbps, the packet switch processing delay is 3 msec, the length of the first link is 5,000 km, the length of the second link is 4,000 km, and the length of the last link is 1,000 km. For these values, what is the end-to-end delay?** **Answer:** packet length = L link i Length = di propagation speed = si

transmission rate = Ri
the first end system requires to transmit the packet onto the first link = L/R1
the first end system requires to transmit the packet onto the second link = L/R2
the first end system requires to transmit the packet onto the third link = L/R3
the packet propagates over the first link = d1/s1
the packet propagates over the second link = d2/s2
the packet propagates over the third link = d3/s3
end to end delay = L/R1+L/R2+L/R3+d1/s1+d2/s2+d3/s3+d(proc)+ d(pro)
packet size = 1500 bytes = 1500*8 bits
propagation speed on both links = 2.5 x 10^8
transmission rate of all three links = 2 Mbps
packet switch processing delay = 3 msec
length of the first link = 5000 km
length of the second link = 4000km
length of the last link = 1000 km
the first end system requires to transmit the packet onto the first link = L/R1 = .006 sec
packet propagates over the first link = .02 sec
the packet switch requires to transmit the packet onto the first link = L/R1 = .006 sec
packet propagates over the second link = .016 sec
the packet switch requires to transmit the packet onto the third link = L/R1 = .006 sec
packet propagates over the third link = .004 sec
end to end delay = .006+.006+.006+.02+.016+.004+.003+.003
= 0.064 sec=64 msec

2) **Suppose two hosts, A and B, are separated by 20,000 kilometers and are connected by a direct link of R = 2 Mbps. Suppose the propagation speed over the link is 2.5* 10^8 meters/sec.**
   a. **Calculate the bandwidth-delay product, R* dprop.**
   b. **Consider sending a file of 800,000 bits from Host A to Host B. Suppose the file is sent continuously as one large message. What is the maximum number of bits that will be in the link at any given time?**
   c. **Provide an interpretation of the bandwidth-delay product.**
   d. **What is the width (in meters) of a bit in the link? Is it longer than a football field?**
   e. **Derive a general expression for the width of a bit in terms of the propagation speed s, the transmission rate R, and the length of the link m.**

f. For what value of R is the width of a bit as long as the length of the link?

Answer:

a) R* (D/V) = 2M *20,000 k / 2.5*10^8 = 160,000 bits

b) 160,000 bits because the link can only has a number of bits equals bandwidth delay product.

c) The bandwidth-delay product of a link is the maximum number of bits that can be in the link.

d) the width of a bit = length of link / bandwidth-delay product, so 1 bit is 125 meters long.

e) *s/R*

3) Consider sending real-time voice from Host A to Host B over a packet-switched network (VoIP). Host A converts analog voice to a digital 64 kbps bit stream on the fly. Host A then groups the bits into 56-byte packets. There is one link between Hosts A and B; its Transmission rate is 2Mbps and its propagation delay is 10 msec. As soon as Host A gathers a packet, it sends it to Host B. As soon as Host B receives an entire packet, it converts the packet's bits to an analog signal. How much time elapses from the time a bit is created (from the original analog signal at Host A) until the bit is decoded (as part of the analog signal at Host B)?

Answer:

Consider the first bit in a packet. Before this bit can be transmitted, all of the bits in the packet must be generated. This requires

1. The time required to make analog- to digital conversion

   56*8/(64*10^3)sec=7msec.

2. The time required to transmit the packet is

   56*8/(2*10^6)sec=224μsec.

3. Propagation delay = 10 msec.

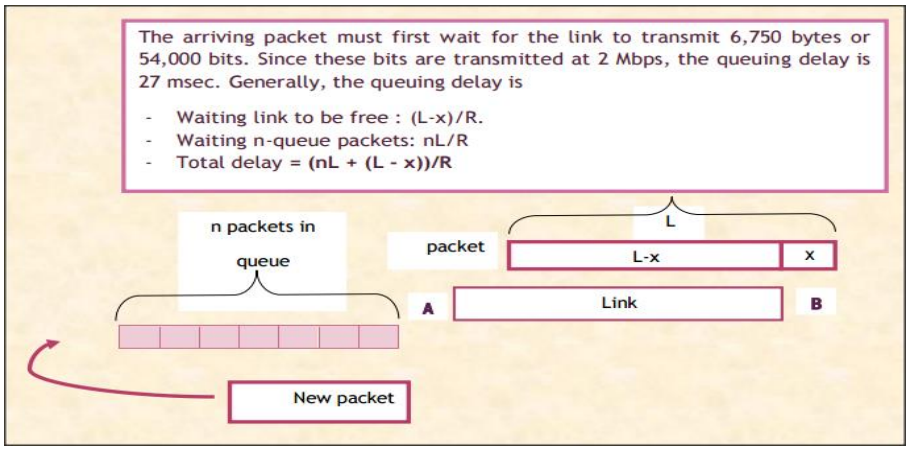
The delay until decoding is

7msec +μ224sec + 10msec = **17.224msec**

4) A packet switch receives a packet and determines the outbound link to which the packet should be forwarded. When the packet arrives, one other packet is halfway done being transmitted on this outbound link and four other packets are waiting to be transmitted. Packets are transmitted in order of arrival. Suppose all packets are 1,500 bytes and the link rate is 2 Mbps. What is the queuing delay for the packet? More generally, what is the queuing delay when all packets have length L, the transmission rate is R, x bits of the currently-being-transmitted packet have been

transmitted, and n packets are already in the queue?

Answer:

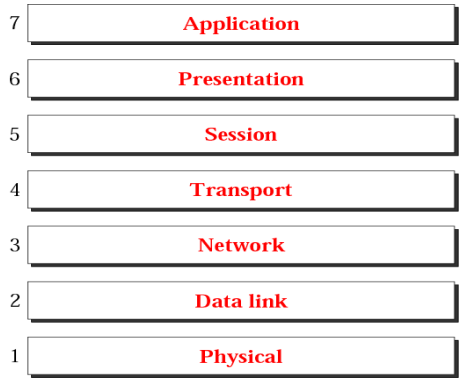


6. **Answer the following:**

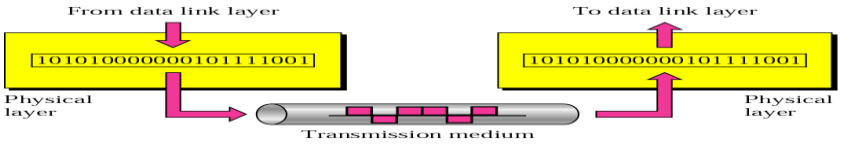
1) **Explain OSI Layers. (Generally two to three layers can be ask in one question)**
2) **Draw the layered architecture of TCP/IP model and write at least two services provided by each layer of the model.**
3) **What are the five layers in the Internet protocol stack? What are the principal responsibilities of each of these layers?**

1) Explain OSI Layers.

- OSI stands for Open System Interconnections.
- OSI model is used for understanding the concept of network architecture.
- OSI consists of **seven** separate but **related layers**, each of which defines a segment of the process of moving information across a network.



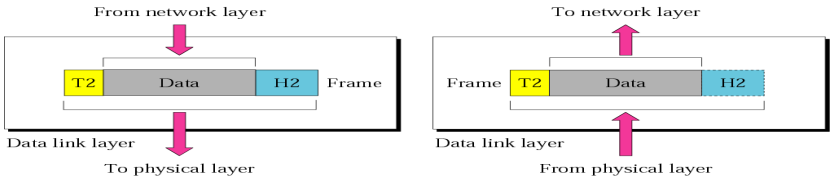
Physical Layer



- **Physical characteristic of interfaces and media**
  - Transmission media between Source and Destination
  - It may be wired or wireless
- **Representation of bits**
  - Bit must be encoded into signals-Electrical or Optical.
  - It is define the type of encoding.
- **Data rate**
  - Rate of data transmission / No. of bits transmitted per second
- **Synchronization**
  - Sender and Receiver must be synchronies at bit level.
  - Clock must be synchronies.
- **Line configuration**
  - Point-to-Point configuration & multipoint configuration
- **Physical topology**
  - Type of topology:-Star, Mesh, Ring, Bus.
- **Transmission mode**
  - Simplex, half-duplex or full-duplex

<u>Data Link Layer</u>

- Transmitting Frame
- Responsible for Node to Node delivery
- Makes the physical layer appear error free to the upper layer (network layer).



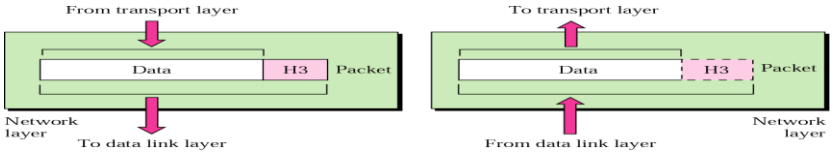
Responsibilities of the data link layer

- **Framing:**
  - the data link layer divides the stream of bits received from the network layer into manageable data units called frames.
- **Physical addressing:**
  - Add MAC Address (Layer 2 address/physical address) of Source and Receiver.
- **Flow control:**
  - Flow of data must be controlled.
  - If sending rate of sender is higher than receiving rate of receiver then flow

control is required.

- **Error Control:**
  - Adds reliability
  - Detect and Retransmit damaged or lost frames.
  - Prevent duplication of frames.
  - Trailer is added to the end of the frame for error controlling.
- **Access Control:**
  - Determine the Controlling of node over the link at any time.
  - HUB and Switch (L2) are operated at Layer2.

Network Layer:



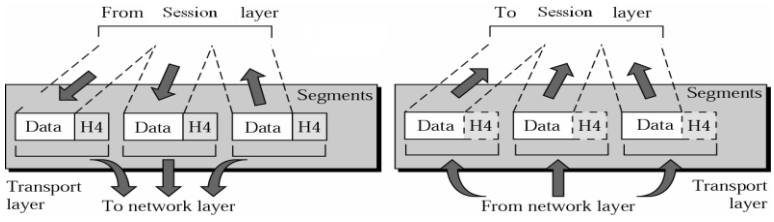
- It is Responsible for the source-to-destination delivery of a packet.
- The network layer ensures that each packet gets from its point origin to its final destination.

➢ **Specific responsibilities of the network layer include the following:**

- **Logical Addressing:**
  - Add IP Address (Layer 3 address/Logical address) of Source and Receiver.
- **Routing:**
  - Route the packets to their final destination
  - Switch (l3) and Router is operated at Layer3.
  - Routing table is maintained by router.

Transport Layer

- Responsible for **source-to-destination** (end-to-end) delivery of the entire message
- Network layer oversees end-to-end delivery of individual packets; it does not recognize any relationship between those packets.
- The transport layer ensures that the whole message arrives in order.
- It is managing both error control and flow control at the source-to-destination level.



➢ **Responsibilities of the transport layer**

- **Service-point Addressing:**
  - Computers often run several programs at the same time.
  - The transport layer header therefore must include a type of address called a service-point address (or port address).
- **Segmentation and reassembly:**
  - A message is divided into transmittable segments.
  - Each segment is containing a sequence number.
  - Reassemble the message correctly upon arriving at the destination and to identify and
  - Replace packets that were lost in transmission.
- **Connection Control:**
  - The transport layer can be either connectionless or connection-oriented.
  - A connectionless transport layer treats each segment as an independent packet and delivers it to the transport layer at the destination machine.
  - A connection-oriented transport layer makes a connection with the transport layer at the destination machine first before delivering the packets. After all the data are transferred, the connection is terminated.
- **Flow control:**
  - It is responsible for flow control for end to end rather than across a single link.
- **Error control:**
  - It is responsible for error control. for end to end rather than across a single link.
  - The sending transport layer makes sure that entire message arrives at the receiving transport layer without error (damage, loss or duplication).
  - Error correction is usually achieved through retransmission.

Session Layer

- The **session layer** is network *dialog controller*.
- It establishes, maintains, and synchronizes the interaction between communicating systems.

➢ **Specific responsibilities of the session layer include the following:**

- **Dialog control:**
  - The session layer allows two systems to enter into a dialog.
  - It allows the communication between two processes to take place either in half-duplex (one way at a time) or full-duplex (two ways at a time). For example, the dialog between a terminal connected to a mainframe can be half-duplex.
- **Synchronization:**

- It allows a process to add checkpoints (synchronization points) into a stream of file.
- For example, if a system is sending a file of 2000 pages, it is advisable to insert checkpoints after every 100 pages to ensure that each 100-page unit is received and acknowledged independently. In this case, if a crash happens during the transmission of page 523, retransmission begins at page 501: pages 1 to 500 need not be transmitted.

Presentation Layer

The presentation layer is concerned with the syntax and semantics of the information exchanged between two systems.

➢ **Specific responsibilities of the presentation layer include the following:**

- **Translation:**
  - The processes (running programs) in two systems are usually exchanging information in the form of character strings, numbers, and so on.
  - The information should be changed to bit streams before being transmitted. Because different computers use different encoding systems, the presentation layer is responsible for interoperability between these different encoding methods.
  - The presentation layer at the sender changes the information from its sender-dependent format into a common format. The presentation layer at the receiving machine changes the common format into its receiver-dependent format.
- **Encryption:**
  - To carry sensitive information, a system must be able to assure privacy.
  - Encryption means that the sender transforms the original information to another form and sends the resulting message but over the network. Decryption reverses the original process to transform the message back its original form.
- **Compression:**
  - Data compression reduces the number of bits to be transmitted.
  - Data compression becomes particularly important in the transmission of multimedia such as text, audio, and video.

Application Layer

- The application layer, enables the user, whether human or software, to access the network.

- It provides user interfaces and support for services such as electronic mail, remote file access and transfer, shared database management, and other types of distributed information services.
- The application layer, enables the user, whether human or software, to access the network.
- It provides user interfaces and support for services such as electronic mail, remote file access and transfer, shared database management, and other types of distributed information services.

➢ **Specific responsibilities of the application layer include the following:**

- **Network virtual terminal:** A network virtual terminal is a software version of a physical terminal and allows a user to log on to a remote host. To do so, the application creates a software emulation of a terminal at the remote host.
- **File transfer, access, and management (FTAM):** This application allows a user to access files in a remote computer (to make changes or read data), to retrieve files from a remote computer; and to manage or control files in a remote computer.
- **Mail services:** This application provides the basis for e-mail forwarding and storage.
- **Directory services:** This application provides distributed database sources and access for global information on about various objects and services.

2) Draw the layered architecture of TCP/IP model and write at least two services provided by each layer of the model.



OSI model TCP/IP model

TCP/IP model

Application Layer

- File Transfer Protocol (FTP)
- Trivial File Transfer Protocol (TFTP)
- Network File System (NFS)
- Simple Mail Transfer Protocol (SMTP)
- Telnet

- Simple Network Management Protocol (SNMP)
- Domain Name System (DNS)

Transport Layer

- TCP (Transmission Control Protocol)
- UDP(User Datagram Protocol)
- TCP is connection oriented & UDP is connection less
- TCP or UDP creates is called a segment or user datagram
- The functions of UDP are as follows:
  - Segment upper-layer application data
  - Send segments from one end device to another
- The functions of TCP are as follows:
  - Establish end-to-end operations
  - Provide flow control through the use of sliding windows
  - Ensure reliability through the use of sequence numbers and acknowledgments

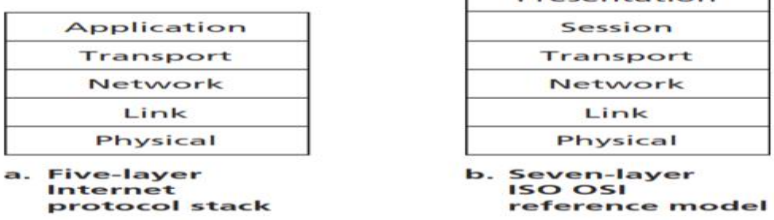
Internet Layer

- IP stands for Internet Protocol
- IP provides connectionless, best-effort delivery routing of packets.
- IP is not concerned with the content of the packets but looks for a path to the destination.
- Four supporting protocols
  - ARP (Address Resolution Protocol)
  - RARP (Reverse Address Resolution Protocol)
  - ICMP (Internet Control Message Protocol)
  - IGMP (Internet Group Message Protocol)

Network Access Layer

- Ethernet, Fast Ethernet and Gigabit Ethernet

3) What are the five layers in the Internet protocol stack? What are the principal responsibilities of each of these layers?



Five Layer Internet Protocol Stack.

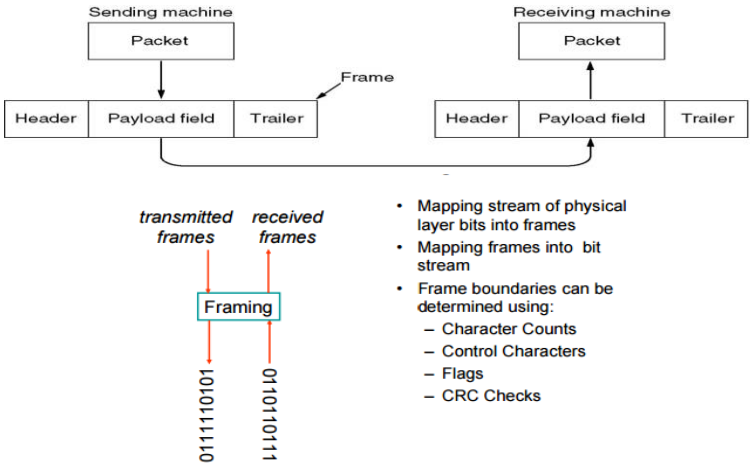
Student have to explain the same thing as given above but in five parts

Note: AS per the book Computer Networking – A Top Down Approach By Kurose & Rose

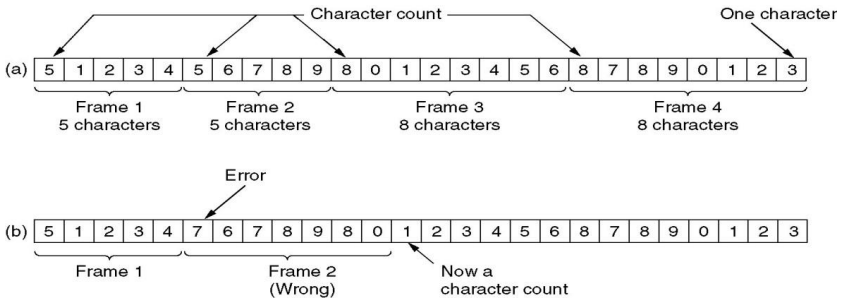
7. Enlist various framing techniques used in data link layer. Explain any one in detail with suitable example.

Framing

In the OSI model of computer networking, a frame is the protocol data unit at the data link layer. Frames are the result of the final layer of encapsulation before the data is transmitted over the physical layer



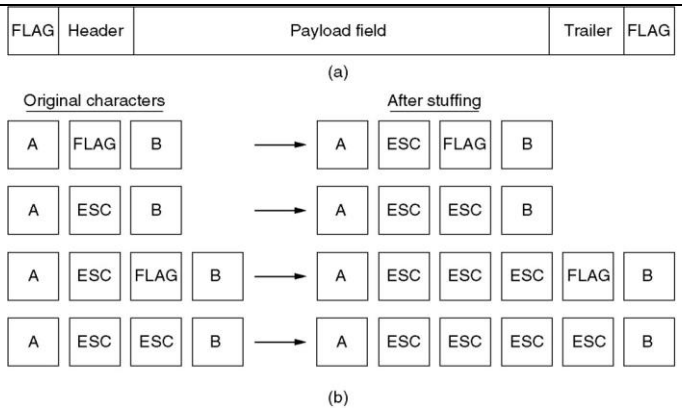
Framing: Using Counters



(a) Without errors. (b) With one error.

Framing: Flag Byte

- *Each frame starts and ends with special bytes: flag bytes.*
- *Two consecutive flag bytes indicate end of frame and beginning on new frame.*
- *Problem?*
  - *What if flag bit pattern occurs in data?*

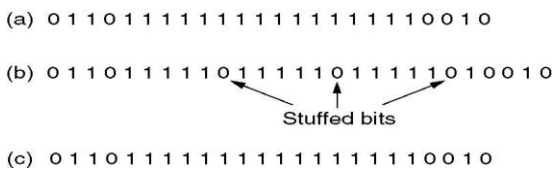


- *(a) A frame delimited by flag bytes.*
- *(b) Four examples of byte sequences before and after stuffing.*
    - *Single ESC: part of the escape sequence.*
    - *Doubled ESC: single ESC is part of data.*
    - *De-stuffing.*
    - *Problem:*
        - *What if character encoding does not use 8-bit characters?*

Bit Stuffing

- *Allows character codes with arbitrary bits per character.*
- *Each frames begins and ends with special pattern.*
- *Example: 01111110.*
- *When sender's DLL finds 5 consecutive 1's in data stream, stuffs 0.*
- *When receiver sees 5 1's followed by 0, de-stuffs.*

Bit Stuffing: Example



(a) Original data.

(b) Data as they appear on the line.

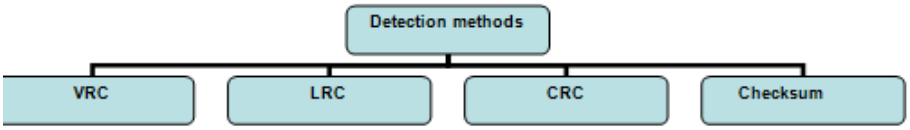
(c) Data after de-stuffing.

8. **Explain with example:**

1) Error detection methods at Data link layer

2) Error correction method at Data link layer

1) Error detection methods at Data link layer (Generally one method can be ask)



[1] VRC –Vertical Redundancy Check

- It's least expensive method.
- It's also called Parity Checking and bit is called parity bit.
- This Parity bit is append to every data unit, So the total number of 1's (Including parity bit) becomes either Even or Odd.
- ***Example:-***

  If data is 10101010

  And if user wants data with Even Parity then Data becomes 10101010**0**{Total number of 1 is Four},

  And if user wants data with Odd Parity then Data becomes 10101010**1**{Total number of 1 is Five}
- VRC can detect all single bit error.
- It also detects burst error as long as the total number of bit changed to Odd (1, 3, 5…) & For Odd parity, and total number of bit changed to Even (2, 4, 6…) For Even parity.

[2] LRC –Longitudinal Redundancy Check

- Blocks of bits are organized in a table format (means In a Row and Column).
- **Example:**

  We have data

  01100111000111010001100100101001

  32 bit block is divided into 4 rows and 8 columns.

  01100111 00011101 00011001 00101001

  Arrange it in table Format.

  01100111
  00011101
  00011001
  00101001

  01001010 ← **LRC**

  Sender sends Original Data plus LRC to the receiver.

  01100111 00011101 00011001 00101001 **01001010**
- It generally used for detect burst error
- Sometimes LRC checker can not be detecting an error.
- Example: If data is 11110000 and 11000011.

  And if the first and last bit in each of them are changed,

  Then Data becomes 01110001 and 01000010 and LRC not find an Error.

[3] CRC –Cyclic Redundancy Check

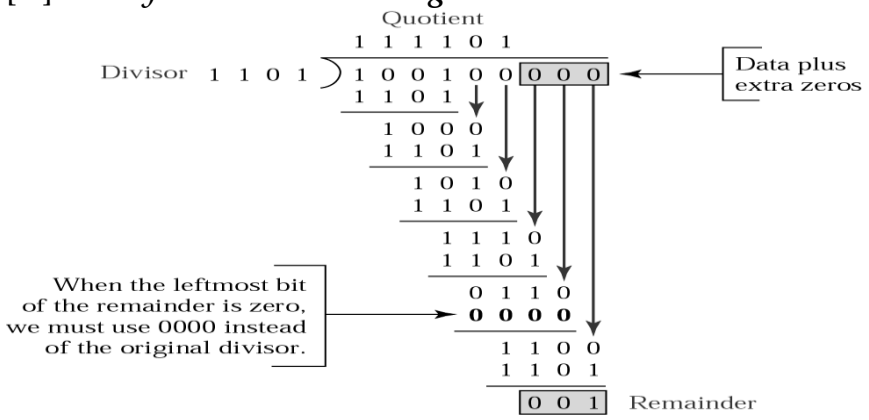
- CRC is based on Binary Division.

- CRC (CRC reminder) is appended to the end of Data unit.
- CRC reminder has exactly one less bit then Divisor.
- Data is also called frame
- Divisor is also called generator.
- Most powerful redundancy checking system.

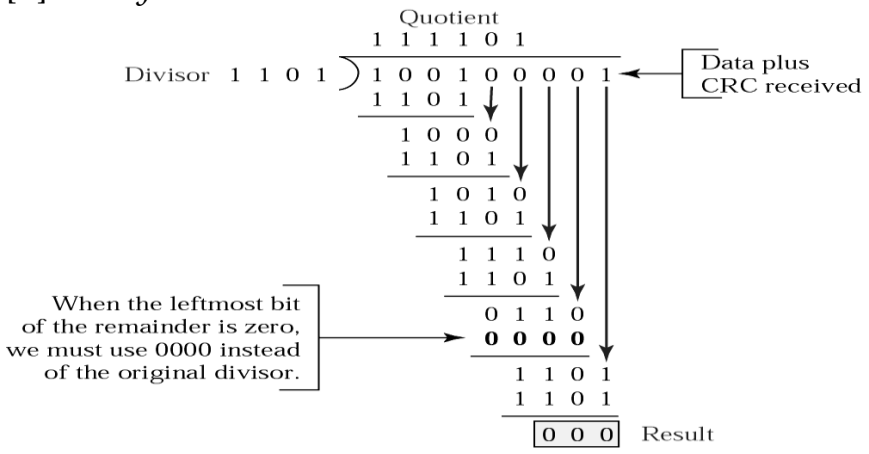
- ***CRC generator and checker***

Example-1

[A] ***Binary division in a CRC generator***



[B] ***Binary division in CRC checker***



[4] Checksum:

- It is an error detection method used by higher layer protocol.
- **The Sender follows these steps.**
  1) Divide whole data unit into k section, each contain n bits.
  2) All sections are added together.
  3) The sum is complemented and become the checksum.
  4) The checksum is send with the data.
- **The Receiver follows these steps.**
  1) Divide whole data unit into k section, each contain n bits.
  2) All sections are added together.
  3) The sum is complemented.

4) If the result is zero then data are accepted: otherwise data is rejected.

- **Example:**

At sender side

Data of 16 bit

10101001 00111001

Addition of this two number (Each contain 8 bit).is:

1 0 1 0 1 0 0 1
0 0 1 1 1 0 0 1

SUM-→ 1 1 1 0 0 0 1 0

CHECKSUM: → 0 0 0 1 1 1 0 1

This pattern is send to the Destination

10101001 00111001 00011101

At Receiver side

Data of 24 bit

10101001 00111001 00011101

1 0 1 0 1 0 0 1
0 0 1 1 1 0 0 1
0 0 0 1 1 1 0 1

SUM-→1 1 1 1 1 1 1 1

And its complement 0 0 0 0 0 0 0 0

If we get all zero then there is not any error in the data are accepted.

The receiver can detect an error but it can't find error in particular data unit

2) Error correction method at Data link layer

Error correction can be done in two ways:

1) Whenever error detect Retransmit entire data unit.

2) Whenever error detect Correct the error using any techniques: Forward Error Correction and Burst Error Correction

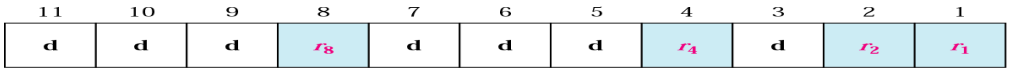
Data redundancy bit

$2^r >= m+r+1$

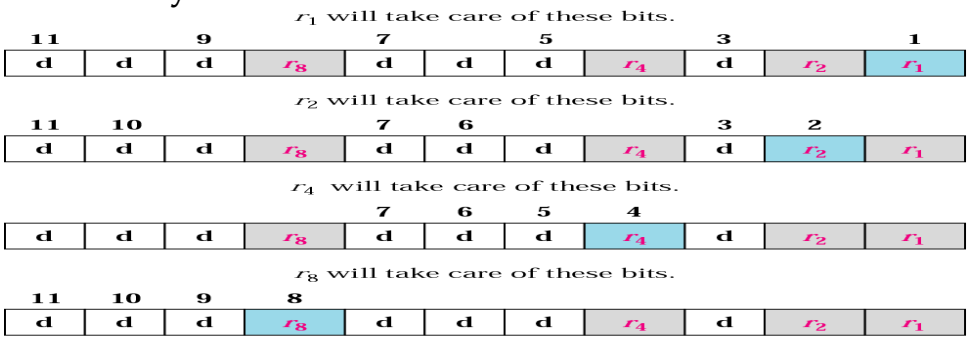
Number of data bits m	Number of redundancy bits r	Total bits m + r
1	2	3
2	3	5
3	3	6
4	3	7
5	4	9
6	4	10
7	4	11

Hamming code

Position of redundant bit

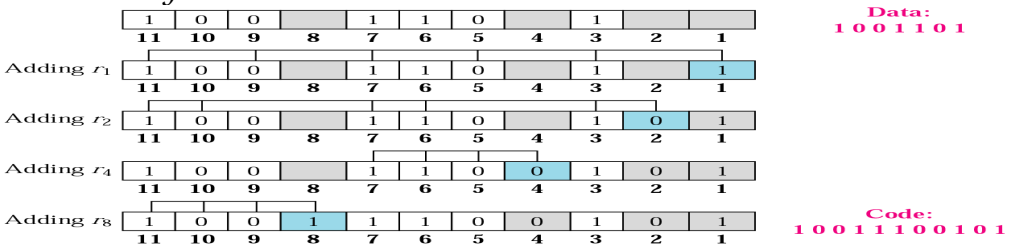


Redundancy bits calculation

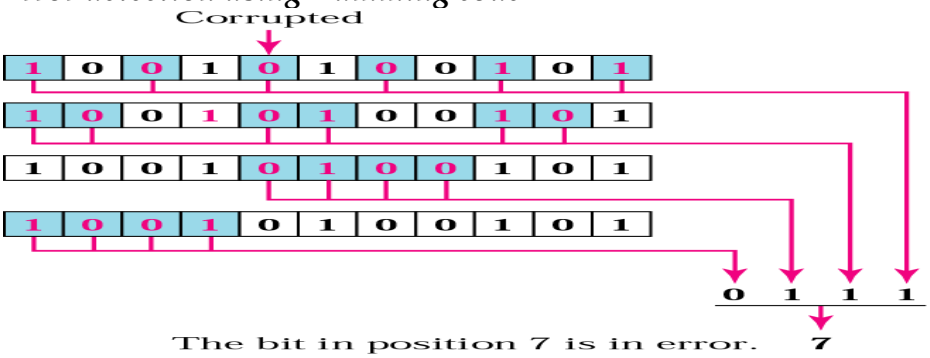


Example:

Redundancy bit calculation



Error detection using Hamming code



9. Solve the following Examples:

What is cyclic redundancy check? Show the calculation polynomial code for a frame 1101011011 using the generator x4 + x + 1.

An error-detection technique used widely in today's computer networks is based on **cyclic redundancy check (CRC) codes**. CRC codes are also known as **polynomial codes**, since it is possible to view the bit string to be sent as a polynomial whose coefficients are the 0 and 1 values in the bit string, with operations on the bit string interpreted as polynomial arithmetic.

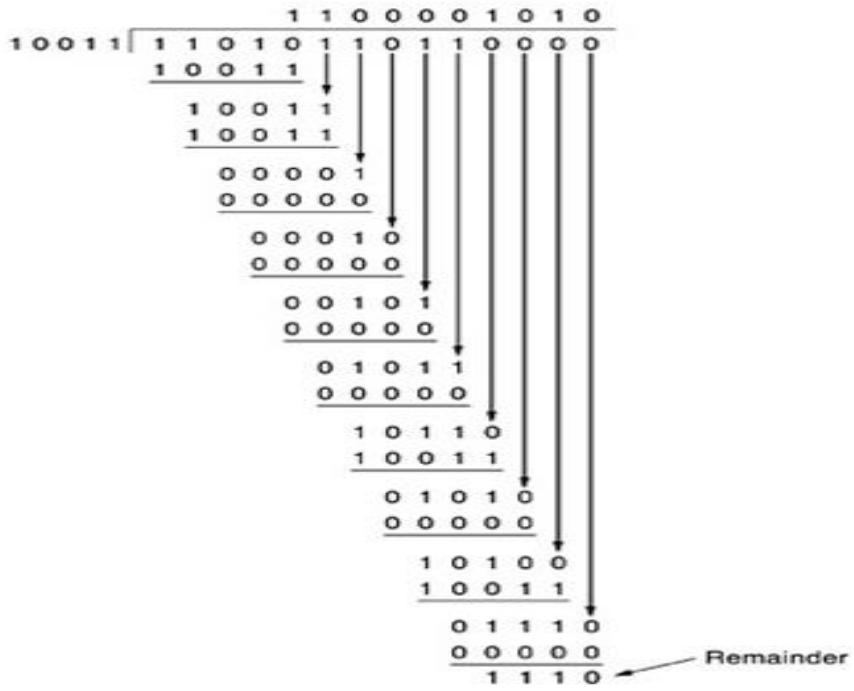
At first convert Polynomial to binary,

Polynomial	Binary generator bits
$x^4 + x + 1$ is equivalent to $(1x^4+0x^3+0x^2+1x^1+1x^0)$	10011 (Power of x represent place of bits and co-efficient of x represent value of bits)

Frame : 1 1 0 1 0 1 1 0 1 1

Generator: 1 0 0 1 1

Message after 4 zero bits are appended: 1 1 0 1 0 1 1 0 1 1 0 0 0 0

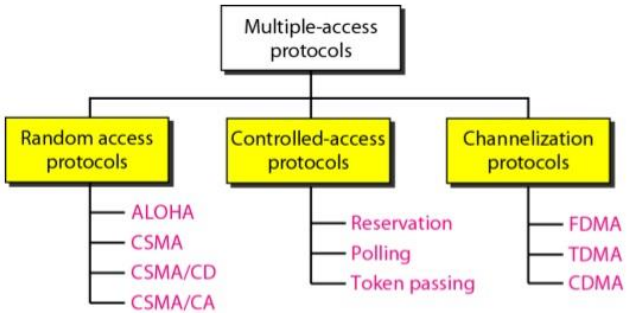


Transmitted frame: 1 1 0 1 0 1 1 0 1 1 1 1 1 0

10. Write a short note on:

1) **ALOHA**
2) **slotted ALOHA**
3) **CSMA**
4) **CSMA/CD Protocol**
5) **Bitmap protocol**

Types of Multiple –access protocol



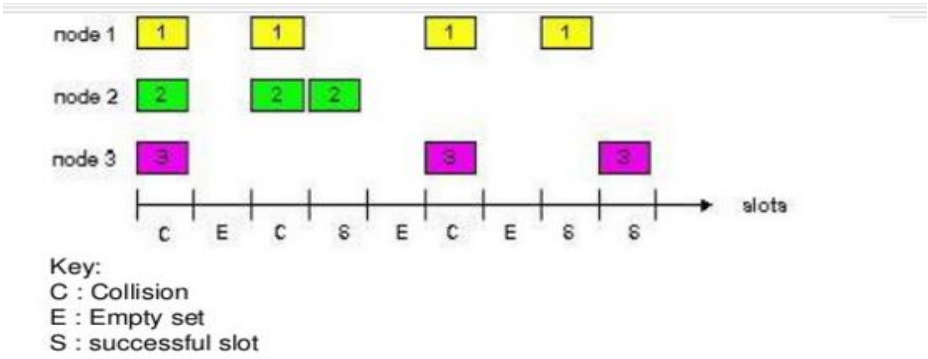
B.1 Slotted ALOHA

Assumptions:

- All frames same size
- Time is divided into equal size slots, time to transmit 1 frame
- Nodes start to transmit frames only at beginning of slots
- Nodes are synchronized
- If 2 or more nodes transmit in slot, all nodes detect collision

Operation:

- When node obtains fresh frame, it transmits in next slot
- No collision, node can send new frame in next slot
- If collision, node retransmits frame in each subsequent slot with prob. p until success



Nodes 1,2 and 3 collide in the first slot. Node 2 finally succeeds in the fourth slot, node 1 n the eight slot, and node 3 in the ninth slot

Pros:

- Single active node can continuously transmit at full rate of channel
- Highly decentralized: only slots in nodes need to be in sync
- Simple

Cons:

- Collisions, wasting slots
- Idle slots
- Nodes may be able to detect collision in less than time to transmit packet
- Clock synchronization

Slotted Aloha efficiency:

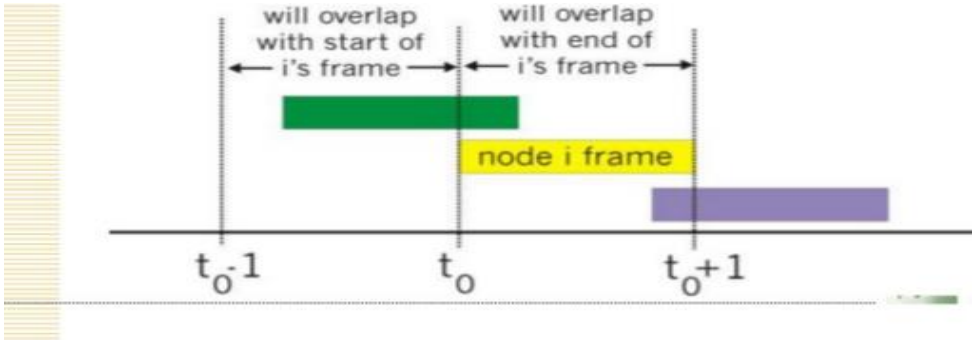
- Efficiency is the long-run fraction of successful slots when there are many nodes, each with many frames to send
- Suppose N nodes with many frames to send, each transmits in slot with probability p
- Prob that node 1 has success in a slot = $p(1-p)^{N-1}$
- Prob that any node has a success = $Np(1-p)^{N-1}$
- For max efficiency with N nodes, find p* that maximizes $Np(1-p)^{N-1}$
- For many nodes, take limit of $Np^*(1-p^*)^{N-1}$ as N goes to infinity, gives 1/e = .37
- **At best: channel used for useful transmissions 37% of time!**

B.2 Pure (Un-Slotted) ALOHA

Introduction:

Un-slotted Aloha: simpler, no synchronization

- when frame first arrives
  - transmit immediately
- collision probability increases:
  - frame sent at t0 collides with other frames sent in [t0-1,t0+1]



Difference between Slotted and Pure ALOHA:

1. Pure Aloha is a Continuous time system whereas Slotted Aloha is discrete time system.
2. Pure ALOHA doesn't check whether the channel is busy before transmission.
3. Slotted ALOHA send the data at the beginning of timeslot.
   Pure aloha not divided in to time .Slotted aloha divided in to time
4. In slotted ALOHA, a frame can be sent only at fixed times, whereas in pure

ALOHA, you can send any time.

5. Pure ALOHA is featured with the feedback property that enables it to listen to the channel and finds out whether the frame was destroyed.

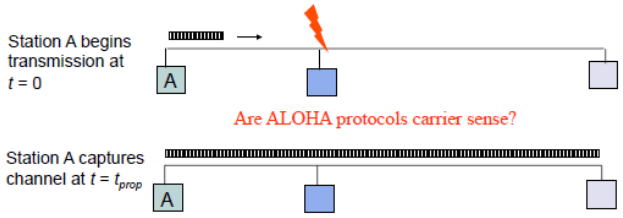
Pure Aloha efficiency:

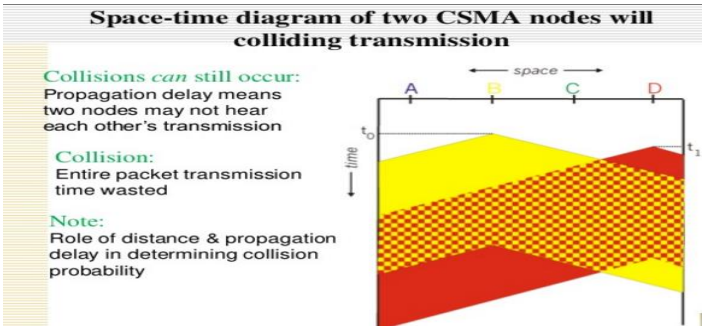
- P(success by given node) = P(node transmits) .
  P(no other node transmits in [p0-1,p0] .
  P(no other node transmits in [p0-1,p0]
  = p. $(1-p)^{N-1}$. $(1-p)^{N-1}$
  = p. $(1-p)^{2(N-1)}$
- **Choosing optimum p and then letting n -> infty ... Even worse ! = 1/(2e) = .18**

B.3 CSMA (Carrier Sense Multiple Access)

Introduction:

- A station senses the channel before it starts transmission
  - If busy, either wait or schedule backoff (different options)
  - If idle, start transmission
  - Vulnerable period is reduced to *tprop* (due to *channel capture* effect)
  - When collisions occur they involve entire frame transmission times
  - **Human analogy: don't interrupt others!**



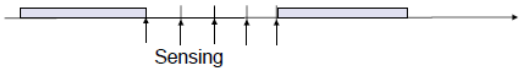


CSMA Options:

- Transmitter behavior when busy channel is sensed
  - **1-persistent CSMA (most greedy)**
    - Start transmission as soon as the channel becomes idle
    - Low delay and low efficiency
  - **Non-persistent CSMA (least greedy)**

- If busy, wait a backoff period, then sense carrier again
- High delay and high efficiency

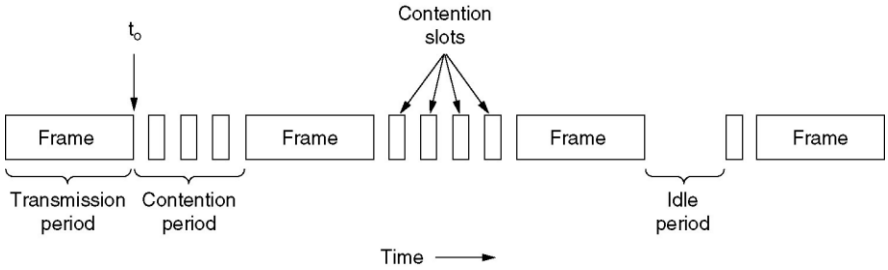
- **p-persistent CSMA (adjustable greedy)**
  - Wait till channel becomes idle, transmit with prob. p; or
  - wait one mini-slot time & re-sense with probability 1-*p*
  - Delay and efficiency can be balanced



B.4 CSMA/CD (Carrier Sense Multiple Access with Collision Detection)

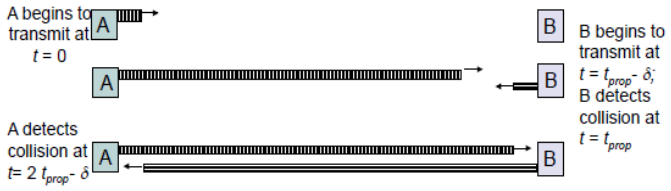
Introduction:

- **Monitor for collisions & abort transmission**
  - Stations with frames to send, first do carrier sensing
  - After beginning transmissions, stations continue **listening to the medium to detect collisions**
- If collisions detected, all stations involved abort transmission, reschedule random backoff times, and try again at scheduled times - quickly terminating a damaged frame saves Time & Bandwidth
- **Binary exponential backoff:** after *k* collisions, a random number between 0 to $2^{k} - 1$ is chosen, that number of slots is skipped



- In CSMA collisions result in wastage of **X seconds spent transmitting an entire frame**
- CSMA-CD **reduces wastage to time** to detect collision and abort transmission
- CSMA/CD can be in one of three states: contention, transmission, or idle.
- **human analogy: the polite conversationalist**

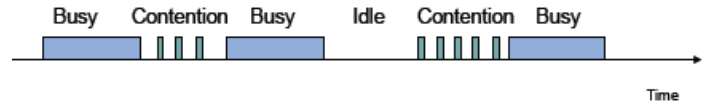
CSMA/CD reaction time:

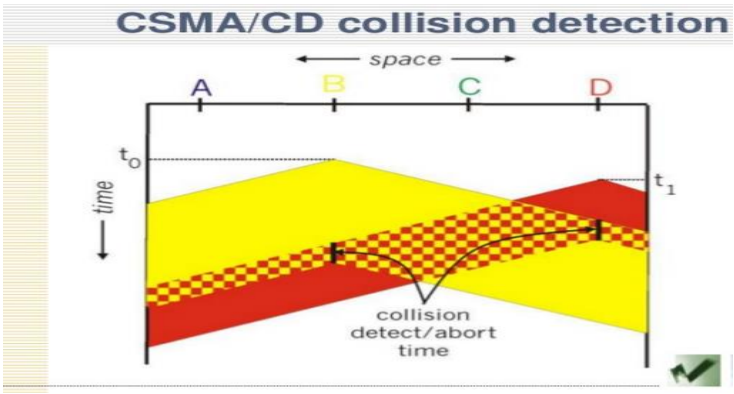


It takes 2 t_{prop} to find out if channel has been captured

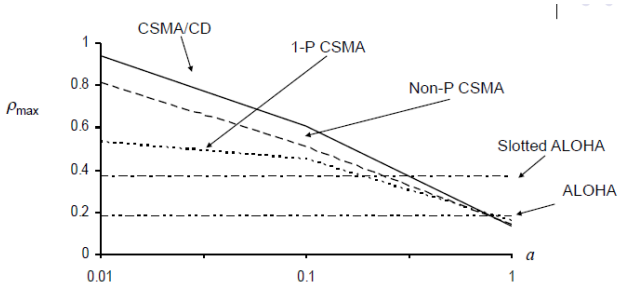
Assumptions:

- Collisions can be detected and resolved in $2t_{prop}$
- Time slotted in $2t_{prop}$ slots during contention periods
- Assume n busy stations, and each may transmit with probability p in each contention time slot
- Once the contention period is over (a station successfully occupies the channel), it takes X seconds for a frame to be transmitted
- It takes t_{prop} before the next contention period starts.





Throughput for Random Access MACs



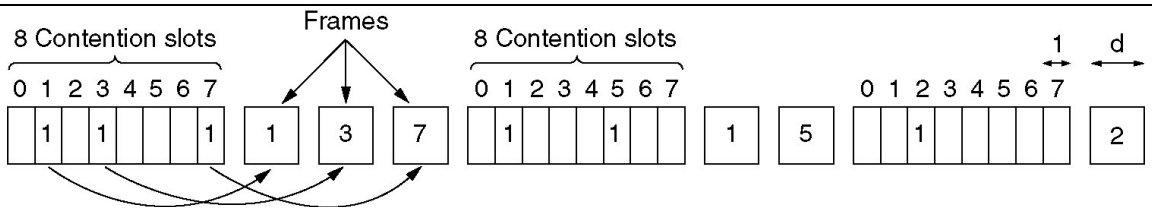
- For small a: CSMA-CD has best *maximum throughput*
- For larger a: Aloha & slotted Aloha better *maximum throughput*, since not dependent on a (reaction time)

B.5 Bitmap Protocol

Reservation Protocol

Bit Map Protocol can be used for Reservation

- A bit-map protocol:
  - A contention period has exactly M slots and a station **j** announces it has a frame to send by inserting 1 into slot **j**



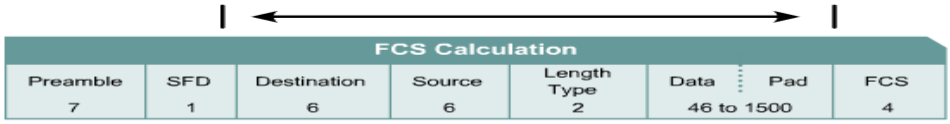
Reservation System Options

- **Centralized or distributed system**
  - *Centralized systems*: A central controller listens to reservation information, decides order of transmission, issues grants
  - *Distributed systems*: Each station determines its slot for transmission from the reservation information
- **Single or Multiple Frames**
  - *Single frame reservation*: Only one frame transmission can be reserved within a reservation cycle
  - *Multiple frame reservation*: More than one frame transmission can be reserved within a frame.

11. Explain Ethernet Frame structure.

OR

Draw and explain Ethernet header.



IEEE 802.3 Ethernet frame fields

Octets	Description
7	Preamble
1	Start Frame Delimeter (SFD)
6	Destination MAC Address
6	Source MAC Address
2	Length/Type Field (Length if less than 0600 in hexadecimal, otherwise protocol Type)
46 to 1500	Data* (If less than 46 octets, then a pad must be added to the end)
4	Frame Check Sequence (CRC Checksum)

The basic frame consists of seven elements split between three main areas:-

- ***Header***
  - *Preamble (PRE)* - This is seven bytes long and it consists of a pattern of alternating ones and zeros, and this informs the receiving stations that a frame is starting as well as enabling synchronisation.
  - *Start Of Frame delimiter (SOF)* - This consists of one byte and contains an alternating pattern of ones and zeros but ending in two ones.
  - *Destination Address (DA)* – It consist of Destination 6 byte Address.
  - *Source Address (SA)* - The source address consists of six bytes, and it is used to identify the sending station

- *Length / Type* - This field is two bytes in length. It provides MAC information and indicates the number of client data types that are contained in the data field of the frame. It may also indicate the frame ID type if the frame is assembled using an optional format.(IEEE 802.3 only).

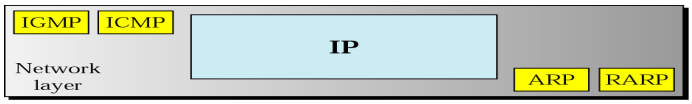
- ***Payload***
  - *Data* - This block contains the payload data and it may be up to 1500 bytes long. If the length of the field is less than 46 bytes, then padding data is added to bring its length up to the required minimum of 46 bytes.
- ***Trailer***
  - *Frame Check Sequence (FCS)* - This field is four bytes long. It contains a 32 bit Cyclic Redundancy Check (CRC) which is generated over the DA, SA, Length / Type and Data fields.

The original Ethernet IEEE 802.3 standard defined the minimum Ethernet frame size as 64 bytes and the maximum as 1518 bytes

12. Explain ARP and RARP

Other Protocols in the N/W Layer

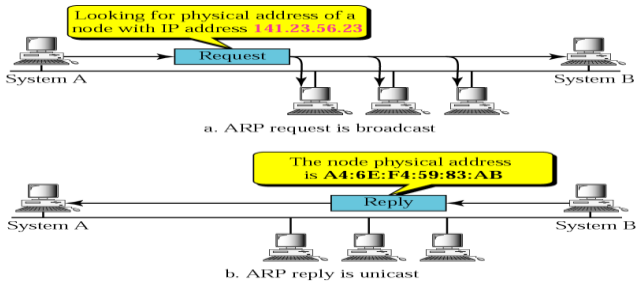
TCP/IP supports four other Protocols in the network Layer: ARP, RARP, ICMP, and IGMP.



ARP [Address Resolution Protocol]

- ARP associates an IP Address with the Physical Address. [IP to MAC Binding]
- ARP is used to find the Physical Address of the node when its internet address is known.
- Anytime a host needs to find the Physical Address of another Host on its N/W, It Send ARP Query packets that includes the IP Address and Broadcast it over the N/W.
- Every Host I the N/W Receive the ARP Packet but only host matching the IP Address Reply it backs and give own Physical Address.
- *ARP* - is a low-level protocol used to bind addresses dynamically.
- ARP allows a host to find a physical address of a target host on the same physical network, given only it's IP address.
- ARP broadcasts special packets with the destination's IP address to ALL hosts.
- The destination host (only) will respond with it's physical address.
- When the response is received, the sender uses the physical address of

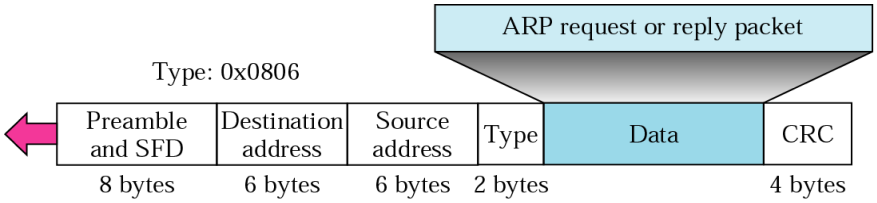
destination host to send all packets.



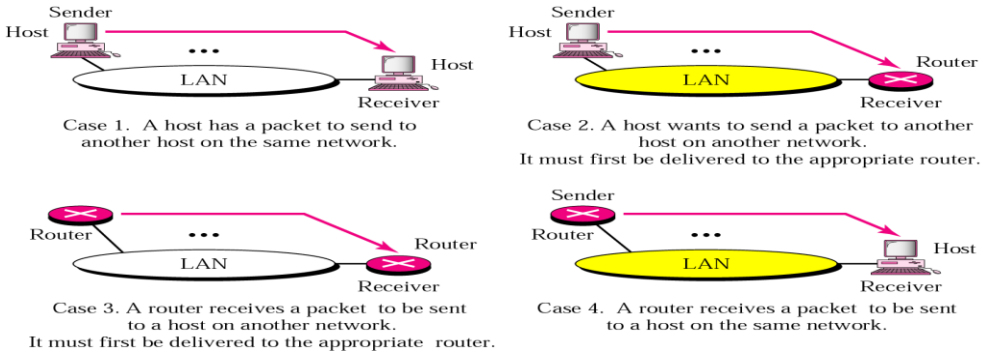
ARP packet

Hardware Type		Protocol Type
Hardware length	Protocol length	Operation Request 1, Reply 2
Sender hardware address (For example, 6 bytes for Ethernet)		
Sender protocol address (For example, 4 bytes for IP)		
Target hardware address (For example, 6 bytes for Ethernet) (It is not filled in a request)		
Target protocol address (For example, 4 bytes for IP)		

Encapsulation of ARP packet

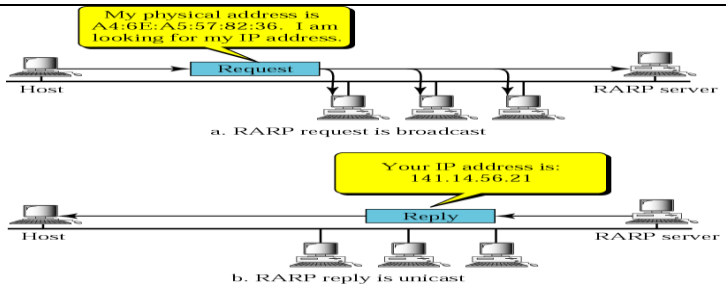


Four cases using ARP



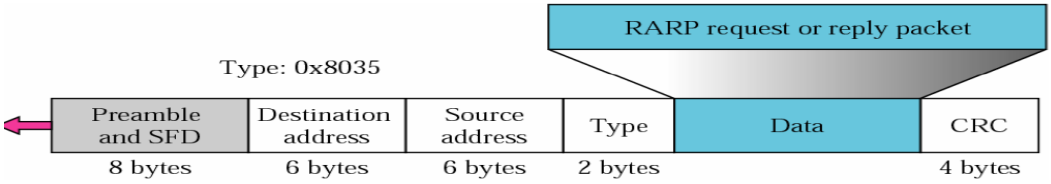
14.4.2 RARP [Reverse Address Resolution Protocol]

- RARP associates a Physical Address with the IP Address. [MAC to IP binding]
- RARP is used to find the Internet Address of the node when its Physical address is known.
- RARP works much like ARP but mapping address differently.
- RARP requests and responses use the same frame format as ARP.



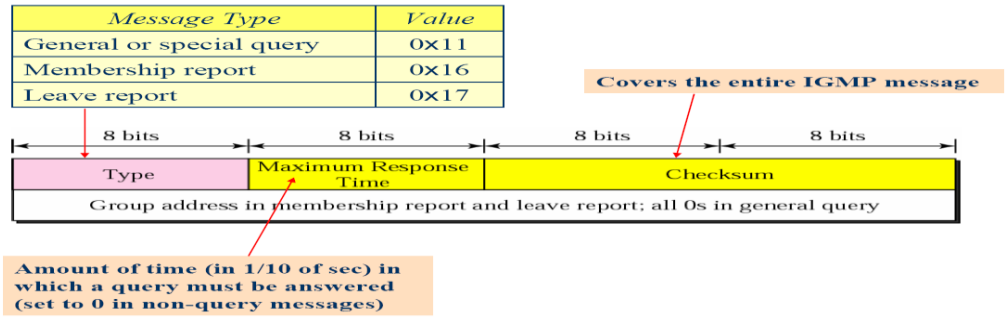
Hardware type		Protocol type
Hardware length	Protocol length	Operation Request 3, Reply 4
Sender hardware address (For example, 6 bytes for Ethernet)		
Sender protocol address (For example, 4 bytes for IP) (It is not filled for request)		
Target hardware address (For example, 6 bytes for Ethernet) (It is not filled for request)		
Target protocol address (For example, 4 bytes for IP) (It is not filled for request)		

Encapsulation of RARP packet

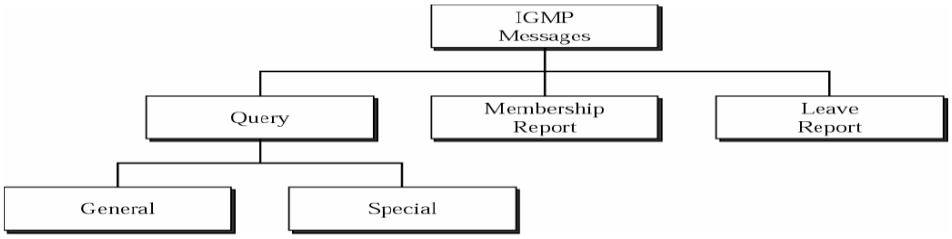


13. Explain the working of IGMP. List its message types.

- The IP protocol can be viewed in three type of communication. –Unicasting, Multicasting and Broadcasting.
- Unicasting is the communication between one sender and one Receiver; it is one to one communication.
- Broadcasting is a one type of communication in which one sender send message to all nodes available in the network.
- Some process is needed to send same message to a large number of receivers simultaneously at a time. This is called Multicasting.
- Ex:-Multiple Stock brokers can simultaneously be informed of changes in price, Video on demand.
- IP address support multicasting addresses.
- All IP Address starts with 1110 (Class D) Multicast Address., 28 remaining are group bits.



IGMP message type



14. Draw the IPV4 & IPV6 datagram format.

OR

Compare the IPv4 and IPv6 header.

IPV4 Datagram

- Packets of IP is called Datagram
- Maximum size of IP Datagram is (65,536 bytes)
- IP datagram containing two part: Header and Data
- Size of Header may be vary between 20 to 60 bytes
- Header containing the information regarding routing and delivery

Description of each field of IP Datagram:

- **VER (4 BITS)**
  - The version field is set to the value '4' in decimal or '0100' in binary.
  - The value indicates the version of IP (4 or 6, there is no version 5).
- **HLEN (4 BITS)**
  - Defines the length of header
  - Length is in a multiple of four bytes
  - The four bit can represent a number between 0 to 15, which multiply by 4 gives a maximum of 60 bytes
- **Service types (8 Bits)**

- Define How the Datagram should be handled.
- Define the Priority.

- **TOTAL LENGTH (16 BITS)**
  - This informs the receiver of the datagram where the end of the data in this datagram is.
  - This is why an IP datagram can be up to **65,535** bytes long, as that is the maximum value of this 16-bit field.
- **IDENTIFICATION (16 bits)**
  - It is used for Fragmentation.
  - Sometimes, a device in the middle of the network path cannot handle the datagram at the size it was originally transmitted, and must break it into fragments.
  - If an intermediate system needs to break up the datagram, it uses this field to aid in identifying the fragments.
- **FLAGS (3 BITS)**
  - The flags field contains single-bit flags that indicate whether the datagram is a fragment, whether it is permitted to be fragmented, and whether the datagram is the last fragment, or there are more fragments.
  - The first bit in this field is always zero.
- **FRAGMENT OFFSET (13 BITS)**
  - When a datagram is fragmented, it is necessary to reassemble the fragments in the correct order.
  - The fragment offset numbers the fragments in such a way that they can be reassembled correctly.
- **TIME TO LIVE (8 BITS)**
  - This field determines how long a datagram will exist.
  - At each hop along a network path, the datagram is opened and it's time to live field is decremented by one (or more than one in some cases).
  - When the time to live field reaches zero, the datagram is said to have 'expired' and is discarded.
  - This prevents congestion on the network that is created when a datagram cannot be forwarded to it's destination.
  - Most applications set the time to live field to 30 or 32 by default.
- **PROTOCOL (8 BITS)**
  - This indicates what type of protocol is encapsulated within the IP datagram. Some of the common values seen in this field include:

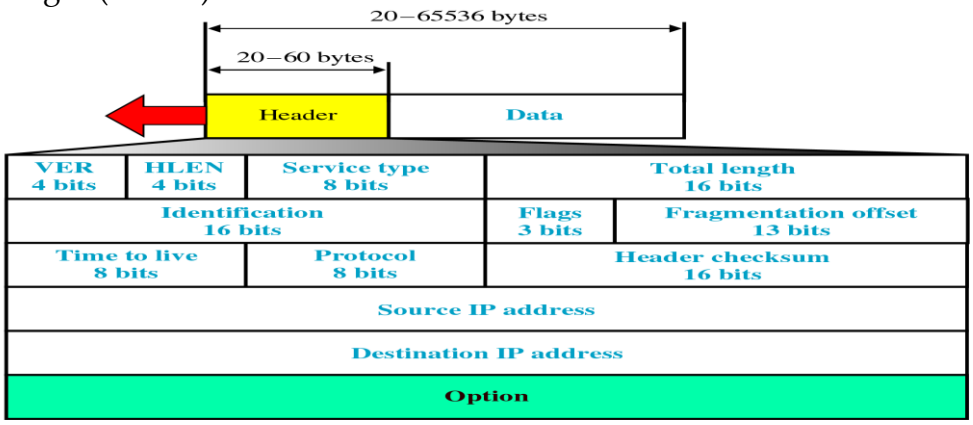
• Protocol	• Number (Decimal)
• ICMP	• 1
• IGMP	• 2
• TCP	• 6
• UDP	• 17

- **HEADER CHECKSUM (16 BITS)**
  - The checksum allows IP to detect datagram with corrupted headers and discard them.
  - Since the time to live field changes at each hop, the checksum must be re-calculated at each hop.

- **SOURCE ADDRESS (32 BITS)**
  This is the IP address of the sender of the IP datagram.

- **DESTINATION ADDRESS (32 BITS)**
  This is the IP address of the intended receiver(s) of the datagram. If the host portion of this address is set to all 1's, the datagram is an 'all hosts' broadcast.

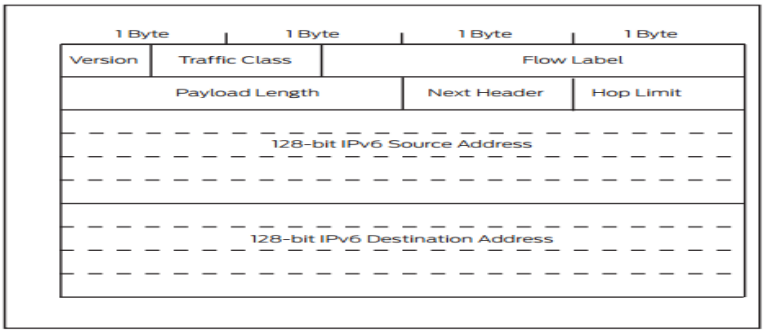
- **OPTIONS & PADDING (VARIABLE)**
  - Various options can be included in the header by a particular vendor's implementation of IP.
  - If options are included, the header must be padded with zeroes to fill in any unused octets so that the header is a multiple of 32 bits, and matches the count of bytes in the Header Length (HLEN) field.



IPV6 Header Format

IPv6 packets have their own frame Ethertype value, 0x86dd, making it easy for receivers that must handle both IPv4 and IPv6 to distinguish the frame content on the same interface. The IPv6 header is comprised of the following fields:

- **Version:** A four-bit field for the IP version number (0x06). „
- **Traffic Class:** An 8-bit field that identifies the major class of the packet content (for example, voice or video packets). The default value is 0, meaning it is ordinary bulk data (such as FTP) and requires no special handling. „
- **Flow Label:** A 20-bit field used to label packets belonging to the same flow (those with the same values in several TCP/IP header parameters). The flow label is normally 0 (flows are detected in other ways).



- **Payload Length:** A 16-bit field giving the length of the packet in bytes, excluding the IPv6 header. „
- **Next Header:** An 8-bit field giving the type of header immediately following the IPv6 header (this serves the same function as the Protocol field in IPv4). „
- **Hop Limit:** An 8-bit field set by the source host and decremented by 1 at each router. Packets are discarded if Hop Limit is decremented to zero (this replaces the IPv4 Time To Live field). Generally, implementers choose the default to use, but values such as 64 or 128 are common.

15. Differentiate between IPv4 and IPv6.

IPv4 and IPv6 Comparisons:

IPv4	**IPv6**
32-bit (4 byte) address supporting 4,294,967,296 address (although many were lost to special purposes, like 10.0.0.0 and 127.0.0.0)	128-bit (16 byte) address supporting 228 (about 3.4 x 1038) addresses
Address Shortages: IPv4 supports 4.3×10^{9} (4.3 billion) addresses, which is inadequate to give one (or more if they possess more than one device) to every living person.	Larger address space: IPv6 supports 3.4×10^{38} addresses, or 5×10^{28}(50 octillion) for each of the roughly 6.5 billion people alive today.
NAT can be used to extend address limitations	No NAT support (by design)
IP addresses assigned to hosts by DHCP or	IP addresses self-assigned to hosts with

static configuration	stateless address auto-configuration or DHCPv6
IPSec support optional	IPSec support required
Options integrated in header fields	Options supported with extensions headers (simpler header format)
Has broadcast addresses for all devices	No such concept in IPv6 (uses multicast groups)
Uses 127.0.0.1 as loopback address	Uses ::1 as loopback address
IPv4 is subdivided into classes <A-E>	IPv6 is classless. IPv6 uses a prefix and an Identifier ID known as IPv4 network
IPv4 header has 20 bytes. IPv4 header has many fields (13 fields)	IPv6 header is the double, it has 40 bytes IPv6 header has fewer fields, it has 8 fields.
IPv4 address uses a subnet mask.	IPv6 uses a prefix length.
IPv4 has lack of security. IPv4 was never designed to be secure - Originally designed for an isolated military network - Then adapted for a public educational & research network	IPv6 has a built-in strong security - Encryption - Authentication

16. Sub netting Examples:

1) One of the addresses in a block is 17.63.110.114/24. Find the first address, and the last address in the block.

Answer:

Solution

The network mask is 255.255.255.0.

a. The number of addresses in the network is $2^{32-24} = 256$.

b. To find the first address, we use the short cut methods discussed early in the chapter. The first address is 17.63.110.0/24.

Address:	17	.	63	.	110	.	114
Network mask:	255	.	255	.	255	.	0
First address (AND):	17	.	63	.	110	.	0

c. To find the last address, we use the complement of the network mask and the first short cut method we discussed before. The last address is 17.63.110.255/24.

Address in binary:	10100111	11000111	10101010	01010010
Complement of network mask:	00000000	00000000	00000000	00011111
Last address:	10100111	11000111	10101010	01011111

2) Consider a router that interconnects three subnets: Subnet 1, Subnet 2, and Subnet 3. Suppose all of the interfaces in each of these three subnets are required to have the

prefix 223.1.17/24. Also suppose that Subnet 1 is required to support at least 60 interfaces, Subnet 2 is to support at least 90 interfaces, and Subnet 3 is to support at least 12 interfaces. Provide three network addresses (of the form a.b.c.d/x) that satisfy these constraints.

Answer:

Recall:

- *The network address cannot be used for an interface (Network prefix + all zeros).*
- *The broadcast address cannot be used for an interface (Network prefix + all ones)*

Then:

Subnet 1 requires 60 interfaces + 2 (network + broadcast) =>
so 64 addresses will sufice

Subnet 2 requires 90 interfaces + 2 (network + broadcast) =>
so 128 addresses will sufice

Subnet 3 requires 12 interfaces + 2 (network + broadcast) =>
so 16 addresses will sufice

One Solution that Works:

Sub 2 230.1.17.0/25 → 223.1.17.00000000/25 → 223.1.17.0 to 223.1.17.127 = 128

Sub 1 230.1.17.128/26 → 223.1.17.10000000/26 → 223.1.17.128 to 223.1.17.191 = 64

Sub 3 230.1.17.192/28 → 223.1.17.11000000/28 → 223.1.17.192 to 223.1.17.207 = 16

3) An ISP is granted a block of addresses starting with 120.60.4.0/20. The ISP wants to distribute these blocks to 100 organizations with each organization receiving 8 addresses only. Design the subblocks and give the slash notation for each subblock. Find out how many addresses are still available after these allocations.

The site has 232−20 = 212 = 4096 addresses.

Each group needs 8 addresses only. That means the prefix length is 32-3=29 so that the slash notation will be /29.

.

1st subnet: 120.60.4.0/29 to 120.60.4.7/29

… … …

32nd subnet: 120.60.4.248/29 to 120.60.4.255/29

33rd subnet: 120.60.5.0/29 to 120.60.5.7/29

… … …

64th subnet: 120.60.5.248/29 to 120.60.5.255/29

… … …

99th subnet: 120.60.7.16/29 to 120.60.7.23/29

100th subnet: 120.60.7.24/29 to 120.60.7.31/29

Subnets:

4096 − 800 = 3296 addresses left

4) An organization is granted the block 211.17.180.0/24. The administrator wants to create 32 subnets.

a. Find the subnet mask.

b. Find the number of addresses in each subnets.

c. Find the first and last addresses in subnet 1.

a. Find the first and last addresses in subnet 32

Answer:

Step 1 : Subnet mask :

This is a class C network. So the default mask is given by, 255.255.255.0

As we need 32 subnets we need 5 extra I's. So the subnet mask will be as follows in the binary form.

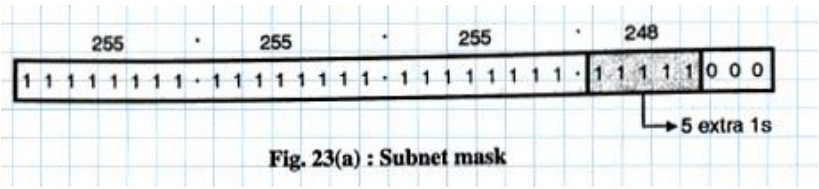


Fig. 23(a) : Subnet mask

Subnet Mask

The subnet mask is therefore given by, 255.255.255.248

Step 2 : Number of addresses in each subnet :

The structure of a class C address is as shown in Fig. 23(b).

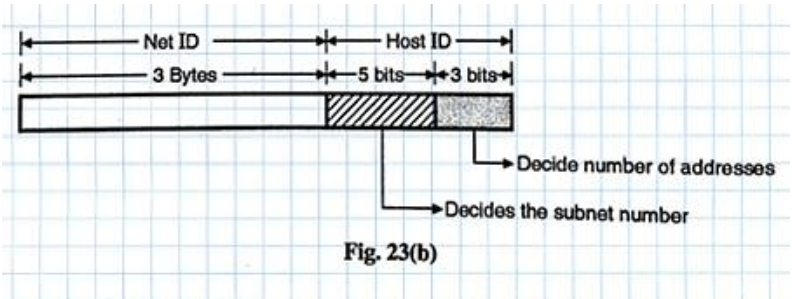


Fig. 23(b)

Number of address in each subnet

We have only 3 bits to decide the number of addresses in each subnet.

No. Of addresses per subnet 23 = 8

Step 3 : First and the last address in subnet 1 :

First address in subnet-I = 211.17.180.0

Last address in subnet-I = 211.17.180.7

Step 4 : First and the last address in subnet 32 :

First address in subnet-32 = 211.17.180.248

Last address in subnet-32 = 211.17.180.255

5) Consider a router that interconnects three subnets: Subnet 1, Subnet 2, and Subnet 3. Suppose all of the interfaces in each of these three subnets are required to have the prefix 223.1.17/24. Also suppose that Subnet 1 is required to support at least 60 interfaces, Subnet 2 is to support at least 90 interfaces, and Subnet 3 is to support at least 12 interfaces. Provide three network addresses (of the form a.b.c.d/x) that satisfy these constraints. (Dec- 2015, May- 2015)

Answer:

1. **Subnet 1**:- 223.1.17.0/26
2. **Subnet 2**:- 223.1.17.128/25
3. **Subnet 3**:- 223.1.17.224/28

Method:

Prefix of network for all:223.1.17/24

The above prefix shows that total network bits here are 24 and so we have remaining 8 bits (32-24) of host at our disposal.

For Subnet 1:-

It should support 60 interfaces(interface is same as number of hosts)

So total host bits required= 6 (As $2^6 = 64$)

Network bits= 32-6=26

So we have,

N=26

H=6

Block Size/IP addresses in each subnet $=2^H = 2^6 = 64$

Total networks= $2^N=2^{26}$

Valid Hosts= Block size – 2 =64-2 = 62

Subnet mask= (Write N(26) times 1's and H(6) times 0's and convert to decimal as we convert IP address)

Subnet mask (Binary) 11111111.11111111.11111111.11000000

Subnet mask (Decimal Equivalent) =255.255.255.192
Subnet mask (CIDR notation- /N): /26
To find network addresses:-
As the prefix must be : 223.1.17
Range can be defined as follows:
(To find the range, make a table as below containing starting addresses and ending addresses. To compute starting addresses: place starting address as all 0's after given network address i.e. 223.1.17.0 as initial address, keep on adding block size to initial starting address until you do not reach the last starting address. Last starting address must be network address followed by number in subnet mask except 255 i.e. 223.1.17.192 here. Ending address will always be next starting address-1. Last ending address will always be network address followed by 255 i.e. 233.1.17.255 here)

(Note: Throughout the answer usable ranges will be shown in orange colour and ***selected range*** will be orange, bold and italics)

Starting addresses	Ending Addresses
223.1.17.0	***223.1.17.63***
223.1.17.64	223.1.17.127
223.1.17.128	223.1.17.191
223.1.17.192	223.1.17.255

Network address in CIDR notation(a.b.c.d/x)= 223.1.17.0/26

(Note: Here to convert in CIDR notation you can use any address from the range defined above and starting address of the range will be used to denote the subnet. For convenience and ease here I have used 1st range of addresses. But remember at least 62 (as valid hosts are 62-2=60 are required number of interfaces) addresses of that range will not be usable by remaining subnets as these subnets are part of same network 223.1.17/24. These addresses can or cannot be taken continuously. For ease let us consider they are taken continuously i.e. for the next subnet you cannot use range 223.1.17.0 to 223.1.17.62. Place the CIDR notation of subnet mask after the network address)

For Subnet 2:-
It should support 90 interfaces(interface is same as number of hosts)
So total host bits required= 7 (As $2^7 = 128$)
Network bits= 32-7=25

So we have,

N=25

H=7

Block Size/IP addresses in each subnet $=2^H = 2^7 = 128$

Total networks= $2^N=2^{25}$

Valid Hosts= Block size – 2 =128-2 = 126

Subnet mask= (Write N(25) times 1's and H(7) times 0's and convert to decimal as we convert IP address)

Subnet mask (Binary) 11111111.11111111.11111111.10000000

Subnet mask (Decimal Equivalent) =255.255.255.128

Subnet mask (CIDR notation- /N): /25

To find network addresses:-

As the prefix must be : 223.1.17

Range can be defined as follows:

Starting addresses	Ending Addresses
223.1.17.0	223.1.17.127
223.1.17.128	***223.1.17.255***

Network address in CIDR notation(a.b.c.d/x)= 223.1.17.128/25

(Note: Here to convert in CIDR notation you can use any address from the range defined above and starting address of the range will be used to denote the subnet. But for this subnet you cannot use range containing addresses 223.1.17.0 to 223.1.17.62 as they are used in creating subnet 1. So we are left only with second option having range 223.1.17.128 to 223.1.17.255. Remember that these addresses will not be usable by next subnet.)

For Subnet 3:-

It should support 12 interfaces(interface is same as number of hosts)

So total host bits required= 4 (As $2^4 = 16$)

Network bits= 32-4=28

So we have,

N=28

H=4

Block Size/IP addresses in each subnet $=2^H = 2^4 = 16$

Total networks= $2^N=2^{28}$

Valid Hosts= Block size – 2 =16-2 = 14

Subnet mask= (Write N(28) times 1's and H(4) times 0's and convert to decimal as we convert IP address)

Subnet mask (Binary) 11111111.11111111.11111111.11110000

Subnet mask (Decimal Equivalent) =255.255.255.240

Subnet mask (CIDR notation- /N): /28

To find network addresses:-

As the prefix must be : 223.1.17

Range can be defined as follows:

	Starting addresses	Ending Addresses
1.	223.1.17.0	223.1.17.15
2.	223.1.17.16	223.1.17.31
3.	223.1.17.32	223.1.17.47
4.	223.1.17.48	223.1.17.63
5.	223.1.17.64	223.1.17.79
6.	223.1.17.80	223.1.17.95
7.	**223.1.17.96**	**223.1.17.111**
8.	223.1.17.112	223.1.17.127
9.	223.1.17.128	223.1.17.143
10.	223.1.17.144	223.1.17.159
11.	223.1.17.160	223.1.17.175
12.	223.1.17.176	223.1.17.191
13.	223.1.17.192	223.1.17.207
14.	223.1.17.208	223.1.17.223
15.	223.1.17.224	223.1.17.239
16.	223.1.17.240	223.1.17.255

Network address in CIDR notation(a.b.c.d/x)= 223.1.17.96/28

(Note: Here to convert in CIDR notation you can use any address from the range defined above but for this subnet you cannot use range containing addresses 223.1.17.0 to 223.1.17.62 as they are used in creating subnet 1. Also you cannot use the range containing addresses 223.1.17.128 to 223.1.17.255 as they are used in creating subnet 2. So we are left only with remaining ranges numbered 5, 6, 7, and 8. You can use any of these ranges and denote the subnet by the starting address of selected range. Here let us use range 7 i.e. 223.1.17.96. Place the CIDR notation of subnet mask after the network address)

[**Extra**:- Some alternate correct answers:

1.

1. **Subnet 1**:- 223.1.17.0/26
2. **Subnet 2**:- 223.1.17.128/25
3. **Subnet 3**:- 223.1.17.80/28

2.

All	223.1.17/24
Subnet 1	223.1.17.0/25
Subnet 2	223.1.17.128/26
Subnet 3	223.1.17.192/26

223.1.17/24 (Network ID)	11011111 00000001 00010001 ????????
Subnet 1	11011111 00000001 00010001 0???????
Subnet 2	11011111 00000001 00010001 10??????
Subnet 3	11011111 00000001 00010001 11??????

3.

1. **Subnet 1**:- 223.1.17.0/26
2. **Subnet 2**:- 223.1.17.128/25
3. **Subnet 3**:- 223.1.17.192/28

Etc]

6) What is the full subnet mask for address 172.16.5.10/28?

Answer: (255.255.255.240)

7) You currently use the default mask for your IP network 192.168.1.0. You need to subnet your network so that you have 30 additional networks, and 4 hosts per network. Is this possible, and what subnet mask should you use?

Answer: (Yes it is possible, using subnet mask 255.255.255.248.)

8) You have sub-netted your class C network 192.168.1.0 with a subnet mask of 255.255.255.240. Please list the following: number of networks, number of hosts per network, the full range of the first three networks, and the usable address range from those first three networks.

(Number of networks = 16

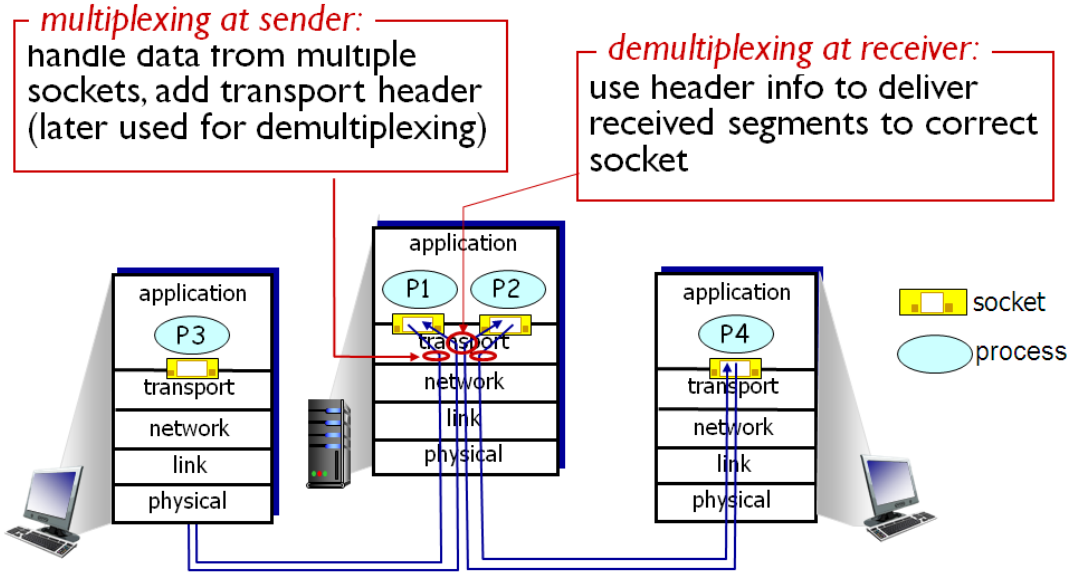
Number of hosts = 14

Full Range for first three networks: 192.168.1.0-15 192.168.1.16-31 192.168.1.32-47

Usable Range for first three networks: 192.168.1.1-14 192.168.1.17-30 192.168.1.33-46)

17. Discuss transport layer multiplexing and Demultiplexing concept.

- Each transport-layer segment has a set of fields in the segment for directing an incoming transport-layer segment to the appropriate socket at the receiving host
- At the receiving end, the transport layer examines these fields to identify the receiving socket and then directs the segment to that socket. This job of delivering the data in a transport-layer segment to the correct socket is called **demultiplexing**.
- The job of gathering data chunks at the source host from different sockets, encapsulating each data chunk with header information (that will later be used in demultiplexing) to create segments, and passing the segments to the network layer is called **multiplexing**.
- Note that the transport layer in the middle host in Figure must demultiplex segments arriving from the network layer below to either process P1 or P2 above; this is done by directing the arriving segment's data to the corresponding process's socket. The transport layer in the middle host must also gather outgoing data from these sockets, form transport-layer segments, and pass these segments down to the network layer.



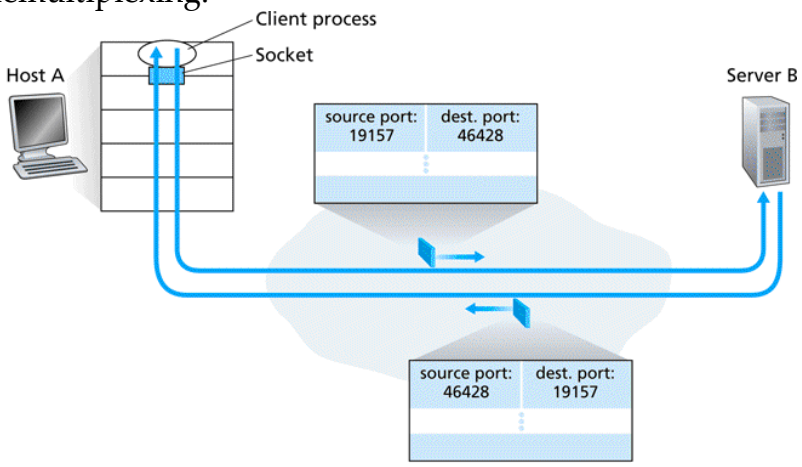
- Transport-layer multiplexing requires (1) that sockets have unique identifiers, and (2) that each segment have special fields that indicate the socket to which the segment is to be delivered. These special fields, illustrated in Figure, are the **source port number field** and the **destination port number field**.
- Each port number is a 16-bit number, ranging from 0 to 65535. The port numbers ranging from 0 to 1023 are called **well-known port numbers** and are restricted, which means that they are reserved for use by well-known application protocols such as HTTP (which uses port number 80) and FTP (which uses port number 21)
- It should now be clear how the transport layer *could* implement the demultiplexing service: Each socket in the host could be assigned a port number, and when a segment arrives at the host, the transport layer examines the destination port number in the segment and directs the segment to the corresponding socket. The segment's data

then passes through the socket into the attached process.

- This is basically how UDP does it.
- However, we'll also see that multiplexing/demultiplexing in TCP is yet more subtle.

1) Connectionless Multiplexing and Demultiplexing

- Python program running in a host can create a UDP socket with the line clientSocket = socket(socket.AF_INET, socket.SOCK_DGRAM)
- When a UDP socket is created in this manner, the transport layer automatically assigns a port number to the socket. In particular, the transport layer assigns a port number in the range 1024 to 65535 that is currently not being used by any other UDP port in the host.
- Alternatively, we can add a line into our Python program after we create the socket to associate a specific port number (say, 19157) to this UDP socket via the socket bind() method:
  clientSocket.bind(('', 19157))
- If the application developer writing the code were implementing the server side of a "well-known protocol," then the developer would have to assign the corresponding well-known port number.
- Typically, the client side of the application lets the transport layer automatically (and transparently) assign the port number, whereas the server side of the application assigns a specific port number.
- With port numbers assigned to UDP sockets, we can now precisely describe UDP multiplexing/demultiplexing.



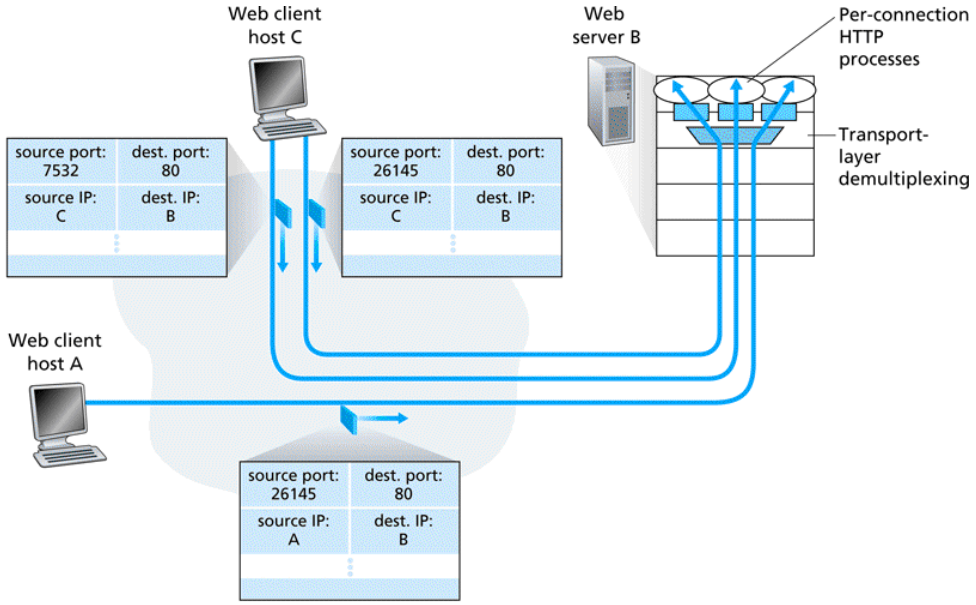
- Suppose a process in Host A, with UDP port 19157, wants to send a chunk of application data to a process with UDP port 46428 in Host B. The transport layer in Host A creates a transport-layer segment that includes the application data, the source port number (19157), the destination port number (46428), and two other values (which will be discussed later, but are unimportant for the current discussion). The transport layer then passes the resulting segment to the network layer. The network layer encapsulates the segment in an IP datagram and makes a best-effort attempt to deliver the segment to the receiving host. If the segment arrives at the

receiving Host B, the transport layer at the receiving host examines the destination port number in the segment (46428) and delivers the segment to its socket identified by port 46428. Note that Host B could be running multiple processes, each with its own UDP socket and associated port number. As UDP segments arrive from the network, Host B directs (demultiplexes) each segment to the appropriate socket by examining the segment's destination port number.

- It is important to note that a UDP socket is fully identified by a two-tuple consisting of a destination IP address and a destination port number. As a consequence, if two UDP segments have different source IP addresses and/or source port numbers, but have the same *destination* IP address and *destination* port number, then the two segments will be directed to the same destination process via the same destination socket.

2) Connection-Oriented Multiplexing and Demultiplexing

- In order to understand TCP demultiplexing, we have to take a close look at TCP sockets and TCP connection establishment. One subtle difference between a TCP socket and a UDP socket is that a TCP socket is identified by a four-tuple: (source IP address, source port number, destination IP address, destination port number).
- Thus, when a TCP segment arrives from the network to a host, the host uses all four values to direct (demultiplex) the segment to the appropriate socket.
- In particular, and in contrast with UDP, two arriving TCP segments with different source IP addresses or source port numbers will (with the exception of a TCP segment carrying the original connection-establishment request) be directed to two different sockets. To gain further insight, let's reconsider the TCP client-server programming
- The TCP server application has a "welcoming socket," that waits for connectionestablishment requests from TCP clients (see Figure) on port number 12000.



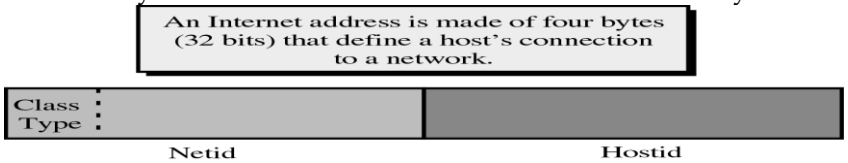
- The TCP client creates a socket and sends a connection establishment request segment with the lines:
  clientSocket = socket(AF_INET, SOCK_STREAM)clientSocket.connect((serverName,12000))
- A connection-establishment request is nothing more than a TCP segment with destination port number 12000 and a special connection-establishment bit set in the TCP header.
- The segment also includes a source port number that was chosen by the client.
- When the host operating system of the computer running the server process receives the incoming connection-request segment with destination port 12000, it locates the server process that is waiting to accept a connection on port number 12000. The server process then creates a new socket:
  connectionSocket, addr = serverSocket.accept()
- Also, the transport layer at the server notes the following four values in the connection- request segment:
  (1) the source port number in the segment, (2) the IP address of the source host, (3) the destination port number in the segment, and (4) its own IP address. The newly created connection socket is identified by these four values; all subsequently arriving segments whose source port, source IP address, destination port, and destination IP address match these four values will be demultiplexed to this socket. With the TCP connection now in place, the client and server can now send data to each other. The server host may support many simultaneous TCP connection sockets, with each socket attached to a process, and with each socket identified by its own fourtuple. When a TCP segment arrives at the host, all four fields (source IP address, source port, destination IP address, destination port) are used to direct (demultiplex) the segment to the appropriate socket.

18. Explain classful IP addressing scheme.

OR

Explain IP Addressing.

- IP address is used to unlikely identify the node in the network.
- IPV4 version contains 32 bit address.
- Divided into main two parts : Net ID and Host ID
- Originally, all IP addresses were classful – they belonged to Class A, B, C or D. Class D is for Multicast and is rarely used. Class E is reserved and is not currently used.

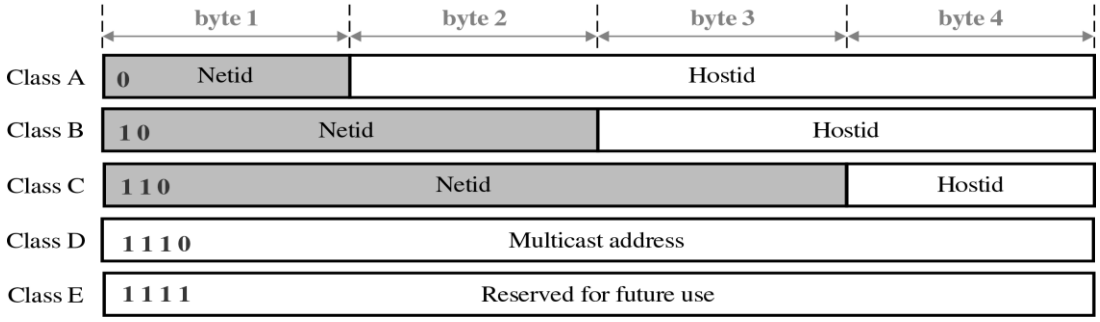


Class Wise Netid and Host ID bits:

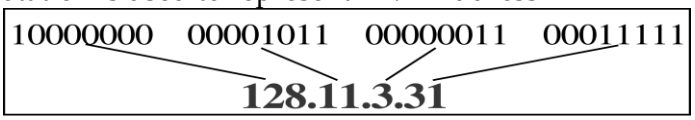
Class	Net ID Bit	Host ID Bit

A	8	24
B	16	16
C	24	8
D	Used for Multicasting	
E	Reserved for Future Use	

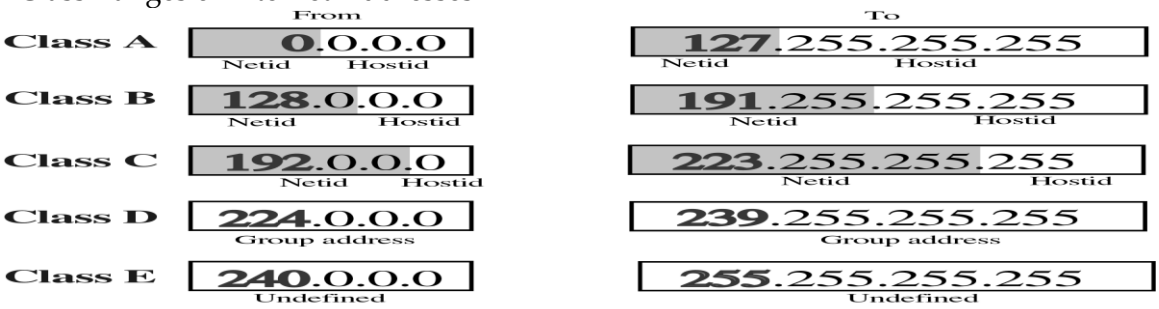
- Classes of IPV4 Address



- Doted decimal notation is used to represent IPV4 Address

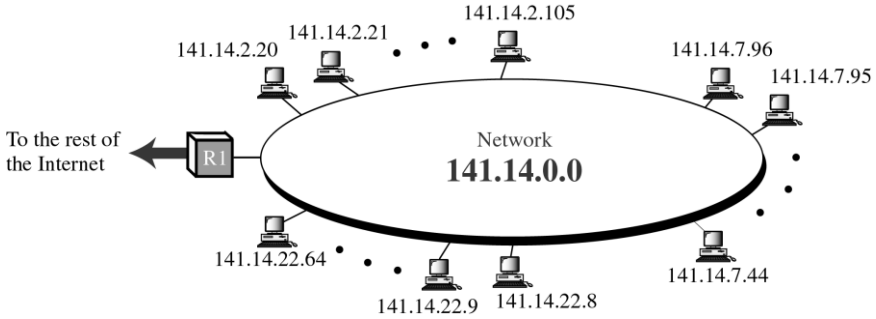


- Class Ranges of Internet Addresses

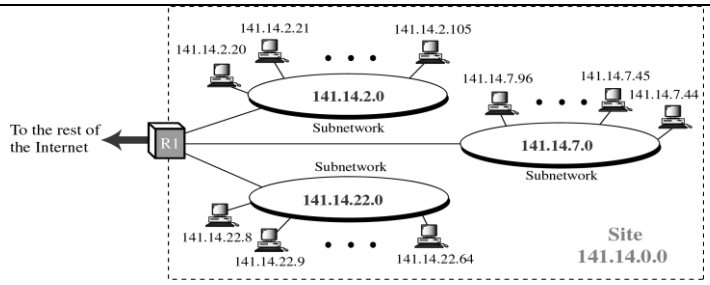


19. With example explain sub netting

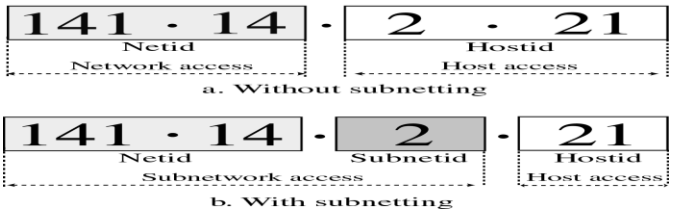
- Subnating is a process to divide network into sub networks.
- Host ID bit is used in Net ID
- A Network with Two Levels of Hierarchy



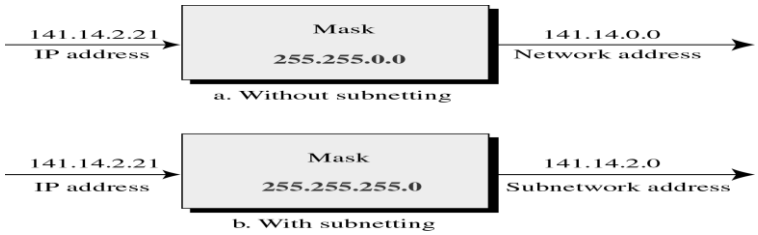
- A Network with Three Levels of Hierarchy



- Addresses with and without Subnetting



- **Masking:** Masking is a process that extracts the address of the physical network from an IP address.



Sub netting Example:

Let's take class B where the default mask is 255.255.0.0 and prefix /16 and allows for 65534 host for that one single network, below is how it works while subneting

Ip address	Prefix	Mask	Subnet bits	Subnets possible	Hosts per subnet
172.16.0.0	/16	255.255.0.0	0	1	65534
172.16.0.0	/17	255.255.128.0	1	2	32766
172.16.0.0	/18	255.255.192.0	2	4	16382
172.16.0.0	/19	255.255.224.0	3	8	8190
172.16.0.0	/20	255.255.240.0	4	16	4094
172.16.0.0	/21	255.255.248.0	5	32	2046
172.16.0.0	/22	255.255.252.0	6	64	1022
172.16.0.0	/23	255.255.254.0	7	128	510
172.16.0.0	/24	255.255.255.0	8	256	254
172.16.0.0	/25	255.255.255.128	9	512	126
172.16.0.0	/26	255.255.255.192	10	1024	62
172.16.0.0	/27	255.255.255.224	11	2048	30
172.16.0.0	/28	255.255.255.240	12	4096	14
172.16.0.0	/29	255.255.255.248	13	8192	6
172.16.0.0	/30	255.255.255.252	14	16384	2

.

20. Explain Distance Vector Routing with count-to-Infinity problem

In distance vector routing, each router maintains a routing table indexed by, and containing one entry for, each router in the subnet. This entry contains two parts: the preferred outgoing line to use for that destination and an estimate of the time or distance to that destination. The metric used might be number of hops, time delay in milliseconds, total number of packets queued along the path, or something similar.

The router is assumed to know the "distance" to each of its neighbors. If the metric is hops, the distance is just one hop. If the metric is queue length, the router simply examines each queue. If the metric is delay, the router can measure it directly with special ECHO packets that the receiver just timestamps and sends back as fast as it can.

As an example, assume that delay is used as a metric and that the router knows the delay to each of its neighbors. Once every T msec each router sends to each neighbor a list of its estimated delays to each destination. It also receives a similar list from each neighbor. Imagine that one of these tables has just come in from neighbor X, with X_i being X's estimate of how long it takes to get to router i. If the router knows that the delay to X is m msec, it also knows that it can reach router i via X in X_i + m msec. By performing this calculation for each neighbor, a router can find out which estimate seems the best and use that estimate and the corresponding line in its new routing table. Note that the old routing table is not used in the calculation.

This updating process is illustrated in Fig. 13-1. Part (a) shows a subnet. The first four columns of part (b) show the delay vectors received from the neighbors of router J. A claims to have a 12-msec delay to B, a 25-msec delay to C, a 40-msec delay to D, etc. Suppose that J has measured or estimated its delay to its neighbors, A, I, H, and K as 8, 10, 12, and 6 msec, respectively.

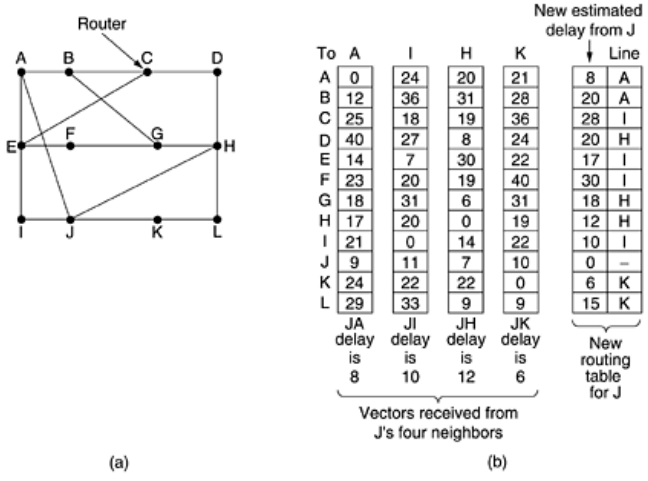


Figure 13-1. (a) A subnet. (b) Input from A, I, H, K, and the new routing table for J.

Consider how J computes its new route to router G. It knows that it can get to A in 8 msec, and A claims to be able to get to G in 18 msec, so J knows it can count on a delay of 26 msec to G if it forwards packets bound for G to A. Similarly, it computes the delay to G via I, H, and K as 41 (31 + 10), 18 (6 + 12), and 37 (31 + 6) msec, respectively. The best of these values is 18, so it makes an entry in its routing table that the delay to G is 18 msec and that the route to use is via H. The same calculation is performed for all the other destinations, with the new routing table shown in the last column of the figure.

The Count-to-Infinity Problem

Distance vector routing works in theory but has a serious drawback in practice: although it converges to the correct answer, it may do so slowly. In particular, it reacts rapidly to good news, but leisurely to bad news. Consider a router whose best route to destination X is large. If on the next exchange neighbor A suddenly reports a short delay to X, the router just switches over to using the line to A to send traffic to X. In one vector exchange, the good news is processed.

To see how fast good news propagates, consider the five-node (linear) subnet of Fig. 13-2, where the delay metric is the number of hops. Suppose A is down initially and all the other routers know this. In other words, they have all recorded the delay to A as infinity.

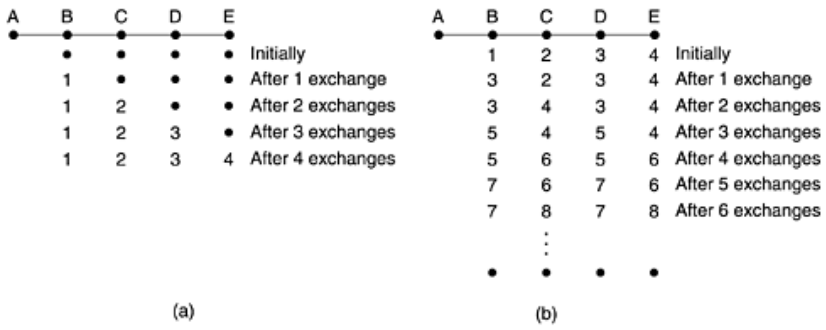


Figure 13-2. The Count to Infinity Problem

When A comes up, the other routers learn about it via the vector exchanges. For simplicity we will assume that there is a gigantic going somewhere that is struck periodically to initiate a vector exchange at all routers simultaneously. At the time of the first exchange, B learns that its left neighbor has zero delay to A. B now makes an entry in its routing table that A is one hop away to the left. All the other routers still think that A is down. At this point, the routing table entries for A are as shown in the second row of Fig. 13-3(a). On the next exchange, C learns that B has a path of length 1 to A, so it updates its routing table to indicate a path of length 2, but D and E do not hear the good news until later. Clearly, the

	good news is spreading at the rate of one hop per exchange. In a subnet whose longest path is of length N hops, within N exchanges everyone will know about newly-revived lines and routers. Now let us consider the situation of Fig. 13-3(b), in which all the lines and routers are initially up. Routers B, C, D, and E have distances to A of 1, 2, 3, and 4, respectively. Suddenly A goes down, or alternatively, the line between A and B is cut, which is effectively the same thing from B's point of view. At the first packet exchange, B does not hear anything from A. Fortunately, C says: Do not worry; I have a path to A of length 2. Little does B know that C's path runs through B itself. For all B knows, C might have ten lines all with separate paths to A of length 2. As a result, B thinks it can reach A via C, with a path length of 3. D and E do not update their entries for A on the first exchange. On the second exchange, C notices that each of its neighbors claims to have a path to A of length 3. It picks one of the them at random and makes its new distance to A 4, as shown in the third row of Fig. 13-3(b). Subsequent exchanges produce the history shown in the rest of Fig. 13-3(b). From this figure, it should be clear why bad news travels slowly: no router ever has a value more than one higher than the minimum of all its neighbors. Gradually, all routers work their way up to infinity, but the number of exchanges required depends on the numerical value used for infinity. For this reason, it is wise to set infinity to the longest path plus 1. If the metric is time delay, there is no well-defined upper bound, so a high value is needed to prevent a path with a long delay from being treated as down. Not entirely surprisingly, this problem is known as the count-to-infinity problem.
21.	**Explain Link State Routing** Distance vector routing was used in the ARPANET until 1979, when it was replaced by link state routing. Two primary problems caused its demise. First, since the delay metric was queue length, it did not take line bandwidth into account when choosing routes. Initially, all the lines were 56 kbps, so line bandwidth was not an issue, but after some lines had been upgraded to 230 kbps and others to 1.544 Mbps, not taking bandwidth into account was a major problem. Of course, it would have been possible to change the delay metric to factor in line bandwidth, but a second problem also existed, namely, the algorithm often took too long to converge (the count-to-infinity problem). For these reasons, it was replaced by an entirely new algorithm, now called link state routing. Variants of link state routing are now

widely used.

The idea behind link state routing is simple and can be stated as **five steps**. Each router must do the following:

1. Discover its neighbors and learn their network addresses.
2. Measure the delay or cost to each of its neighbors.
3. Construct a packet telling all it has just learned.
4. Send this packet to all other routers.
5. Compute the shortest path to every other router.

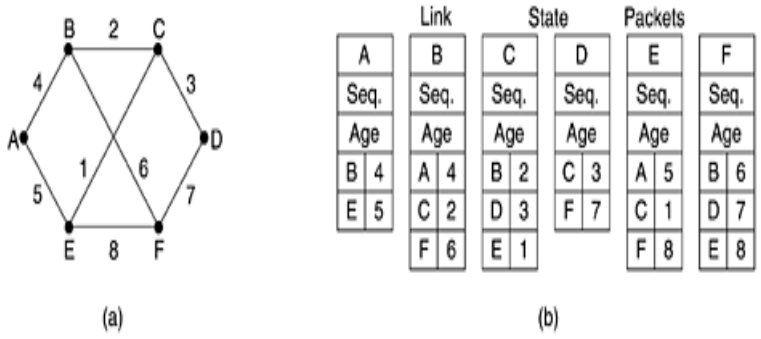


Figure 13-3. a) Router b) Routing Information

In effect, the complete topology and all delays are experimentally measured and distributed to every router. **Then Dijkstra's algorithm can be run to find the shortest path to every other router.**

			Send flags			ACK flags			
Source	Seq.	Age	A	C	F	A	C	F	Data
A	21	60	0	1	1	1	0	0	
F	21	60	1	1	0	0	0	1	
E	21	59	0	1	0	1	0	1	
C	20	60	1	0	1	0	1	0	
D	21	59	1	0	0	0	1	1	

Figure 13-4. The Packet Buffer for Router B

Link-state routing protocols were developed to alleviate the convergence and loop issues of distance-vector protocols. Link-state protocols maintain three separate tables:

- **Neighbor table** – contains a list of all neighbors, and the interface each neighbor is connected off of. Neighbors are formed by sending **Hello** packets.
- **Topology table** – otherwise known as the "link-state" table, contains a map of all links within an **area**, including each link's status.

	• **Shortest-Path table** – contains the *best* routes to each particular destination (otherwise known as the "routing" table")
22.	**Explain Shortest Path Routing** Let us begin our study of feasible routing algorithms with a technique that is widely used in many forms because it is simple and easy to understand. The idea is to build a graph of the subnet, with each node of the graph representing a router and each arc of the graph representing a communication line (often called a link). To choose a route between a given pair of routers, the algorithm just finds the shortest path between them on the graph. The concept of a shortest path deserves some explanation. One way of measuring path length is the number of hops. Using this metric, the paths ABC and ABE in Fig. 13-5 are equally long. Another metric is the geographic distance in kilometers, in which case ABC is clearly much longer than ABE (assuming the figure is drawn to scale). However, many other metrics besides hops and physical distance are also possible. For example, each arc could be labelled with the mean queueing and transmission delay for some standard test packet as determined by hourly test runs. With this graph labeling, the shortest path is the fastest path rather than the path with the fewest arcs or kilometers. In the general case, the labels on the arcs could be computed as a function of the distance, bandwidth, average traffic, communication cost, mean queue length, measured delay, and other factors. By changing the weighting function, the algorithm would then compute the "shortest" path measured according to any one of a number of criteria or to a combination of criteria.

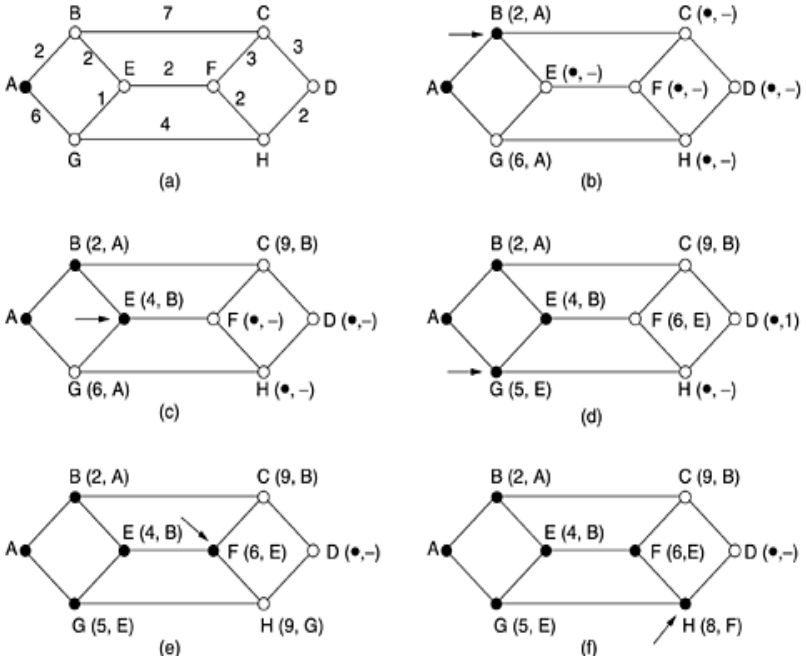


Figure 13-5. The first five steps used in computing the shortest path from A to D. The arrows indicate the working node.

Several algorithms for computing the shortest path between two nodes of a graph are known. This one is due to Dijkstra (1959). Each node is labeled (in parentheses) with its distance from the source node along the best known path. Initially, no paths are known, so all nodes are labeled with infinity. As the algorithm proceeds and paths are found, the labels may change, reflecting better paths. A label may be either tentative or permanent. Initially, all labels are tentative. When it is discovered that a label represents the shortest possible path from the source to that node, it is made permanent and never changed thereafter.

To illustrate how the labeling algorithm works, look at the weighted, undirected graph of Fig. 13-5(a), where the weights represent, for example, distance. We want to find the shortest path from A to D. We start out by marking node A as permanent, indicated by a filled-in circle. Then we examine, in turn, each of the nodes adjacent to A (the working node), relabeling each one with the distance to A. Whenever a node is relabeled, we also label it with the node from which the probe was made so that we can reconstruct the final path later. Having examined each of the nodes adjacent to A, we examine all the tentatively labeled nodes in the whole graph and make the one with the smallest label permanent, as shown in Fig. 13-5(b). This one becomes the new working node.

We now start at B and examine all nodes adjacent to it. If the sum of the label on B and the

distance from B to the node being considered is less than the label on that node, we have a shorter path, so the node is relabeled.

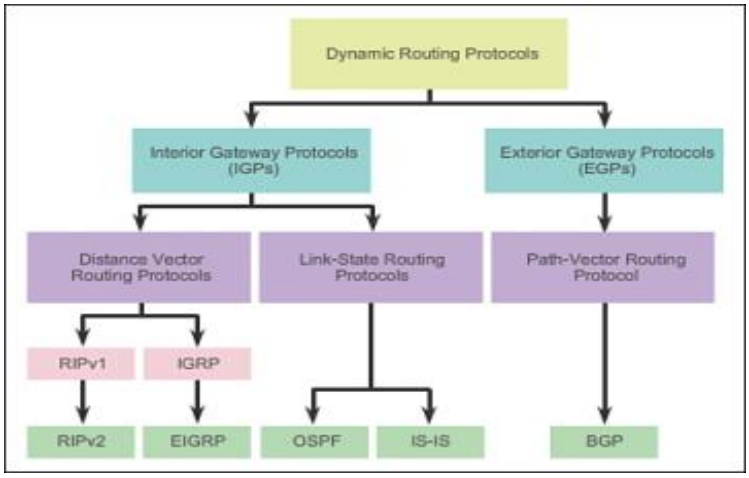
After all the nodes adjacent to the working node have been inspected and the tentative labels changed if possible, the entire graph is searched for the tentatively-labeled node with the smallest value. This node is made permanent and becomes the working node for the next round. Figure 13-5 shows the first five steps of the algorithm.

To see why the algorithm works, look at Fig. 13-5(c). At that point we have just made E permanent. Suppose that there were a shorter path than ABE, say AXYZE. There are two possibilities: either node Z has already been made permanent, or it has not been. If it has, then E has already been probed (on the round following the one when Z was made permanent), so the AXYZE path has not escaped our attention and thus cannot be a shorter path.

Now consider the case where Z is still tentatively labeled. Either the label at Z is greater than or equal to that at E, in which case AXYZE cannot be a shorter path than ABE, or it is less than that of E, in which case Z and not E will become permanent first, allowing E to be probed from Z.

23. Explain Any one Dynamic Routing Protocol.

Dynamic Routing Protocols



	RIPv1	RIPv2	IGRP	EIGRP	OSPF	IS-IS	BGP
Interior/	Interior	Interior	Interior	Interior	Interior	Interior	Exterior

Exterior							
Type	Distance vector	Distance vector	Distance vector	Distance vector	Link state	Link state	Path Vwctor
Speed of Convergence	Slow	Slow	Slow	Fast	Fast	Fast	Average
Scalability – Size of Network	Small	Small	Small	Large	Large	Large	Large
Use of VLSM	No	Yes	No	Yes	Yes	Yes	Yes
Resource Usage	Low	Low	Low	Medium	High	High	High
Implementation and Maintenance	Simple	Simple	Simple	Complex	Complex	Complex	Complex
Classful/ Classless	Classful	Classless	Classful	Classless	Classless	Classless	Classless
Metric	Hop	Hop	Bandwidth and delay (also can include reliability, load, and MTU)	Composite	Cost	Cost	Multiple Attributes
Time Period	30 sec	30 sec	90 sec	Internal-90 sec External-170 Sec	Only When Changes occur	Only When Changes occur	Only When Changes occur
Administrative Distance(AD)	120	120	100	100	110	115	200
Algorithm	Bellman-Ford	Bellman-Ford	Bellman-Ford	DUAL	Dijkstra	Dijkstra	Best Path Algorithm
Updates	Full Table	Full Table	Full Table	Only Changes	Only Changes	Only Changes	Only Changes

24. List out different Sliding Window Protocols and explain any one in details.

OR

Explain the following protocols: Stop and Wait Protocol.

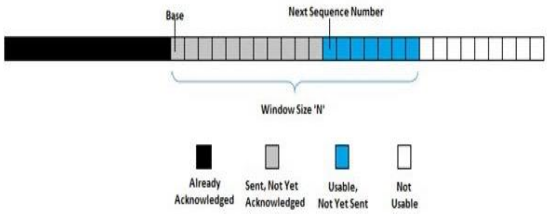
Sliding window protocol

By placing limits on the number of packets that can be transmitted or received at any given time, a sliding window protocol allows an unlimited number of packets to be communicated using fixed-size sequence numbers. The term "window" on the transmitter side represents the logical boundary of the total number of packets yet to be acknowledged by the receiver. The receiver informs the transmitter in each acknowledgment packet the current maximum receiver buffer size (window boundary). The TCP header uses a 16 bit field to report the receive window size to the sender. Therefore, the largest window that can be used is $2^{16} = 64$ kilobytes.

The sliding window method ensures that traffic congestion on the network is avoided. The application layer will still be offering data for transmission to TCP without worrying about the network traffic congestion issues as the TCP on sender and receiver side implement sliding windows of packet buffer. The window size may vary dynamically depending on network traffic.

In any communication protocol based on automatic repeat request for error control, the receiver must acknowledge received packets. If the transmitter does not receive an acknowledgment within a reasonable time, it re-sends the data.

Data in sender buffer are sent in chunks instead of the entire data in buffer. **Why**? Suppose the sender buffer has the capacity of 1 MB and the receiver buffer has the capacity of 512KB, then 50 % of the data will be lost at the receiver end and this will unnecessarily cause re-transmission of packets from the sender end. Therefore, the sender will send data in chunks less than 512. This will be decided by the help of window size. Window size caters the capacity of the receiver. Flow control is the receiver related problem, we do not want the receiver to be overwhelmed, thus in order to avoid overwhelming situation, we need to control the flow by using window size "N".



The above figure shows the buffer at the sender end.

"Base" indicates the sequence number of the packet that has not been acknowledged (the first unacknowledged packet).

Window shifts when the packets starting from the base gets acknowledged and it starts moving, this process continues till all acknowledgements are received, so it is also known as sliding window protocol. When window size is reached the packet will not be sent further till the window slides further.

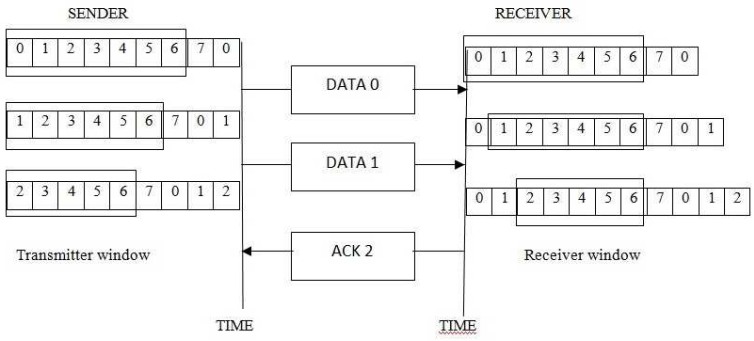
How will the sender be aware of the size of the receiver's buffer or what should be the window size at the sender end?

Before the data is transmitted from one host to another, a connection is first established between the two. During this establishment, such kinds of information are share between the two hosts, like window size, buffer size, etc. After which the data transmission begins.

Sliding window protocol assumes full duplex communication. It uses two types of frames, first data and second acknowledgment. One of important features of all the sliding windows protocol is that each outbound frame contains a sequence number, ranging from 0 to 2n -1,

where the value of n can be arbitrary. Sliding window refers to imaginary boxes at the transmitter and receiver. This window provides the upper limit on the number of frames that can be transmitted before acknowledgment requirement. Window holds the number of frame to provide above mention limit. The frames which are being transmitted to send are falling in sending window similarly frames to be accepted are store in the receiving window.

Significance of Sender's and Receiver's Windows:



The sequence no. within the sender's window gives the number of frame sent but not yet acknowledge. The frames in the sender's window are stored so that they can be possibly retransmitted in the case of damage while travelling to receiver.

The receiver window represents not the number of frame receive but the no. of frame that may still be received before an ACK is sent. Because sliding window of receiver strings from left when frame of data are received and expand to right when ACK is sent. The receiver window contains (n-1) spaces for frame.

Sliding Window

• Sliding window refers to an imaginary boxes that hold the frames on both sender and receiver side.

• It provides the upper limit on the number of frames that can be transmitted before requiring an acknowledgment.

• Frames may be acknowledged by receiver at any point even when window is not full on receiver side.

• Frames may be transmitted by source even when window is not yet full on sender side.

• The windows have a specific size in which the frames are numbered modulo- n, which means they are numbered from 0 to n-l. For e.g. if n = 8, the frames are numbered 0, 1,2,3,4,5,6, 7, 0, 1,2,3,4,5,6, 7, 0, 1, ...

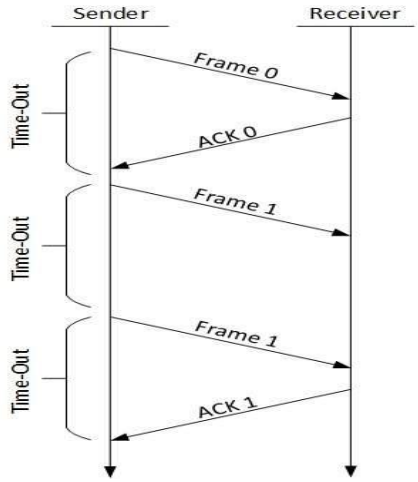
• The size of window is n-1. For e.g. In this case it is 7. Therefore, a maximum of n-l frames may be sent before an acknowledgment.

• When the receiver sends an ACK, it includes the number of next frame it expects to receive. For example in order to acknowledge the group of frames ending in frame 4, the receiver sends an ACK containing the number 5. When sender sees an ACK with number 5, it comes to know that all the frames up to number 4 have been received.

A One Bit Sliding Window Protocol (Stop and Wait ARQ):

In this case n=1 and uses stop and wait technique. Sender waits for ACK after each frame transmission. The operation of this protocol is based on the ARQ(automatic repeat request) principle, which hold the next frame will be transmitted when positive ACK is received and when negative ACK is received, it retransmit the same frame.

Stop and wait ARQ becomes inefficient when the propagation delay is much greater than the time tool retransmit for example let us assume that frame of 800 bits is transmitted over channel with speed 1mbps and let time for transmission if from end ACK is 30 ms. The number of bits that can be transmitted over this channel is 30,000 bits. But in stop and wait ARQ only 800 bits can be transmitted as it waits for ACK. The product of bit rate and delay is called delay bandwidth product. It helps in measuring last opportunity in transmitted bits.



The following transition may occur in Stop-and-Wait ARQ:

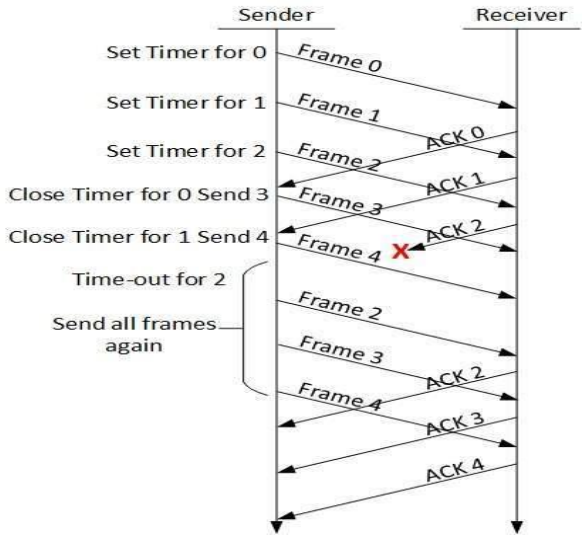
- The sender maintains a timeout counter.
- When a frame is sent, the sender starts the timeout counter.
- If acknowledgement of frame comes in time, the sender transmits the next frame in queue.

- If acknowledgement does not come in time, the sender assumes that either the frame or its acknowledgement is lost in transit. Sender retransmits the frame and starts the timeout counter.
- If a negative acknowledgement is received, the sender retransmits the frame.

25. Explain Go Back N Protocol.

A Protocol Using Go Back n ARQ:

The sender in this case does not waits for the ACK signal for transmission of next frame. The sender continuously transmits the frame so that the channel should be kept busy rather that wasting time in waiting for it ACK. Because in stop and protocol system does not transmit anything while it is waiting So channel remain idle for considerable time period But in this case the system does depends on only NACK(negative feedback). It symbolizes error in a particular frame. But as NACK signal will take same time to reach sender, the sender will continue to transmit. On the reception of the NACK signal, the transmitter will retransmit all the frames 3 onwards. The receivers discard all the frames it has received after 3. Example: suppose the frame is being transmitted end at frame bit 3 error occurs and NACK is transmitted at the receiver. But this takes some time to reach the transmitter. By the time upto frame 7 has all ways been transmitted.



If the transmitter frame is lost or acknowledgement is lost then only error occurs. In case of damaged or lost frames the receiver transmits NACK to transmitter and the transmitter retransmits all the frames sent since the last frame acknowledged. The disadvantage of go back ARQ protocol is that its efficiency decreases in noisy channel as it does not wait for ACK after every frame transmitted.

Stop and wait ARQ mechanism does not utilize the resources at their best. When the acknowledgement is received, the sender sits idle and does nothing. In Go-Back-N ARQ method, both sender and receiver maintain a window.

The sending-window size enables the sender to send multiple frames without receiving the acknowledgement of the previous ones. The receiving-window enables the receiver to receive multiple frames and acknowledge them. The receiver keeps track of incoming frame's sequence number.

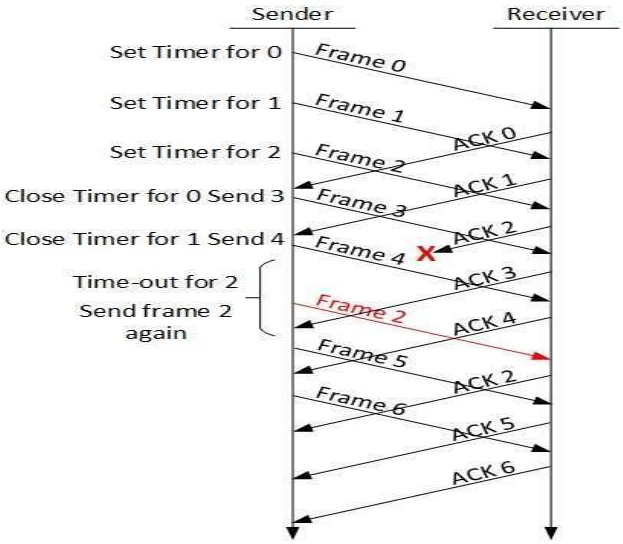
When the sender sends all the frames in window, it checks up to what sequence number it has received positive acknowledgement. If all frames are positively acknowledged, the sender sends next set of frames. If sender finds that it has received NACK or has not receive any ACK for a particular frame, it retransmits all the frames after which it does not receive any positive ACK.

Go-Back-N ARQ is a specific instance of the automatic repeat request (ARQ) protocol, in which the sending process continues to send a number of frames specified by a *window size* even without receiving an acknowledgement (ACK) packet from the receiver. It is a special case of the general sliding window protocol with the transmit window size of N and receive window size of 1. It can transmit N frames to the peer before requiring an ACK.

The receiver process keeps track of the sequence number of the next frame it expects to receive, and sends that number with every ACK it sends. The receiver will discard any frame that does not have the exact sequence number it expects (either a duplicate frame it already acknowledged, or an out-of-order frame it expects to receive later) and will resend an ACK for the last correct in-order frame. [1] Once the sender has sent all of the frames in its *window*, it will detect that all of the frames since the first lost frame are *outstanding*, and will go back to the sequence number of the last ACK it received from the receiver process and fill its window starting with that frame and continue the process over again.

Go-Back-N ARQ is a more efficient use of a connection than Stop-and-wait ARQ, since unlike waiting for an acknowledgement for each packet, the connection is still being utilized as packets are being sent. In other words, during the time that would otherwise be spent waiting, more packets are being sent. However, this method also results in sending frames multiple times – if any frame was lost or damaged, or the ACK acknowledging them was lost or damaged, then that frame and all following frames in the window (even if they were received without error) will be re-sent. To avoid this, Selective Repeat ARQ can be used.

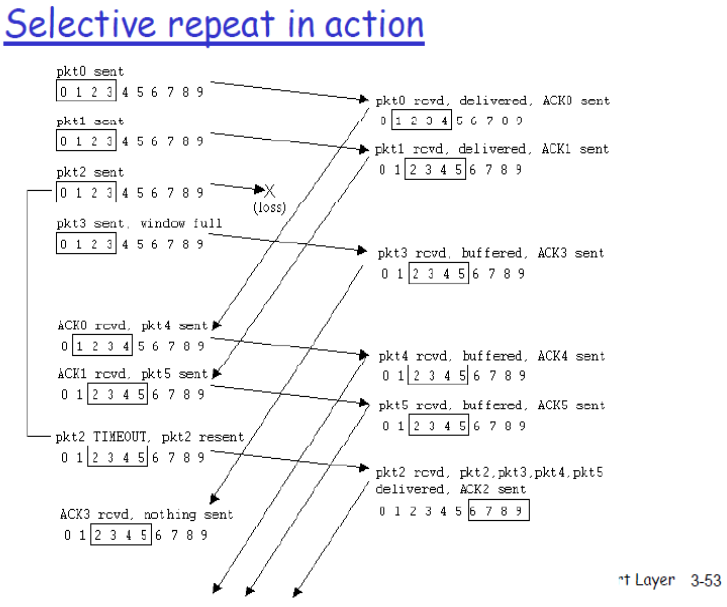
26. Explain Selective Repeat ARQ.



In Go-back-N ARQ, it is assumed that the receiver does not have any buffer space for its window size and has to process each frame as it comes. This enforces the sender to retransmit all the frames which are not acknowledged.

In Selective-Repeat ARQ, the receiver while keeping track of sequence numbers, buffers the frames in memory and sends NACK for only frame which is missing or damaged.

The sender in this case, sends only packet for which NACK is received.



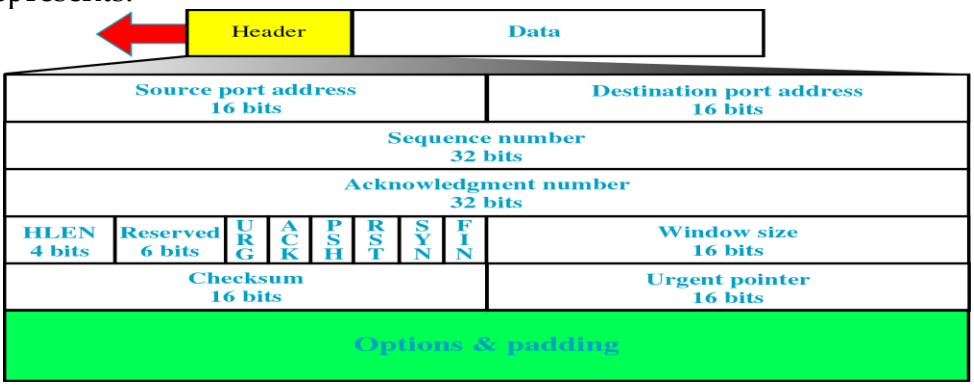
27. Explain TCP header fields

OR

Draw and explain TCP packet header.

The format of the TCP header is as follows:

1. **Source port number** (16 bits)
2. **Destination port number** (16 bits)
3. **Sequence number** (32 bits) - The byte in the data stream that the first byte of this packet represents.



4. **Acknowledgement number** (32 bits) –Acknowledge the receipt of data, and Contains the next sequence number that the sender of the acknowledgement expects to receive which is the sequence number plus 1 (plus the number of bytes received in the last message?). This number is used only if the ACK flag is on.
5. **Header length** (4 bits) – It shows the length of the header.
6. **Reserved** (6 bits) –Reserved for future use.
7. **Control bits**
    a) **URG** (1 bit) - The urgent pointer is valid.
    b) **ACK** (1 bit) - Makes the acknowledgement number valid.
    c) **PSH** (1 bit) - High priority data for the application.
    d) **RST** (1 bit) - Reset the connection.
    e) **SYN** (1 bit) - Turned on when a connection is being established and the sequence number field will contain the initial sequence number chosen by this host for this connection.
    f) **FIN** (1 bit) - The sender is done sending data.
8. **Window size** (16 bits) - The maximum number of bytes that the receiver will to accept. (it's maximum size of sliding window)
9. **TCP checksum** (16 bits) – Use to detect error.
10. **Urgent pointer** (16 bits) - It is only valid if the URG bit is set. The urgent mode is a way to transmit emergency data to the other side of the connection.
11. **Options and Padding** - (variable length) Convey additional Information for alignment purpose.

28. Explain Congestion Control.

Congestion:

An important issue in a packet-switched network is congestion. Congestion in a network may occur if the load on the network-the number of packets sent to the network-is greater than the *capacity* of the network-the number of packets a network can handle. Congestion control refers to the mechanisms and techniques to control the congestion and keep the load below the capacity.

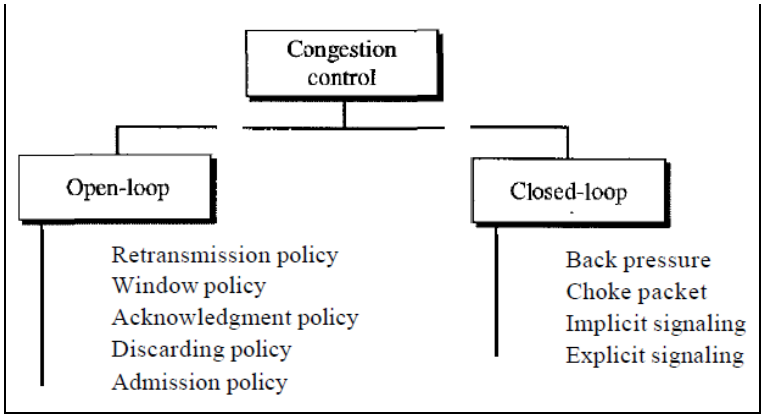
We may ask why there is congestion on a network. Congestion happens in any system that involves waiting. For example, congestion happens on a freeway because any abnonnality in the flow, such as an accident during rush hour, creates blockage.

Congestion in a network or internetwork occurs because routers and switches have queues-buffers that hold the packets before and after processing. A router, for example, has an input queue and an output queue for each interface.

Congestion Control:

<u>Open-Loop Congestion Control</u>

In open-loop congestion control, policies are applied to prevent congestion before it happens. In these mechanisms, congestion control is handled by either the source or the destination. We give a brief list of policies that can prevent congestion.



Congestion Control *categories*

Retransmission Policy

Retransmission is sometimes unavoidable. If the sender feels that a sent packet is lost or corrupted, the packet needs to be retransmitted. Retransmission in general may increase congestion in the network. However, a good retransmission policy can prevent congestion. The retransmission policy and the retransmission timers must be designed to optimize efficiency and at the same time prevent congestion. For example, the retransmission policy used by TCP (explained later) is designed to prevent or alleviate congestion.

Window Policy

The type of window at the sender may also affect congestion. The Selective Repeat window is better than the Go-Back-N window for congestion control. In the *Go-Back-N* window, when the timer for a packet times out, several packets may be resent, although some may have arrived safe and sound at the receiver. This duplication may make the congestion worse. The Selective Repeat window, on the other hand, tries to send the specific packets that have been lost or corrupted.

Acknowledgment Policy

The acknowledgment policy imposed by the receiver may also affect congestion. If the receiver does not acknowledge every packet it receives, it may slow down the sender and help prevent congestion. Several approaches are used in this case. A receiver may send an acknowledgment only if it has a packet to be sent or a special timer expires. A receiver may decide to acknowledge only *N* packets at a time. We need to know that the acknowledgments are also part of the load in a network. Sending fewer acknowledgments means imposing less load on the network.

Discarding Policy

A good discarding policy by the routers may prevent congestion and at the same time may not harm the integrity of the transmission. For example, in audio transmission, if the policy is to discard less sensitive packets when congestion is likely to happen, the quality of sound is still preserved and congestion is prevented or alleviated.

Admission Policy

An admission policy, which is a quality-of-service mechanism, can also prevent congestion in virtual-circuit networks. Switches in a flow first check the resource requirement of a flow before admitting it to the network. A router can deny establishing a virtualcircuit connection if there is congestion in the network or if there is a possibility of future congestion.

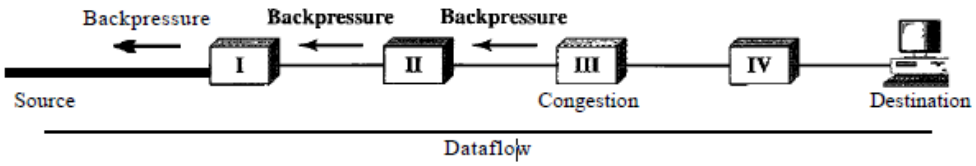
Closed-Loop Congestion Control

Closed-loop congestion control mechanisms try to alleviate congestion after it happens.

Several mechanisms have been used by different protocols. We describe a few of them here.

Backpressure

The technique of *backpressure* refers to a congestion control mechanism in which a congested node stops receiving data from the immediate upstream node or nodes. This may cause the upstream node or nodes to become congested, and they, in turn, reject data from their upstream nodes or nodes. And so on. Backpressure is a node-to-node congestion control that starts with a node and propagates, in the opposite direction of data flow, to the source. The backpressure technique can be applied only to virtual circuit networks, in which each node knows the upstream node from which a flow of data is corning. Figure shows the idea of backpressure.

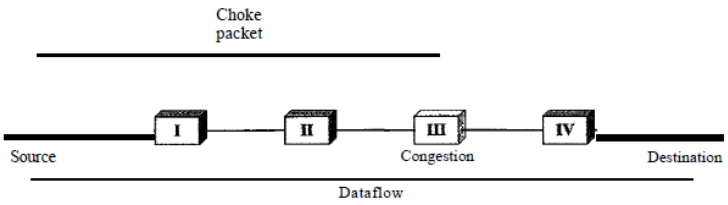


Backpressure method for alleviating congestion

Node III in the figure has more input data than it can handle. It drops some packets in its input buffer and informs node II to slow down. Node II, in turn, may be congested because it is slowing down the output flow of data. If node II is congested, it informs node I to slow down, which in turn may create congestion. If so, node I informs the source of data to slow down. This, in time, alleviates the congestion. Note that the *pressure* on node III is moved backward to the source to remove the congestion.

Choke Packet

A choke packet is a packet sent by a node to the source to inform it of congestion. Note the difference between the backpressure and choke packet methods. In backpressure, the warning is from one node to its upstream node, although the warning may eventually reach the source station. In the choke packet method, the warning is from the router, which has encountered congestion, to the source station directly. The intermediate nodes through which the packet has traveled are not warned. We have seen an example of this type of control in ICMP. When a router in the Internet is overwhelmed with IP datagram, it may discard some of them; but it informs the source host, using a source quench ICMP message. The warning message goes directly to the source station; the intermediate routers, and does not take any action. Figure shows the idea of a choke packet.



Implicit Signaling

In implicit signaling, there is no communication between the congested node or nodes and the source. The source guesses that there is a congestion somewhere in the network from other symptoms. For example, when a source sends several packets and there is no acknowledgment for a while, one assumption is that the network is congested. The delay in receiving an acknowledgment is interpreted as congestion in the network; the source should slow down.

Explicit Signaling

The node that experiences congestion can explicitly send a signal to the source or

destination. The explicit signaling method, however, is different from the choke packet method. In the choke packet method, a separate packet is used for this purpose; in the explicit signaling method, the signal is included in the packets that carry data. Explicit signaling, as we will see in Frame Relay congestion control, can occur in either the forward or the backward direction.

Backward Signaling A bit can be set in a packet moving in the direction opposite to the congestion. This bit can warn the source that there is congestion and that it needs to slow down to avoid the discarding of packets.

Forward Signaling

A bit can be set in a packet moving in the direction of the congestion. This bit can warn the destination that there is congestion. The receiver in this case can use policies, such as slowing down the acknowledgments, to alleviate the congestion.

29. Give differences between TCP and UDP.

	TCP	**UDP**
Acronym for	Transmission Control Protocol	User Datagram Protocol or Universal Datagram Protocol
Connection	TCP is a connection-oriented protocol.	UDP is a connectionless protocol.
Function	As a message makes its way across the internet from one computer to another. This is connection based.	UDP is also a protocol used in message transport or transfer. This is not connection based which means that one program can send a load of packets to another and that would be the end of the relationship.
Usage	TCP is suited for applications that require high reliability, and transmission time is relatively less critical.	UDP is suitable for applications that need fast, efficient transmission, such as games. UDP's stateless nature is also useful for servers that answer small queries from huge numbers of clients.
Examples	HTTP, HTTPs, FTP, SMTP, Telnet	DNS, DHCP, TFTP, SNMP, RIP, VOIP.

Ordering of data packets	TCP rearranges data packets in the order specified.	UDP has no inherent order as all packets are independent of each other. If ordering is required, it has to be managed by the application layer.
Speed of transfer	The speed for TCP is slower than UDP.	UDP is faster because there is no error-checking for packets.
Reliability	There is absolute guarantee that the data transferred remains intact and arrives in the same order in which it was sent.	There is no guarantee that the messages or packets sent would reach at all.
Header Size	TCP header size is 20 bytes	UDP Header size is 8 bytes.
Common Header Fields	Source port, Destination port, Check Sum	Source port, Destination port, Check Sum
Streaming of data	Data is read as a byte stream, no distinguishing indications are transmitted to signal message (segment) boundaries.	Packets are sent individually and are checked for integrity only if they arrive. Packets have definite boundaries which are honored upon receipt, meaning a read operation at the receiver socket will yield an entire message as it was originally sent.
Weight	TCP is heavy-weight. TCP requires three packets to set up a socket connection, before any user data can be sent. TCP handles reliability and congestion control.	UDP is lightweight. There is no ordering of messages, no tracking connections, etc. It is a small transport layer designed on top of IP.
Data Flow Control	TCP does Flow Control. TCP requires three packets to set up a socket connection, before any user data can be sent. TCP handles reliability and congestion control.	UDP does not have an option for flow control

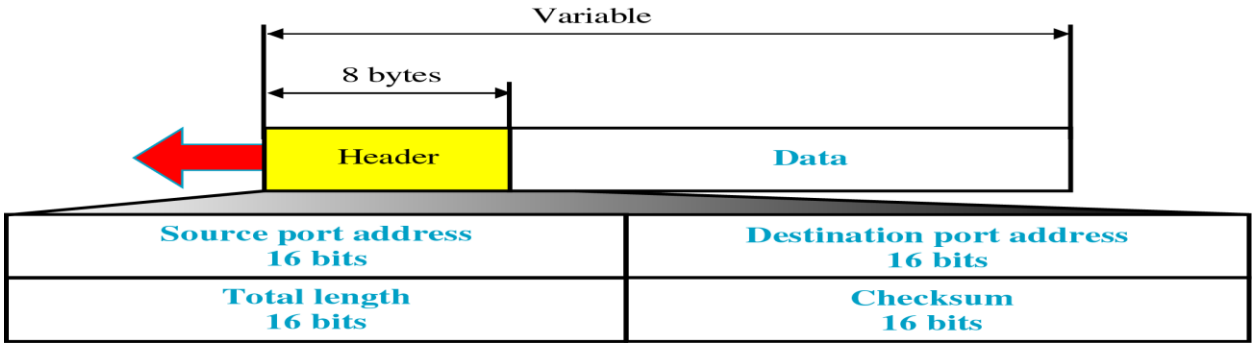
Error Checking	TCP does error checking	UDP does error checking, but no recovery options.
Fields	1. Sequence Number, 2. AcK number, 3. Data offset, 4. Reserved, 5. Control bit, 6. Window, 7. Urgent Pointer 8. Options, 9. Padding, 10. Check Sum, 11. Source port, 12. Destination port	1. Length, 2. Source port, 3. Destination port, 4. Check Sum
Acknowledgement	Acknowledgement segments	No Acknowledgment
Handshake	SYN, SYN-ACK, ACK	No handshake (connectionless protocol)
Checksum	checksum	to detect errors

30. Explain UDP with its header format, Features and Applications. (Any Two can be ask)

User Datagram Protocol

- User Datagram Protocol (UDP) supports the network at the transport layer.
- User Datagram Protocol (UDP) is an unreliable connection-less protocol
- There is no guarantee that the data will reach its destination.
- UDP is meant to provide service with very little transmission overhead.
- It adds very little to IP data packets except for some error checking and port direction (Remember, UDP encapsulates IP packets).
- The Packet produced by UDP is called **"User Datagram".**

UDP Message Format /User Datagram Format



The UDP header includes:

1. **Source port number** (16 bits) - An optional field, The Address of Application Program that has created the message.
2. **Destination port number** (16 bits) –The Address of Application Program that will

receive the message.

3. **UDP length** (16 bits) –Gives the total Data Length of the user Datagram.
4. **UDP checksum** (16 bits)-Used in error detection.

Use of UDP

The following lists some uses of the UDP protocol:

- UDP is suitable for a process that requires simple request-response communication with little concern for flow and error control. It is not usually used for a process such as FrP that needs to send bulk data.
- UDP is suitable for a process with internal flow and error control mechanisms. For example, the Trivial File Transfer Protocol (TFTP) process includes flow and error control. It can easily use UDP.
- UDP is a suitable transport protocol for multicasting. Multicasting capability is embedded in the UDP software but not in the TCP software.
- UDP is used for management processes such as SNMP.
- UDP is used for some route updating protocols such as Routing Information Protocol (RIP).

Features

- UDP is used when acknowledgement of data does not hold any significance.
- UDP is good protocol for data flowing in one direction.
- UDP is simple and suitable for query based communications.
- UDP is not connection oriented.
- UDP does not provide congestion control mechanism.
- UDP does not guarantee ordered delivery of data.
- UDP is stateless.
- UDP is suitable protocol for streaming applications such as VoIP, multimedia streaming.

UDP application

Here are few applications where UDP is used to transmit data:

- Domain Name Services
- Simple Network Management Protocol
- Trivial File Transfer Protocol
- Routing Information Protocol

31. Explain the concept of congestion control with example.

Congestion Window

We said that the sender window size is determined by the available buffer space in the receiver *(rwnd).*

The sender has two pieces of information: the receiver-advertised window size and the congestion window size. The actual size of the window is the minimum of these two.

Actual window size = minimum (rwnd, cwnd)

Congestion Policy

TCP's general policy for handling congestion is based on three phases:

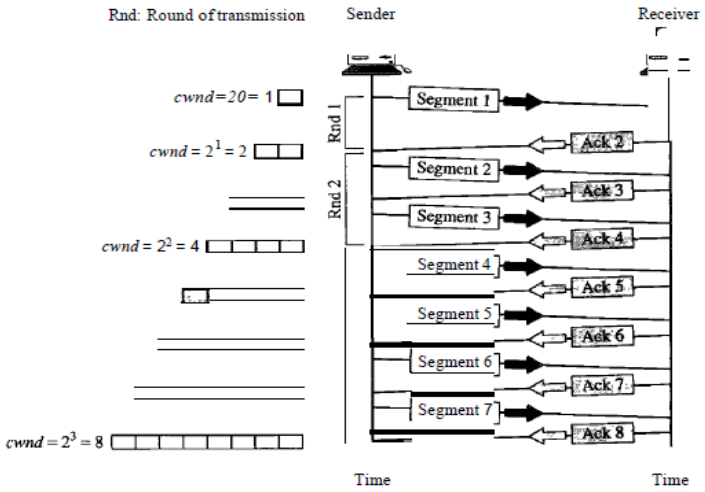
- Slow start (*exponential increase*),
- Congestion avoidance (Additive Increase), and
- Congestion detection (Multiplicative Decrease)

Slow start:

In the slow-start phase, the sender starts with a very slow rate of transmission, but increases the rate rapidly to reach a threshold. When the threshold is reached, the data rate is reduced to avoid congestion. Finally if congestion is detected, the sender goes back to the slow-start or congestion avoidance phase based on how the congestion is detected.

Exponential Increase One of the algorithms used in TCP congestion control is called slow start. This algorithm is based on the idea that the size of the congestion window *(cwnd)* starts with one maximum segment size (MSS). The MSS is determined during connection establishment by using an option of the same name. The size of the window increases one MSS each time an acknowledgment is received. As the name implies, the window starts slowly, but grows exponentially. To show the idea, let us look at Figure

We have used segment numbers instead of byte numbers (as though each segment contains only 1 byte). We have assumed that *rwnd* is much higher than *cwnd,* so that the sender window size always equals *cwnd.* We have assumed that each segment is acknowledged individually.



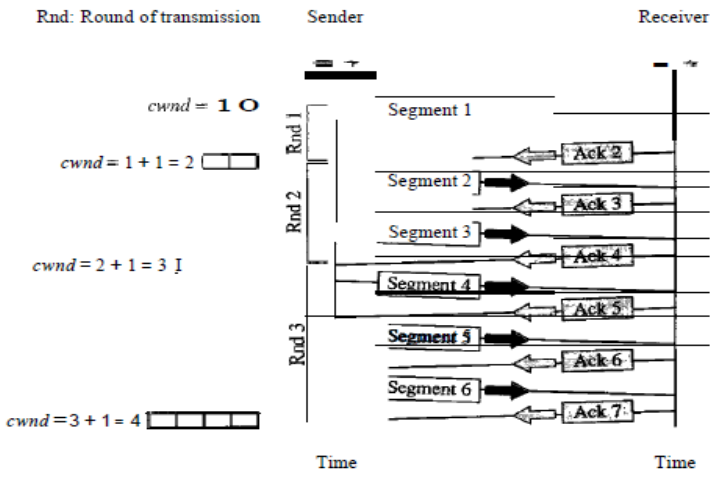
Slow start, exponential increase

The sender starts with *cwnd* =1 MSS. This means that the sender can send only one segment. After receipt of the acknowledgment for segment 1, the size of the congestion window is increased by 1, which means that *cwnd* is now 2. Now two more segments can be sent. When each acknowledgment is received, the size of the window is increased by 1 MSS. When all seven segments are acknowledged, *cwnd* = 8.

In the slow-start algorithm, the size of the congestion window increases exponentially until it reaches a threshold.

Congestion Avoidance: Additive Increase

If we start with the slow-start algorithm, the size of the congestion window increases exponentially. To avoid congestion before it happens, one must slow down this exponential growth. TCP defines another algorithm called congestion avoidance, which undergoes an additive increase instead of an exponential one. When the size of the congestion window reaches the slow-start threshold, the slow-start phase stops and the additive phase begins. In this algorithm, each time the whole window of segments is acknowledged (one round), the size of the congestion window is increased by 1. To show the idea, we apply this algorithm to the same scenario as slow start, although we will see that the congestion avoidance algorithm usually starts when the size of the window is much greater than 1. Figure shows the idea.



Congestion avoidance, additive increase

In the congestion avoidance algorithm, the size of the congestion window increases additively until congestion is detected.

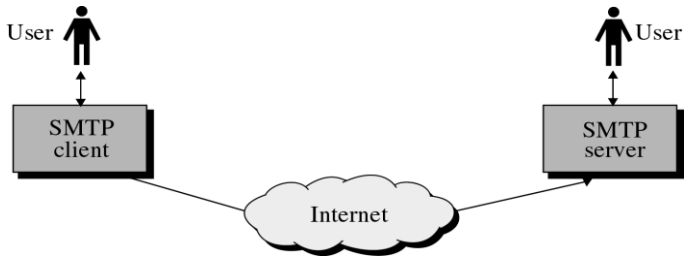
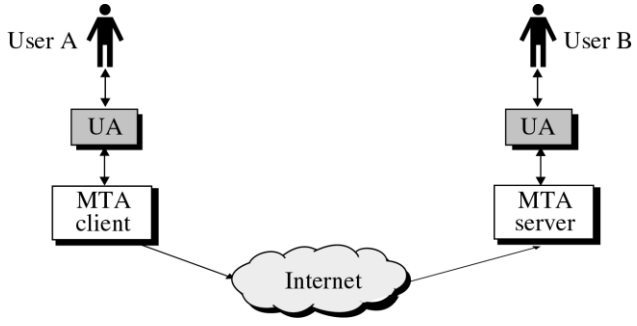
Congestion Detection: Multiplicative Decrease

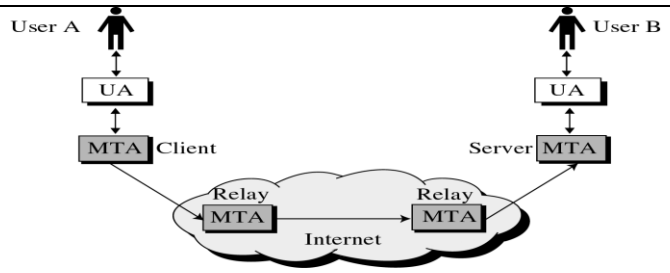
If congestion occurs, the congestion window size must be decreased. The only way the sender can guess that congestion has occurred is by the need to retransmit a segment. However, retransmission can occur in one of two cases: when a timer times out or when three ACKs are received. In both cases, the size of the threshold is dropped to one-half, a multiplicative decrease. Most TCP implementations have two reactions:

I. If a time-out occurs, there is a stronger possibility of congestion; a segment has probably been dropped in the network, and there is no news about the sent segments.
In this case TCP reacts strongly:
a. It sets the value of the threshold to one-half of the current window size.
b. It sets *cwnd* to the size of one segment.
c. It starts the slow-start phase again.

2. If three ACKs are received, there is a weaker possibility of congestion; a segment may have been dropped, but some segments after that may have arrived safely since three ACKs are received. This is called fast transmission and fast recovery. In this case, TCP has a weaker reaction: a. It sets the value of the threshold to one-half of the current window size. b. It sets *cwnd* to the value of the threshold (some implementations add three segment sizes to the threshold). c. It starts the congestion avoidance phase.

Implementation reacts to congestion detection in one of the following ways:

	o If detection is by time-out, a new *slow-start* phase starts. **o If detection is by three ACKs, a new *congestion avoidance* phase starts.**
32.	**Explain SMTP and POP3.** **SMTP (Simple Mail Transfer Protocol):** • One of the most popular network services is **electronic mail (e-mail)**. • The TCP/IP protocol that supports electronic mail on the Internet is called **Simple Mail transfer protocol (SMTP).** • It is a system for sending messages to other computer users based in e-mail addresses. • SMTP provides for mail exchange between users on the same or different computers and supports:  o Sending a single message to one or more recipients. o Sending messages that include text, voice, video, or graphics. o Sending messages to users on networks outside the Internet. • SMTP system is a combination of SMTP client and SMTP server. • SMTP client and server into two components: **user agent (UA)** and **mail transfer agent (MTA)**. • The **UA** prepares the message, creates the envelope, and puts the message in the envelope. • The **MTA** transfers the mail across the Internet.  • Instead of just one MTA at the sender site and one at the receiving site, other MTAs, acting either as client or server, can relay the mail.



- Specific format of E-mail Address:

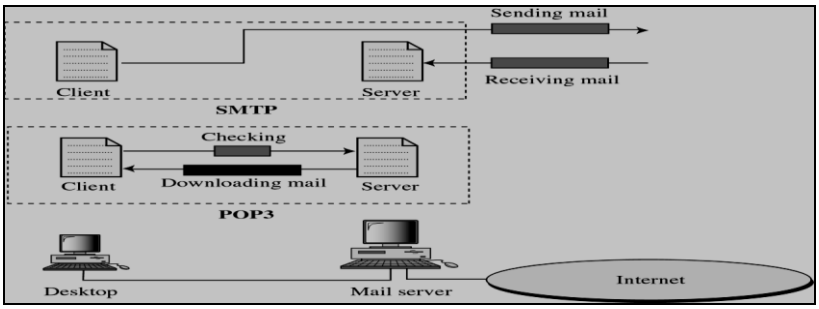


Local Part: The local part defines the name of a special file, called the user mailbox, where all the mail received for a user is stored for retrieval by the message access agent.

Domain Name: The second part of the address is the domain name. An organization usually selects one or more hosts to receive and send e-mail; the hosts are sometimes called *mail servers* or *exchangers.* The domain name assigned to each mail exchanger either comes from the DNS database or is a logical name (for example, the name of the organization).

Mailing List

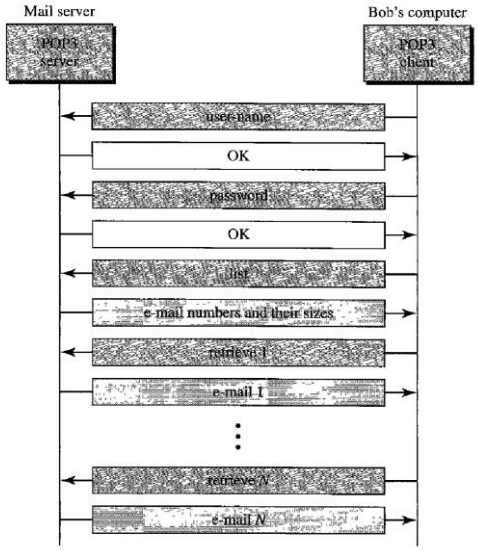
Electronic mail allows one name, an alias, to represent several different e-mail addresses; this is called a mailing list. Every time a message is to be sent, the system checks the recipient's name against the alias database; if there is a mailing list for the defined alias, separate messages, one for each entry in the list, must be prepared and handed to the MTA. If there is no mailing list for the alias, the name itself is the receiving address and a single message is delivered to the mail transfer entity.



POP3 (Post Office Protocol version 3):

- SMTP expects the destination host, the mail server receiving the mail, to be on-line all the time; otherwise, a TCP connection cannot be established.

- For this reason, it is not practical to establish an SMTP session with a desktop computer because desktop computers are usually powered down at the end of the day.
- In many organizations, mail is received by an SMTP server that is always on-line. This SMTP server provides a mail-drop service.
- The server receives the mail on behalf of every host in the organization.
- Workstations interact with the SMTP host to retrieve messages by using a client-server protocol such as **Post office protocol version 3 (POP3)**.



- Although POP3 is used to download messages form the server, the SMTP client is still needed on the desktop to forward messages from the workstation user to its SMTP mail server
- Post Office Protocol, version 3 (POP3) is simple and limited in functionality. The client POP3 software is installed on the recipient computer; the server POP3 software is installed on the mail server.
- Figure shows an example of downloading using POP3
- POP3 has two modes: the delete mode and the keep mode. In the delete mode, the mail is deleted from the mailbox after each retrieval. In the keep mode, the mail remains in the mailbox after retrieval. The delete mode is normally used when the user is working at her permanent computer and can save and organize the received mail after reading or replying. The keep mode is normally used when the user accesses her mail away from her primary computer (e.g., a laptop). The mail is read but kept in the system for later retrieval and organizing.

33. Explain IMAP4 and MIME.

IMAP4

- Another mail access protocol is Internet Mail Access Protocol, version 4 (IMAP4). IMAP4 is similar to POP3, but it has more features; IMAP4 is more powerful and

more complex.

- POP3 is deficient in several ways. It does not allow the user to organize her mail on the server; the user cannot have different folders on the server. (Of course, the user can create folders on her own computer.) In addition, POP3 does not allow the user to partially check the contents of the mail before downloading.
- IMAP4 provides the following extra functions:
    - A user can check the e-mail header prior to downloading.
    - A user can search the contents of the e-mail for a specific string of characters prior to downloading.
    - A user can partially download e-mail. This is especially useful if bandwidth is limited and the e-mail contains multimedia with high bandwidth requirements.
    - A user can create, delete, or rename mailboxes on the mail server.
    - A user can create a hierarchy of mailboxes in a folder for e-mail storage.

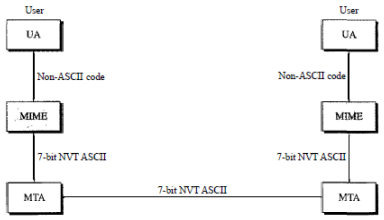
MIME

Electronic mail has a simple structure. Its simplicity, however, comes at a price. It can send messages only in NVT 7-bit ASCII format. In other words, it has some limitations.

For example, it cannot be used for languages that are not supported by 7-bit ASCII characters (such as French, German, Hebrew, Russian, Chinese, and Japanese). Also, it cannot be used to send binary files or video or audio data.

Multipurpose Internet Mail Extensions (MIME) is a supplementary protocol that allows non-ASCII data to be sent through e-mail. MIME transforms non-ASCII data at the sender site to NVT ASCII data and delivers them to the client MTA to be sent through the Internet. The message at the receiving side is transformed back to the original data

We can think of MIME as a set of software functions that transforms non-ASCII data (stream of bits) to ASCII data and vice versa, as shown in Figure

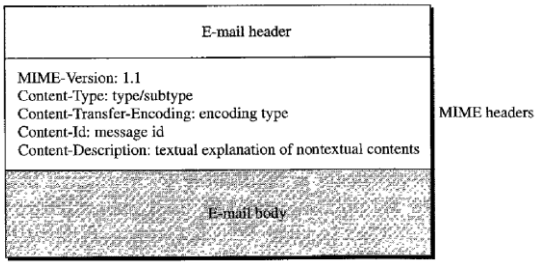


MIME defines five headers that can be added to the original e-mail header section to define the transformation parameters:

1. MIME-Version
2. Content-Type
3. Content-Transfer-Encoding

4. Content-Id

5. Content-Description



34. What is HTTP? Explain with respect to persistent and non-persistent connections.

- The HyperText Transfer Protocol (HTTP), the Web's application-layer protocol, is at the heart of the Web. It is defined in [RFC 1945] and [RFC 2616].
- HTTP is implemented in two programs: a client program and a server program.
- The client program and server program, executing on different end systems, talk to each other by exchanging HTTP messages. HTTP defines the structure of these messages and how the client and server exchange the messages.
- HTTP defines how Web clients request Web pages from Web servers and how servers transfer Web pages to clients.

Non Persistent HTTP

- Further suppose the URL for the base HTML file is

http://www.someSchool.edu/someDepartment/home.index

- Here is what happens:

1. The HTTP client process initiates a TCP connection to the server www.someSchool.edu on port number 80, which is the default port number for HTTP. Associated with the TCP connection, there will be a socket at the client and a socket at the server.

2. The HTTP client sends an HTTP request message to the server via its socket. The request message includes the path name /someDepartment/home.index.

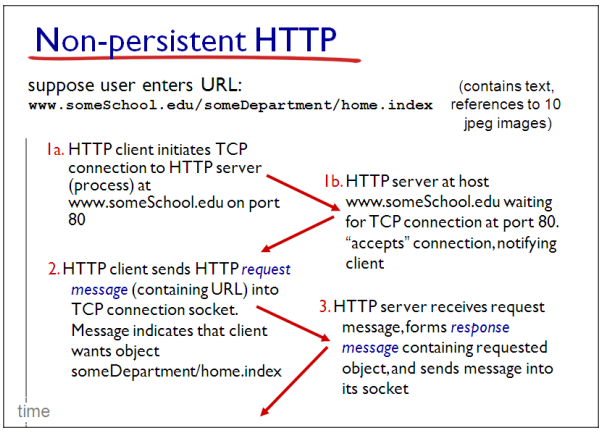
3. The HTTP server process receives the request message via its socket, retrieves the object /someDepartment/home.index from its storage (RAM or disk), encapsulates the object in an HTTP response message, and sends the response message to the client via its socket.

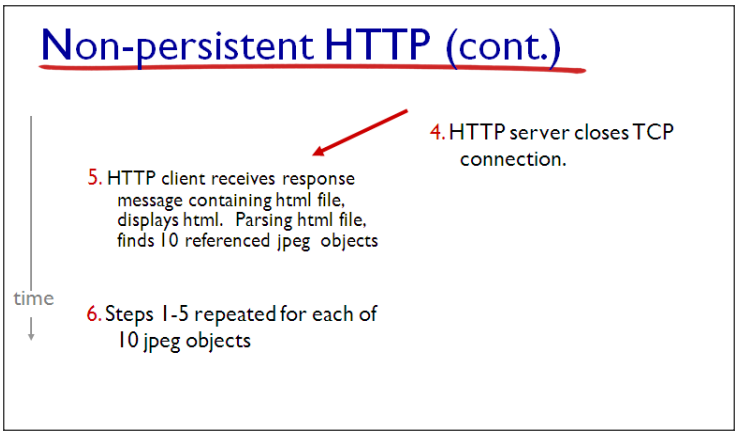
4. The HTTP server process tells TCP to close the TCP connection. (But TCP doesn't actually terminate the connection until it knows for sure that the client has received the response

message intact.)

5. The HTTP client receives the response message. The TCP connection terminates. The message indicates that the encapsulated object is an HTML file. The client extracts the file from the response message, examines the HTML file, and finds references to the 10 JPEG objects.

6. The first four steps are then repeated for each of the referenced JPEG objects.

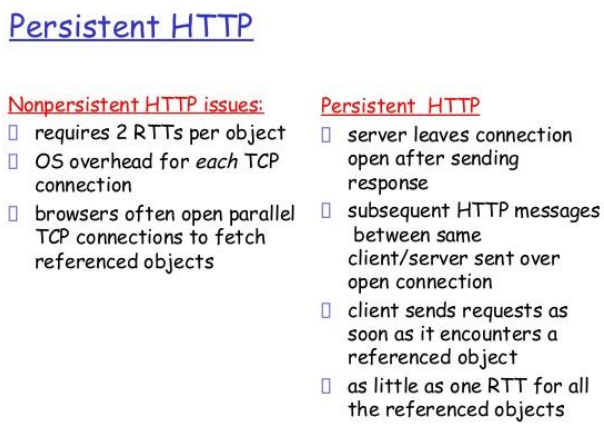




Persistent HTTP

- With persistent connections, the server leaves the TCP connection open after sending a response.
- Subsequent requests and responses between the same client and server can be sent over the same connection.
- In particular, an entire Web page (in the example above, the base HTML file and the 10 images) can be sent over a single persistent TCP connection.
- Moreover, multiple Web pages residing on the same server can be sent from the server to the same client over a single persistent TCP connection.
- These requests for objects can be made back-to-back, without waiting for replies to pending requests (pipelining).

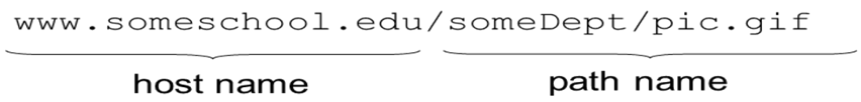
- Typically, the HTTP server closes a connection when it isn't used for a certain time (a configurable timeout interval).
- When the server receives the back-to-back requests, it sends the objects back-to-back.
- The default mode of HTTP uses persistent connections with pipelining.



35. Explain WWW and HTTP

WWW

- In the early 1990s, a major new application arrived on the scene—the World Wide Web [Berners-Lee 1994].
- The Web was the first Internet application that caught the general public's eye. It dramatically changed, and continues to change, how people interact inside and outside their work environments. It elevated the Internet from just one of many data networks to essentially the one and only data network.
- *web page* consists of *objects*
- object can be HTML file, JPEG image, Java applet, audio file,…
- web page consists of *base HTML-file* which includes *several referenced objects*
- each object is addressable by a *URL*, e.g.,



HTTP

The Hypertext Transfer Protocol (HTTP) is a protocol used mainly to access data on the World Wide Web. HTTP functions as a combination of FTP and SMTP. It is similar to FTP because it transfers files and uses the services of TCP. However, it is much simpler

than FTP because it uses only one TCP connection. There is no separate control connection; only data are transferred between the client and the server. HTTP is like SMTP because the

data transferred between the client and the server look like SMTP messages. In addition, the format of the messages is controlled by MIME-like headers. Unlike SMTP, the HTTP messages are not destined to be read by humans; they are read and interpreted by the HTTP server and HTTP client (browser). SMTP messages are stored and forwarded, but HTTP messages are delivered immediately. The commands from the client to the server are embedded in a request message. The contents of the requested file or other information are embedded in a response message. HTTP uses the services of TCP on well-known port 80.

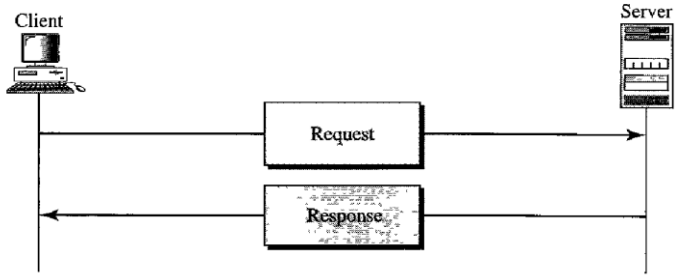
HTTP uses the services of TCP on well-known port 80.

HTTP Transaction

Figure illustrates the HTTP transaction between the client and server. Although

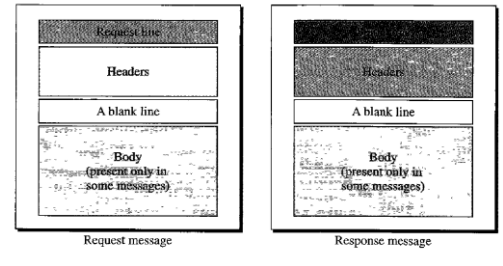
HTTP uses the services of TCP, HTTP itself is a stateless protocol. The client initializes

the transaction by sending a request message. The server replies by sending a response.



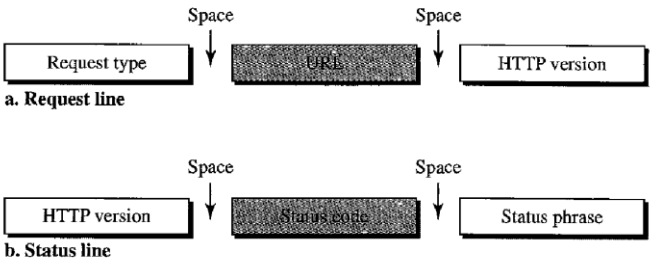
Messages

The formats of the request and response messages are similar; both are shown in Figure A request message consists of a request line, a header, and sometimes a body. A response message consists of a status line, a header, and sometimes a body.



Request and response messages

Request and Status Lines The first line in a request message is called a request line; the first line in the response message is called the status line. There is one common field, as shown in Figure.



Request and status lines

Request type. This field is used in the request message. In version 1.1 of HTTP, several request types are defined. The request type is categorized into *methods* as defined in Table

Method	*Action*
GET	Requests a document from the server
HEAD	Requests information about a document but not the document itself
POST	Sends some information from the client to the server
PUT	Sends a document from the server to the client
TRACE	Echoes the incoming request
CONNECT	Reserved
OPTION	Inquires about available options

36. Consider the following HTTP message and answer the following questions:

```
GET /home.asp HTTP/1.1
Host: gtu.ac.in
Accept-Encoding: gzip, deflate, sdch
Accept-Language: en-US,en;q=0.8
Cookie: OGPC=5061921-11:5061952-13:5061985-24:5061983-27:5061968-13:5062004-
7:5062009-6:5062022-12:;
SID=DQAAALgBAAA3RAje9UlLOOSuH0G91uzL5JOJNUYU2aVOml6jEWVTCo9-
User-Agent: Chrome/49.0.2623.110
X-Client-Data: CIS2yQEIprbJAQjDtskBCP2VygE
Connection:keep-alive
```

i. From which browser URL is requested?

Google Chrome

ii. Does browser request a non-persistent or a persistent connection?
Persistent

iii. Which is the (complete) URL of the document requested by the user?
www.gtu.ac.in/home.asp

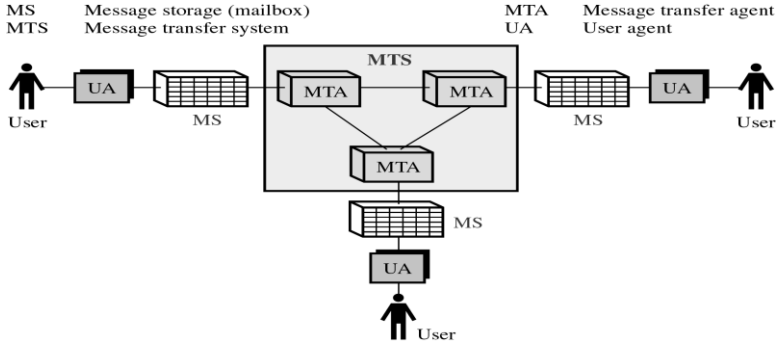
iv. Which HTML method is used to retrieve the requested URL?
GET

37. Explain the high-level view of Internet e-mail system and its major components.

Message Handling System (MHS):

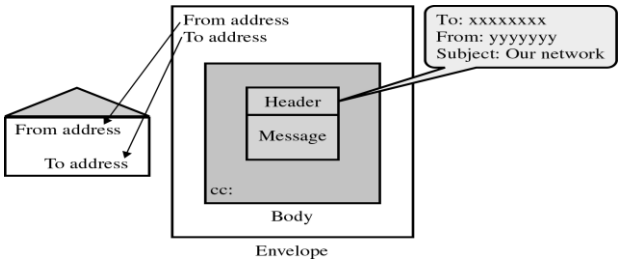
- MHS is the OSI protocol that underlines electronic mail and store-and-forward handling.
- It is derived from the ITU-T **X.400** series.
- MHS is the system used to send any message (including copies of data and files) that can be delivered in a store-and-forward manner.
    - Store-and-forward delivery means that, instead of opening an active channel between the sender and receiver, the protocol provides a delivery service that forwards the message when a link becomes available.
    - In most information-sharing protocols, both sender and the receiver must be able to participate in the exchange concurrently.
    - In a store-and-forward system, the sender passes the message to delivery system.
    - The delivery system may not be able to transmit the message immediately, in which case it stores the message until conditions change.
    - When the message is delivered, it is stored in the recipient's mailbox until called for. For example, the regular postal system.
- **The structure of the MHS:**
    - The structure of the OSI message handling system is shown in figure.
    - Each user communicates with a program or process called a **user agent (UA)**.
    - The UA is unique for each user (each user receives a copy of the program or process).
    - An example of a UA is electronic mail program associated with a specific operating system that allows a user to type and edit messages.
    - Each user has **message storage (MS)**, which consists of disk space in a mail storage system and is usually referred to as a mailbox.
    - Message storage can be used for storing, sending, or receiving messages.
    - The message storage communicates with a series of processes called **message transfer agents** (**MTA**s).

- MTAs are like the different departments of a post office.
- The combined MTAs make up the **message transfer system (MTS)**.



[MHS]

- **Message Format**
  - The MHS standard defines the format of a message.
  - The body of the message corresponds to the material (like a letter) that goes inside of the envelope of a conventional mailing.
  - Every message can include the address (name) of the sender, the address (name) of the recipient, the subject of the message, and a list of anyone other than the primary recipient who is to receive a copy.



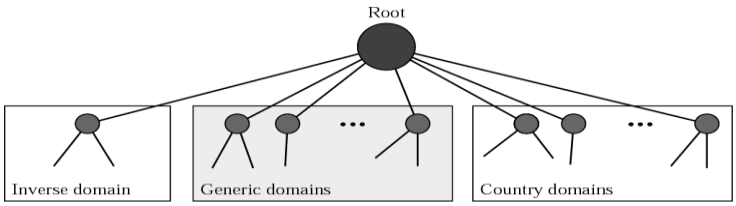
[Message format in MHS]

38. Explain DNS.

Introduction

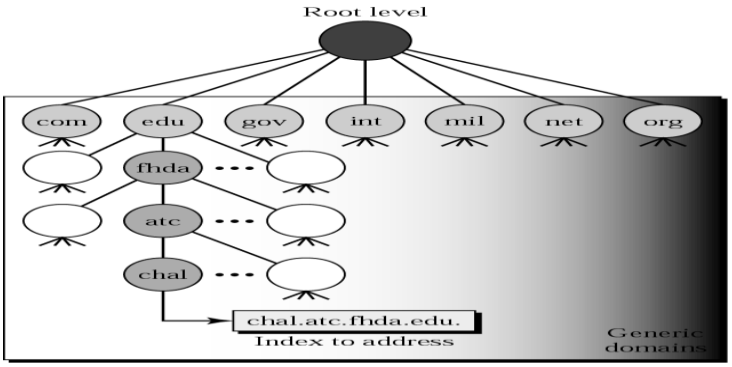
- To identify an entity, TCP/IP protocols use the IP address, which uniquely identifies the connection of a host to the internet.
- However, people prefer to use names instead of addresses.
- Therefore, we need a system that can map a name to an address and conversely an address to a name. In TCP/IP, this is the Domain Name System (DNS).
- DNS is a protocol that can be used in different platforms.
- In the internet, the domain name space (tree) is divided into three different sections:
  1) Generic domains

2) Country domains
3) Inverse domain.



1) Generic Domains:

The generic domains define registered hosts according to their behavior.



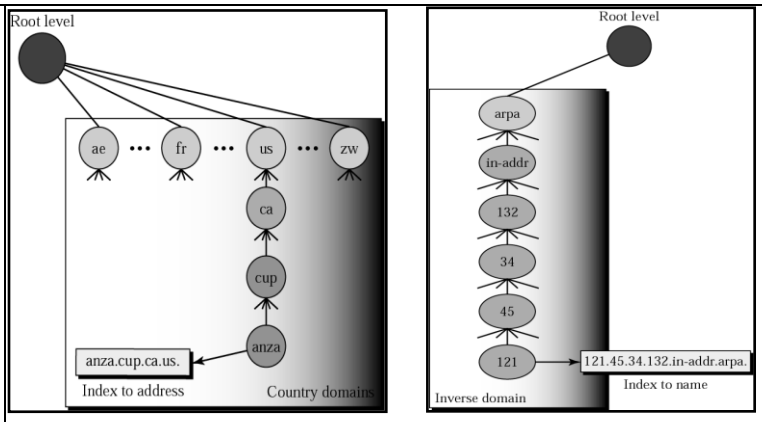
Label	Description
com	**Commercial organizations**
edu	**Educational institutions**
gov	**Government institutions**
int	**International organizations**
mil	**Military groups**
net	**Network support centers**
org	**Nonprofit organizations**

2) Country Domains:

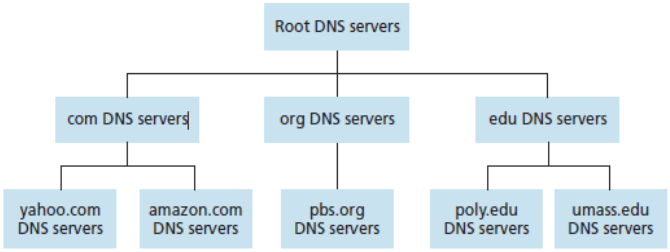
The country domain section follows the same format as the generic domains but uses two-character country abbreviations (eg., "in" for India).

3) Inverse Domains:

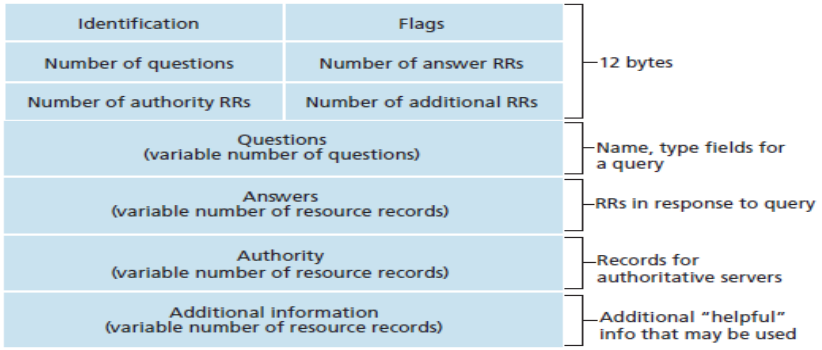
The inverse domain is used to map an address to a name.



Hierarchy of DNS servers

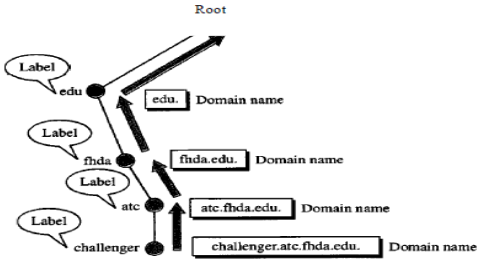


DNS Message



DNS Message Format

Domain names and labels



Fully Qualified Domain Name:

If a label is terminated by a null string, it is called a fully qualified domain name

(FQDN). An FQDN is a domain name that contains the full name of a host. It contains all labels, from the most specific to the most general, that uniquely define the name of the host. For example, the domain name

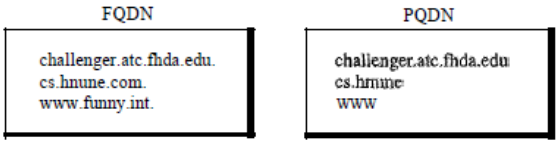
challenger.ate.tbda.edu.

is the FQDN of a computer named *challenger* installed at the Advanced Technology Center (ATC) at De Anza College. A DNS server can only match an FQDN to an address. Note that the name must end with a null label, but because null means nothing, the label ends with a dot (.).

Partially Qualified Domain Name

If a label is not terminated by a null string, it is called a partially qualified domain name (PQDN). A PQDN starts from a node, but it does not reach the root. It is used when the name to be resolved belongs to the same site as the client. Here the resolver can supply the missing part, called the suffix, to create an FQDN. For example, if a user at the *jhda.edu.* site wants to get the IP address of the challenger computer, he or she can define the partial name Challenger.

The DNS client adds the suffix *atc.jhda.edu.* before passing the address to the DNS server

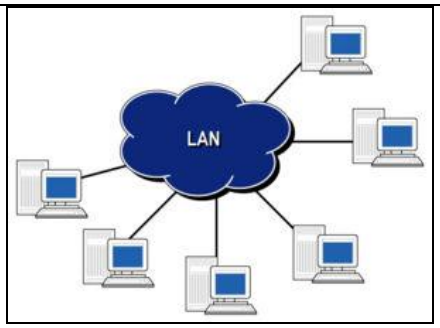


39. Explain the terms :

1) LAN
2) MAN
3) WAN
4) Internet

1. LAN:

- A **Local Area Network (LAN)** is a **collection of networking equipment** located geographically **close together.** **E.g.** Single room, campus etc.
- **Data transferred** in High speed which **ranges** from **100 Mbps to gigabit** for **system development** and have a **low implementation cost**.
- Upper limit: **1 km**
- **Twisted pair cable or Co-axial cable** connects the **plug in cards** to form a network.
- Designed to **share resources** between **PCs and workstation** such as **hardware or data.**



MERITS:

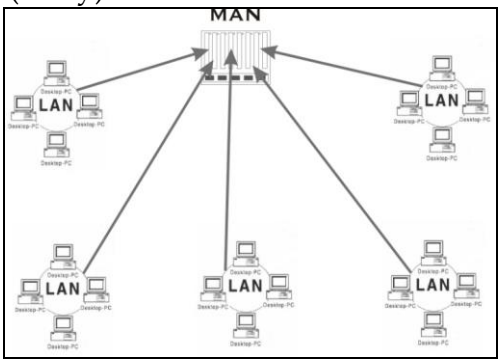
- **Cost reductions** through **sharing** of **information and databases, resources and network services.**
- Increased **information exchange** between different **departments in** an **organization**, or between **individuals.**

DEMERITS

- Special security measures are needed to stop users from using programs and data that they should not have access to;
- Networks are difficult to set up and need to be maintained by skilled technicians.
- If the file server develops a serious fault, all the users are affected, rather than just one user in the case of a stand-alone machine.

2. MAN:

- It is in between LAN & WAN technology that covers the entire city.
- It uses similar technology as LAN.
- It can be a single network such as cable TV network, or a measure of connecting a number of LAN's o a large network so that resources can be shared LAN to LAN as well as device to device.
- It uses DUAL BUS for data transmissions
- Range: Within **5 to 50 km** (a city).



MERITS:

- The dual bus used in MAN helps the transmission of data in both direction simultaneously.

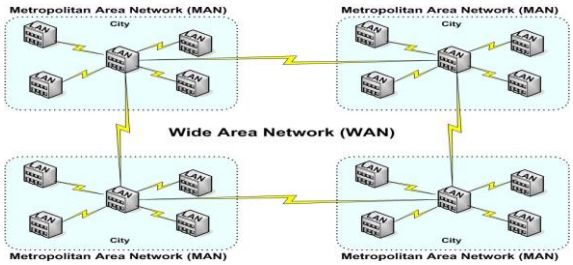
- A Man usually encompasses several blocks of a city or an entire city.

DEMERITS

- More cable required for a MAN connection from one place to another.
- It is difficult to make the system secure from hackers

3. WAN

- When network spans over a large distance or when the computers to be connected to each other are at widely separated locations a local area network cannot be used. A wide area network (WAN) is installed.
- The communication between different users of WAN is established using leased telephone lines, satellite links and similar channels.
- It is cheaper and more efficient to use the phone network for the link.
- Most WAN networks are used to transfer large blocks of data between its users.



MERITS:

- Covers a large geographical area so long distance businesses can connect on the one network.
- Shares software and resources with connecting workstations.
- Messages can be sent very quickly to anyone else on the network. These messages can have pictures, sounds, or data included with them (called attachments).

DEMERITS:

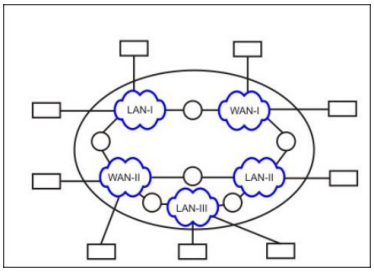
- **Need a good firewall to restrict outsiders** from entering and disrupting the network
- Setting up a **network** can be an **expensive, slow and complicated.** The **bigger the network** the **more expensive** it is.

PARAMETERS	LAN	WAN	MAN
Ownership of network	Private	Private or public	Private or public
Geographical area covered	Small	Very large	Moderate

Design and maintenance	Easy	Not easy	Not easy
Communication medium	Coaxial cable	PSTN or satellite links	Coaxial cables, PSTN, optical fibre, cables, wireless
Bandwidth	Low	High	moderate
Data rates(speed)	High	Low	moderate

4. Internet

- Internet is a collection of networks or network of networks.
- Various networks such as LAN and WAN connected through suitable hardware and software to work in a seamless manner.
- It allows various applications such as e-mail, file transfer, remote log-in, World Wide Web, Multimedia, etc run across the internet.
- The basic difference between WAN and Internet is that WAN is owned by a single organization while internet is not so. But with the time the line between WAN and Internet is shrinking, and these terms are sometimes used interchangeably.



40. Explain the terms :

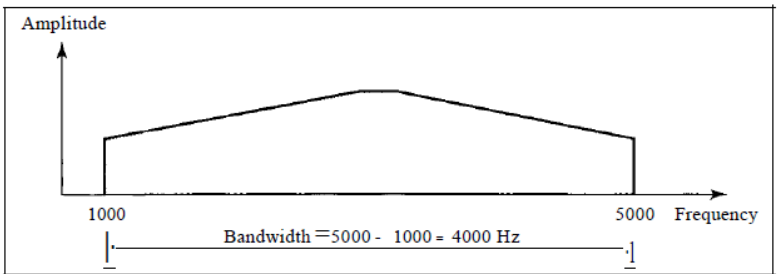
1) Jitter

Jitter is defined as a variation in the delay of received packets. The sending side transmits packets in a continuous stream and spaces them evenly apart. Because of network congestion, improper queuing, or configuration errors, the delay between packets can vary instead of remaining constant.

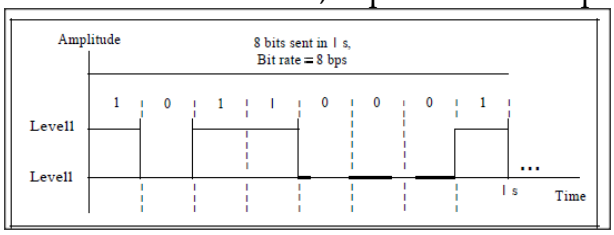
2) Bandwidth

In networking, we use the term bandwidth in two contexts.

1. The first, in **analog signals**, bandwidth in hertz, refers to the range of frequencies in a composite signal or the range of frequencies that a channel can pass.
    - The bandwidth is normally a difference between two numbers. For example, if a composite signal contains frequencies between 1000 and 5000, its bandwidth is 5000 - 1000, or 4000.



2. The second, in **digital signals** bandwidth in bits per second, refers to the speed of bit transmission in a channel or link. Often referred to as Capacity.
   - Another term-bit rate (instead of frequency)-is used to describe digital signals. The bit rate is the number of bits sent in 1 sec, expressed in bits per second (bps).

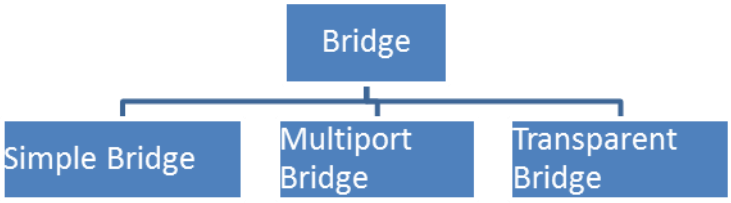


3) Throughput

In data transmission, network throughput is the amount of data moved successfully from one place to another in a given time period, and typically measured in bits per second (bps), as in megabits per second (Mbps) or gigabits per second (Gbps).

4) Bridge

- Bridge operates at both the physical layer and data link layer.
- Bridge divides a larger network onto smaller segments.
- Maintain traffic for each segment.
- Maintain physical address of each node of each segment.
- Store address into look-up table.
- Types of bridge

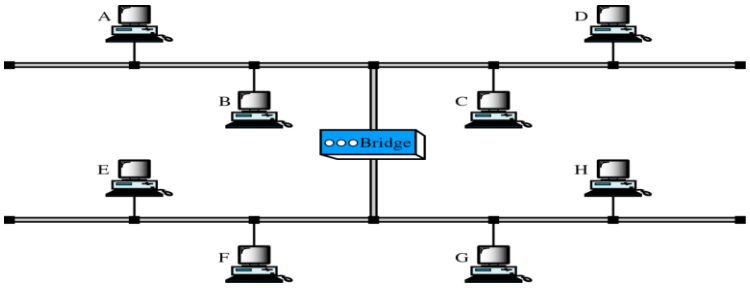


1. **Simple Bridge**
   - Used to connect two segments

- Least expensive
- Address of each node entered manually
- Installation and maintenance is high & time consuming

2. Multiport Bridge

- Used to connect more than two LAN
- Maintain physical address of each station
- If three segment is connected then it maintain three tables
- Address of each node entered manually
- Installation and maintenance is high & time consuming

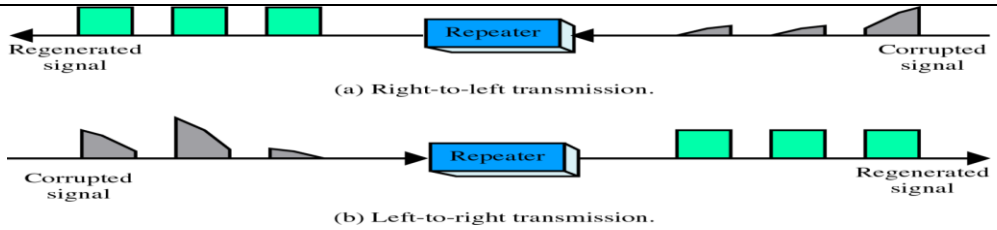


3. Transparent Bridge

- It's also known as a learning Bridge
- Maintain physical address of each station
- Manually entry of each node is not required
- Maintain address by its own table is empty
- At initial level
- After each process its storing details of each node
- Its self updating

5) Repeater

- Repeater is also known as a Regenerator.
- It's an electronic device.
- Used at Physical layer of OSI model.
- Used for carry information at longer distance.
- Not work as an amplifier, it's not amplifying the signal.
- It regenerates the original bit patterns from weak signal patterns.
- So, Repeater is a regenerator not an amplifier.
- It's not an intelligent device.



6) Hub and Switch

Hub

- Used at physical layer
- Dump device
- Not maintaining any type of node details
- Each time broadcast the packets
- Cheaper
- Used to connect two or more number of computers

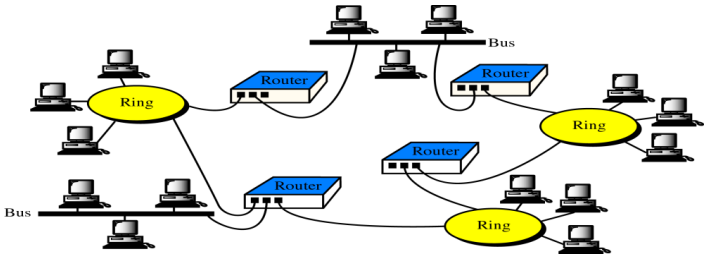
Switch

- Two type of switch: Layer two and Layer three switch
- Layer two switch is operates at second layer and layer three switch is operates at third layer
- We can configure layer three switch
- Switch(l2) maintain physical address of each node
- L3 switch handle logical address
- Not manually handling is required for tables
- For the first time broadcast the message if destination address is not available in the table.
- Two different strategies of switch
    - **Store and forward Switch:** store the frame in the input buffer until the whole packet has arrived.
    - **Cut through Switch:** Not wait for other frames; just transmit it towards the destination

7) Router

- It is used at network layer of OSI model
- Maintain logical address of each node
- It's an intelligent device
- It has its own software
- Determines the best path among available different path
- Used to connect two different networks

- Manually handling is not required
- We can configure the router according to our requirement
- Maintain address by its own
- At initial level table is empty
- After each process its storing details of each node
- Its self updating



8) Gateway

- Operates In all seven layer of OSI model
- Gateway is a protocol converter
- Both network is working on different protocol then accept a packet from one network and transmit packet to the different network.
- It convert's packet into suitable form
- A gateway is software installed within a router.
- A gateway adjusts the data rate size and format of each packet.

